5G RAN

Synchronization Feature Parameter Description

Issue Draft B

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Contents

1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

  None

- Editorial changes

  Revised the descriptions of frequency synchronization. For details, see **Frequency Synchronization**.

  Revised the descriptions of manually updating the start week of a satellite card using an MML command. For details, see **4.1 Principles**.

  Added descriptions of the position check function for satellite cards. For details, see **4.1 Principles** and **4.2.2 Impacts**.

  Revised the descriptions of the hardware requirements for synchronization with GPS. For details, see **4.3.3 Hardware**.

  Revised the descriptions of data preparation for synchronization with BeiDou. For details, see **5.4.1.1 Data Preparation**.

  Revised the principles of the dual-reference-clock backup mode of IEEE 1588V2. For details, see **Dual-Reference-Clock Backup Mode of IEEE 1588V2** and **Dual-Reference-Clock Backup Mode of IEEE 1588V2**.

  Added the conditions for a base station to support the priority clock class function. For details, see **7.1.4 Interworking Between IEEE 1588V2-compliant Equipment from Different Manufacturers**.

  Revised the principles of automatic synchronization source selection and switching. For details, see **10.1.1 Automatic Synchronization Source Selection and Switching**.

  Added the versions corresponding to the profile types for IEEE 1588V2 clock synchronization. For details, see **7.1.4 Interworking Between IEEE 1588V2-compliant Equipment from Different Manufacturers**.

  Revised descriptions in this document.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 04 (2025-08-30).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added the UMPThb series boards and related descriptions. For details, see: • **3.1.2 Clock Quality Levels** • **3.1.4.4 Holdover** • **4.3.3 Hardware** • **6.3.3 Hardware**	None	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added an MML command to directly modify the configuration of a specified IP clock link. For details, see: • **7.4.1.1 Data Preparation** • **7.4.1.2 Using MML Commands**	Added the **MOD IPCLKLINK** command.	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added support for querying the historical status of **Grandmaster Quality Class** of an IP clock server. For details, see **7.4.2 Activation Verification**.	None	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

<table>
<tr><th>Change Description</th><th>Parameter Change</th><th>RAT</th><th>Base Station Model</th></tr>
<tr><td>Added support for automatic switchback between the IEEE 1588V2 and 1PPS +TOD clock sources, and between the IEEE 1588V2 and peer clock sources. For details, see:
10.1 Principles
10.1.2 Automatic Switchback
10.4.1.1 Data Preparation
10.4.1.2 Using MML Commands
10.4.2 Activation Verification</td><td>Modified parameter: Added the values IPCLK +TOD and IPCLK+PEERCLK to the TASM.SWITCHPOLICY parameter.</td><td>Low-frequency TDD
High-frequency TDD</td><td>3900 and 5900 series base stations (macro base stations)
DBS3900 LampSite and DBS5900 LampSite</td></tr>
<tr><td>Added the synchronization source selection policy after the maximum holdover time of the base station elapses. For details, see 10.1.1 Automatic Synchronization Source Selection and Switching.</td><td>None</td><td>Low-frequency TDD
High-frequency TDD</td><td>3900 and 5900 series base stations (macro base stations)
DBS3900 LampSite and DBS5900 LampSite</td></tr>
</table>

Change Description	Parameter Change	RAT	Base Station Model
Added support for clock out-of-synchronization detection in cells covering subways or high-speed railways. For details, see: • **11.1.2 Clock Out-of-Synchronization Detection 2.0** • **11.3.2.2 Mutually Exclusive Functions** • **11.3.4 Networking** • **11.4.1.1 Data Preparation** • **11.4.1.2 Using MML Commands**	Modified parameter: Added the **CLK_DETECT_IN_COMPLEX_SW** option to the **gNodeBParam**.***ClkOutofsyncDetSwitch*** parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between clock out-of-synchronization detection and RF module fast deep dormancy. For details, see **11.3.2.3 Function Impacts**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between ultra-long holdover for sites with abnormal clock sources and RF module fast deep dormancy. For details, see **12.4.3.2.3 Function Impacts**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

- Editorial changes

  Revised the descriptions of the impact relationships between clock out-of-synchronization detection (low-frequency TDD) and some power saving functions. For details, see **11.3.2.3 Function Impacts**.

  Revised other descriptions in this document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FBFD-010020	Synchronization	4 Synchronization with GPS 5 Synchronization with BeiDou 6 Synchronization with Galileo 8 1PPS+TOD Clock 10 Synchronization Source Switching 11 Clock Out-of-Synchronization Detection (Low-Frequency TDD) 12 Intelligent Clock Fault Joint Diagnosis and Self-healing (Low-Frequency TDD)
FOFD-010070	Network Synchronization	7 IEEE 1588V2 Clock Synchronization 9 Combined Synchronization Sources 10 Synchronization Source Switching

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Synchronization with GPS	None	4 Synchronization with GPS
Synchronization with BeiDou	None	5 Synchronization with BeiDou
Synchronization with Galileo	None	6 Synchronization with Galileo
IEEE 1588V2 clock synchronization	None	7 IEEE 1588V2 Clock Synchronization

Function Name	Difference	Chapter/Section
1PPS+TOD clock	None	**8 1PPS+TOD Clock**
Combined synchronization sources	None	**9 Combined Synchronization Sources**
Synchronization source switching	None	**10 Synchronization Source Switching**
Clock out-of-synchronization detection (low-frequency TDD)	None	**11 Clock Out-of-Synchronization Detection (Low-Frequency TDD)**
Intelligent clock fault joint diagnosis and self-healing (low-frequency TDD)	None	**12 Intelligent Clock Fault Joint Diagnosis and Self-healing (Low-Frequency TDD)**

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Synchronization with GPS	None	**4 Synchronization with GPS**
Synchronization with BeiDou	None	**5 Synchronization with BeiDou**
Synchronization with Galileo	None	**6 Synchronization with Galileo**
IEEE 1588V2 clock synchronization	None	**7 IEEE 1588V2 Clock Synchronization**
1PPS+TOD clock	None	**8 1PPS+TOD Clock**
Combined synchronization sources	None	**9 Combined Synchronization Sources**
Synchronization source switching	None	**10 Synchronization Source Switching**

Function Name	Difference	Chapter/Section
Clock out-of-synchronization detection (low-frequency TDD)	Supported only in low frequency bands	**11 Clock Out-of-Synchronization Detection (Low-Frequency TDD)**
Intelligent clock fault joint diagnosis and self-healing (low-frequency TDD)	Supported only in low frequency bands	**12 Intelligent Clock Fault Joint Diagnosis and Self-healing (Low-Frequency TDD)**

3 Overview

This chapter consists of the sections "General Principles" and "Document Outline". "General Principles" describes the basic principles of clocks, and "Document Outline" describes the document architecture and function introduction.

3.1 General Principles

In a digital communications network, synchronization ensures that the clock time or frequency difference among communications equipment across the entire network is within a reasonable error range. In this way, synchronization prevents the transmission performance deterioration caused by the incorrect timing of receiving or transmitting signals in the transmission system. Clock synchronization maintains a certain relationship between two or more signals in terms of time or frequency. Clock synchronization consists of time synchronization and frequency synchronization.

NR TDD networks adopt time division multiplexing and therefore require time synchronization to minimize inter-base-station and inter-UE interference.

This document uses global navigation satellite system (GNSS) as a collective name for Global Positioning System (GPS), Global Navigation Satellite System (GLONASS), BeiDou Navigation Satellite System (BDS), and Galileo Satellite Navigation System (Galileo).

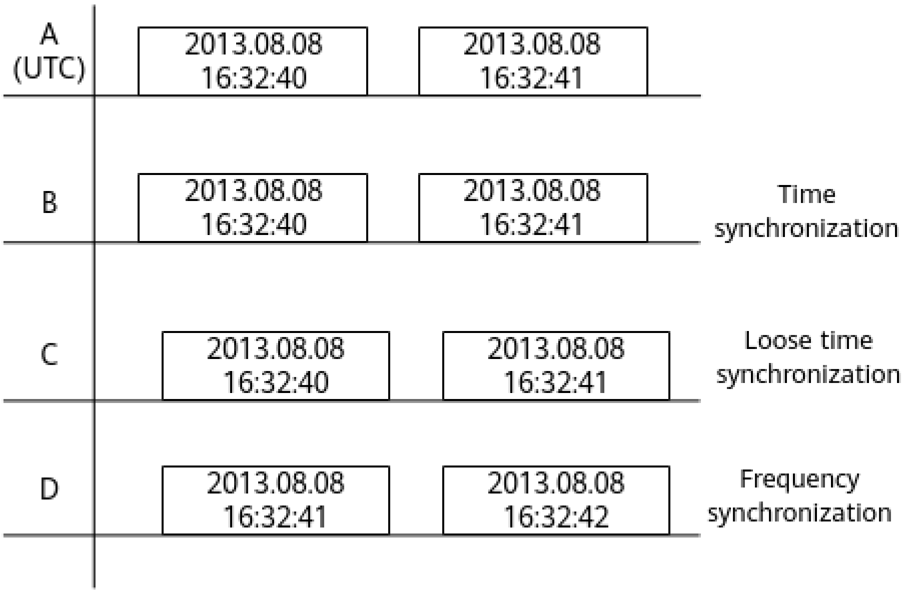
3.1.1 Time Synchronization and Frequency Synchronization

Time Synchronization

Time signals are clock signals that contain time information, including year, month, date, hour, minute, and second. Currently, Coordinated Universal Time (UTC), a universal timing standard, is used to represent time information.

Time synchronization is also known as moment synchronization, and implies the synchronization of the absolute time. It requires that the starting time of the clock signals for a device keeps consistent with that of UTC time. **Figure 3-1** shows a time synchronization example, in which signals A and B are time synchronization signals while signals A, C, and D are not.

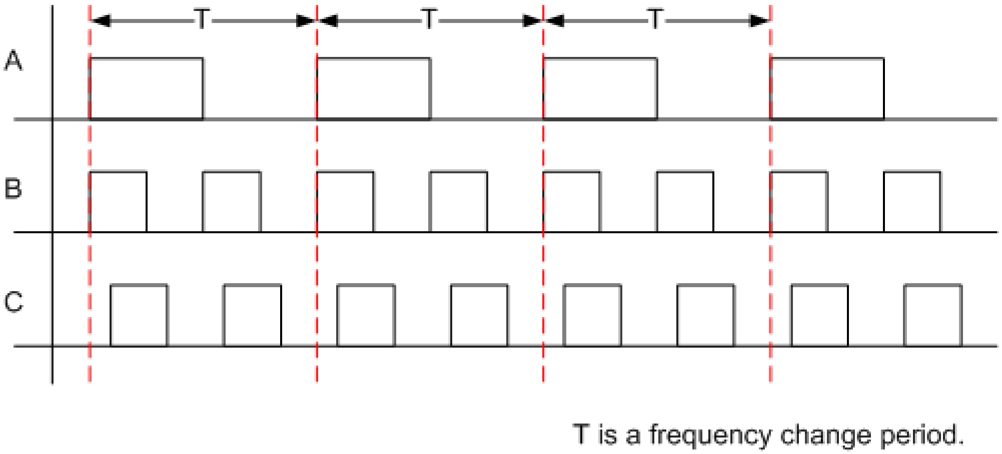
Figure 3-1 Time synchronization



Frequency Synchronization

Frequency synchronization means the same change in frequencies of two signals or a fixed ratio of frequency changes, making allowance for different signal phases. Signals change periodically and do not contain time information. **Figure 3-2** shows a frequency synchronization example, in which signals A, B, and C are frequency synchronization signals.

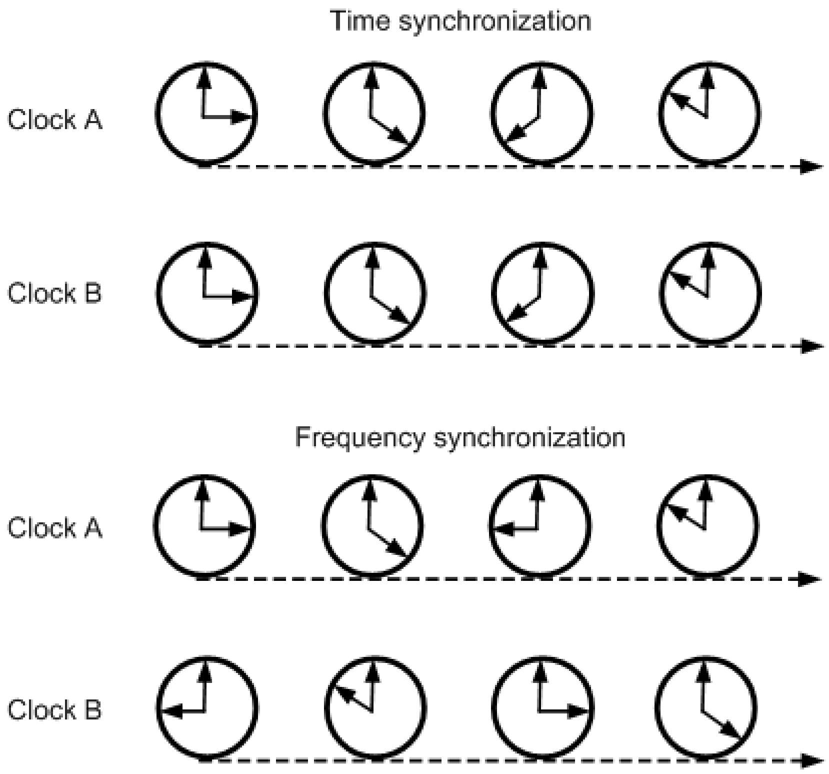
Figure 3-2 Frequency synchronization



Difference Between Time Synchronization and Frequency Synchronization

Figure 3-3 shows an example of the difference between time synchronization and frequency synchronization.

Figure 3-3 Difference between time synchronization and frequency synchronization



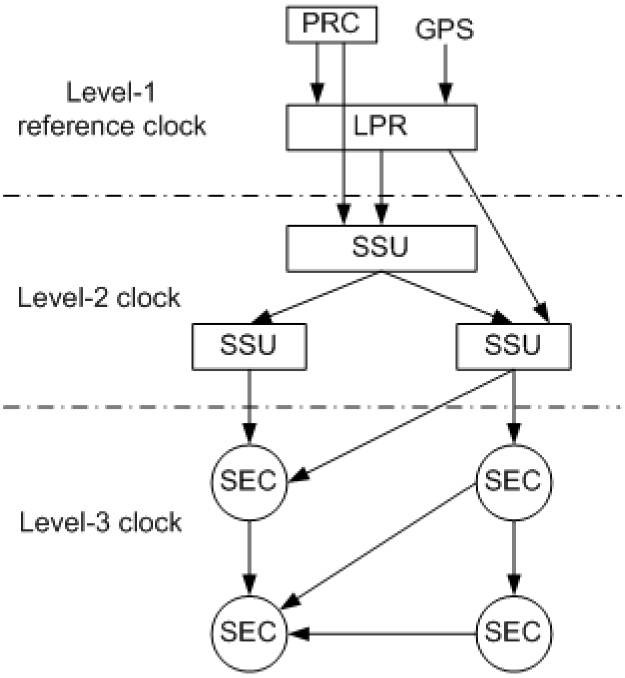
As shown in **Figure 3-3**:

- If the time of clock A and clock B is the same at every moment, they are in time synchronization.
- If the time of clock A and clock B is different but the time difference maintains a fixed value (for example, six hours), they are in frequency synchronization.

3.1.2 Clock Quality Levels

Clocks can be classified into level-1 reference clock, level-2 clock, and level-3 clock, based on the clock quality level. Clock equipment is layered according to the clock quality level and constitutes a hierarchical clock synchronization network through transmission links, as shown in **Figure 3-4**.

Figure 3-4 Clock quality levels



As shown in **Figure 3-4**:

- Level-1 reference clock: primary reference clock (PRC)/local primary reference (LPR) clock
  - PRC: usually a cesium or hydrogen clock providing a high-accuracy synchronization source
  - LPR clock: usually a rubidium clock plus global navigation satellite system (GNSS) providing a high-accuracy synchronization source
- Level-2 clock: synchronization supply unit (SSU). A level-2 clock is equipped with a digital phase locked loop (DPLL) based on a rubidium clock, providing excellent tracing, filtering, and holdover performance. The level-2 clock is divided into:
  - Tandem office slave clock in SSU-A (primary SSU) level.
  - End office slave clock in SSU-B (second-level SSU) level.
- Level-3 clock: Synchronous Digital Hierarchy (SDH) equipment clock (SEC). A level-3 clock is equipped with a DPLL based on a high-performance crystal oscillator. The holdover performance of the level-3 clock is lower than that of the level-2 clock. Clocks provided by network equipment are in SEC level.

The clock server belongs to a level-1 reference clock, while the base station clock is level-3, and its upper-level clock must meet the requirements of level-2. In principle, a lower-level device must synchronize with its upper-level device to achieve clock synchronization.

Transmission equipment is usually connected in a ring formation. To prevent higher-level node clocks from tracing the synchronization source of lower-level node clocks, synchronization sources use Synchronization Status Messages (SSMs) to indicate their clock quality levels.

Main control boards other than the UMPTe/UMPTg/UMPTga/UMPTha/UMPThb do not support NR. In multimode scenarios, when clock mutual lock is used, the clock used by NR may be on the main control boards of other modes. When clock mutual lock is used in multimode scenarios, other main control boards also support mutual lock with the NR clock. As a result, all boards (not only NR-capable main control boards) can report the clock quality level of NR.

3.1.3 Clock Startup Mode

Depending on whether the crystal oscillator of the synchronization source needs to be warmed up before it works normally, the clock startup modes are classified into the following:

- **Cold Startup**
- **Warm Startup**

Cold Startup

In cold startup mode, the crystal oscillator needs some time to warm up before working normally after the system clock enters the startup mode. When the system clock starts up:

- If there are available synchronization sources, the system clock enters the locked mode after the warm-up. For details about the locked mode, see **3.1.4.3 Locked**.
- If there are no available synchronization sources, the system clock enters the free running mode after the warm-up. For details about the free running mode, see **3.1.4.1 Free Running**.

Warm Startup

In warm startup mode, the system clock immediately enters the locked mode after the system restarts without the need of turning off power supply. Warm startup is available only when the system clock works in locked mode before the warm startup. For details about the locked mode, see **3.1.4.3 Locked**.

3.1.4 Clock Working Mode

After the crystal oscillator warms up, the clock enters one of the following working modes:

- **3.1.4.1 Free Running**
- **3.1.4.2 Fast Tracking**
- **3.1.4.3 Locked**
- **3.1.4.4 Holdover**

3.1.4.1 Free Running

A clock enters the free running mode in the following scenarios:

- There are no available reference clocks after the crystal oscillator warms up.
- Network reference clocks become unavailable and the unavailability duration exceeds the threshold specified by the holdover mode.

No synchronization source is configured for a base station before delivery, and therefore the base station directly enters the free running mode. When the base station re-obtains and locks a network reference clock, the base station clock exits the free running mode.

An NR TDD base station cannot work in free running mode. After the base station enters the free running mode, cells served by the base station are automatically deactivated. These cells are automatically activated only after the base station re-enters the locked mode.

3.1.4.2 Fast Tracking

A clock enters the fast tracking mode if the base station obtains a reference clock or if reference clocks become available but the phase offset exceeds the locking threshold.

- If synchronization sources become unavailable, the clock switches to the free running mode.
- When the frequency offset of synchronization sources is less than the traceable threshold but exceeds the locking threshold, the clock remains in fast tracking mode.
- When the frequency offset of synchronization sources is less than the locking threshold, the clock switches to locked mode.

3.1.4.3 Locked

The clock enters the locked mode in either of the following scenarios:

- The base station obtains reference clocks and the frequency offset is less than the locking threshold.
- The reference clocks become available when the base station clock runs in holdover mode, and the phase offset is less than the locking threshold.

The locked mode is the normal working mode of the clock for base stations. When the clock enters the locked mode, its jitter, frequency offset, and accuracy meet the requirements for the normal operation of the system. In addition, the frequency control values of the crystal oscillator are updated only when the frequency of reference clock signals is within the normal range. If the frequency is beyond the normal range, the update process stops.

3.1.4.4 Holdover

A clock enters the holdover mode when reference clocks become faulty, or the phase offset or frequency offset exceeds the locking threshold. The clock can work in holdover mode for a maximum of 90 days.

- In frequency synchronization mode, services can work properly for 90 days when the clock is in holdover mode.
- In time synchronization mode, the period of time during which services can work properly when the clock is in holdover mode differs depending on the main control board type, external clock source, and external environment.

NOTE

The clock holdover duration after the reference clock is lost can be queried by running the **DSP CLKSTAT** command and observing the output item **Holding Time(min)** (which corresponds to the **TASM**.***HOLDTIME*** parameter). This output item indicates the clock holdover duration rather than the period for the base station to provide services properly.

If the base station is reset when the clock is in the holdover mode, the clock immediately enters the free-running mode.

For NR TDD base stations, if the clock deviation between base stations exceeds a predefined threshold, neighboring base stations are interfered. With the clock in holdover mode, base stations can use the internal crystal oscillator to achieve synchronization for NR services if certain conditions are met. When the holdover duration exceeds the maximum duration supported for providing services in different scenarios, cells are shut down to avoid interfering neighboring functional base stations. The conditions are as follows:

- The clock source signals are normal and stable for the three days before the clock enters the holdover mode.
- The internal components (such as the crystal oscillator and phase-locked loop) work properly and stably.
- The BBU works at room temperature, the temperature is stable, and the voltage is stable without fluctuation.

When the holdover duration is too long, and the auto oscillation of the crystal oscillator cannot satisfy the holdover capability requirements posed by base station services, inter-base-station interference occurs.

The **gNodeBParam**.***GnssFailLfHoldoverDuration*** parameter can be used to specify the maximum duration for providing services after the GNSS clock source or a non-GNSS clock source (such as IEEE 1588V2 as well as the IEEE 1588V2 and synchronous Ethernet combination) is lost. The default value of this parameter is **HOLDOVER_24_HOURS**. A shorter holdover duration can also be configured by setting this parameter to **HOLDOVER_8_HOURS**. **Table 3-1** lists the maximum service durations corresponding to different values of **gNodeBParam**.***GnssFailLfHoldoverDuration*** after a clock source is lost.

Table 3-1 Service durations corresponding to different values of the **gNodeBParam**.***GnssFailLfHoldoverDuration*** parameter

Parameter Value	Synchronization Source	Main Control Board	Maximum Duration for Providing Low-Frequency Services (Hour)	Maximum Duration for Providing High-Frequency Services (Hour)
HOLDOVER_24_HOURS	GNSS	UMPTg, UMPTga, UMPTha, or UMPThb	24	24
		UMPTe	24	N/A

Parameter Value	Synchronization Source	Main Control Board	Maximum Duration for Providing Low-Frequency Services (Hour)	Maximum Duration for Providing High-Frequency Services (Hour)
	Non-GNSS clock sources such as IEEE 1588V2	UMPTg, UMPTga, UMPTha, or UMPThb	8	2
		UMPTe	8	N/A
HOLDOVER_8_HOURS	GNSS	UMPTg, UMPTga, UMPTha, or UMPThb	8	8
		UMPTe	8	N/A
	Non-GNSS clock sources such as IEEE 1588V2	UMPTg, UMPTga, UMPTha, or UMPThb	4	2
		UMPTe	4	N/A

NOTE

If the UMPTe is used, it is recommended that this parameter be set to **HOLDOVER_8_HOURS**. If this parameter is set to **HOLDOVER_24_HOURS**, inter-base-station interference may occur due to an excessive inter-base-station synchronization deviation.

If the UMPTg, UMPTga, UMPTha, or UMPThb is used, the default value **HOLDOVER_24_HOURS** is recommended. If this parameter is set to **HOLDOVER_8_HOURS**, NR cells become unavailable at an earlier time.

The **CLK_UNAVBL_TRP_ALM_SW** option of the **gNBOamParam**.***AlarmSwitch*** parameter specifies whether to report the NR DU Cell TRP Unavailable alarm when the clock resource is unavailable. When the clock resource of the base station is unavailable, the cells served by the base station are shut down. In this case, a cell reports ALM-29870 NR DU Cell TRP Unavailable only if this option is selected for the cell.

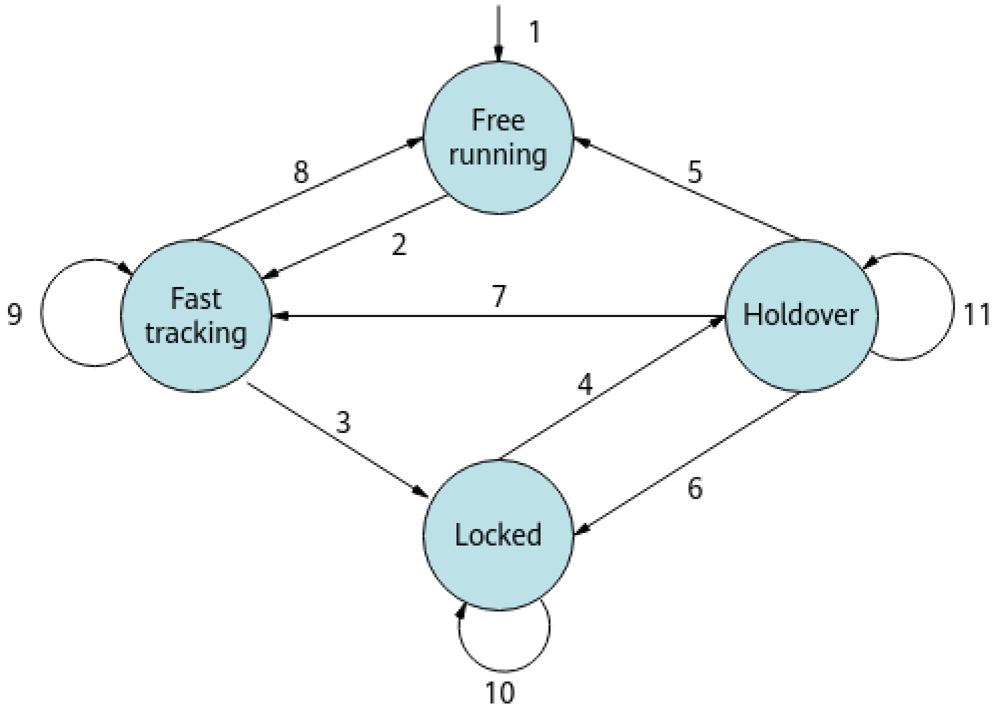
NOTE

The cell can report ALM-29870 NR DU Cell TRP Unavailable only if the **CLK_UNAVBL_TRP_ALM_SW** option of the **gNBOamParam**.***AlarmSwitch*** parameter is selected.

3.1.4.5 Working Mode Transition

The clock working mode can be switched when certain conditions are met. **Figure 3-5** shows the transition of the base station clock working modes.

Figure 3-5 Transition of the clock working modes



1. When there are no available reference clocks after the crystal oscillator warms up, the clock enters free running mode.
2. When a base station receives available clock signals or reference clocks become available, the clock switches from free running mode to fast tracking mode.
3. When the frequency offset of the reference clocks for the base station is less than the locking threshold, the clock switches from fast tracking mode to locked mode.
4. The clock enters holdover mode when any of the following conditions is true:
   – The clock source is faulty.
   – The phase offset exceeds the locking threshold.
   – The frequency offset exceeds the locking threshold.
5. When the duration of the clock in holdover mode exceeds the specified threshold, the clock switches from holdover mode to free running mode.
6. When reference clocks become available and the phase offset is less than the locking threshold, the clock switches from holdover mode to locked mode.
7. When reference clocks become available but the phase offset exceeds the locking threshold, the clock switches from holdover mode to fast tracking mode.
8. When the base station detects that the reference clocks become unavailable, the clock switches from fast tracking mode to free running mode.

9. When the base station detects that the frequency offset is less than the traceable threshold but greater than the locking threshold, the clock remains in fast tracking mode.
10. When the base station detects that the frequency offset is less than the locking threshold, the clock remains in locked mode.
11. When the duration of the clock in holdover mode does not exceed the specified threshold, the clock remains in holdover mode.

3.1.5 Locking Duration

If the crystal oscillator frequency of a base station is significantly deviated from that of a synchronization source or the quality of the transport network is poor, the maximum fast tracking durations for synchronization sources are described as follows:

- For the IEEE 1588V2 synchronization source: Considering jitters in IP transport networks, the reference clock can be locked within 30 minutes in normal cases. If the reference clock cannot be locked within 3200 minutes, it is considered to be faulty. If the fault persists for 200 seconds, the base station reports ALM-26262 External Clock Reference Problem.
- For GPS, BeiDou, or other synchronization sources: The reference clock can be locked within 2 minutes in normal cases. If the reference clock cannot be locked within 6 minutes, it is considered to be faulty. If the fault persists for 200 seconds, the base station reports ALM-26262 External Clock Reference Problem.

3.1.6 Synchronization Accuracy

The clock accuracy denotes the maximum offset of a base station clock from a reference synchronization source. 3GPP specifications define the following clock accuracy requirements of base station synchronization:

- If time synchronization is used, base stations must be time-synchronized within the accuracy of ±1.5 μs.
- If frequency synchronization is used, macro base stations must be frequency-synchronized within the accuracy of ±0.05 ppm.

3.1.7 Clock Synchronization Mode Adjustment

Some features require base stations to use time synchronization. If the base station's synchronization source uses frequency synchronization, you must change the synchronization mode from the synchronization source using frequency synchronization to a synchronization source using time synchronization before enabling these features.

The following clock sources support time synchronization:

- GPS
- BeiDou
- Galileo
- 1PPS+TOD clock
- IEEE 1588V2

NOTICE

Changing the synchronization mode causes base station frame numbers and frame phases to change, leading to temporary service interruptions. Therefore, you are advised to change the synchronization mode during off-peak hours.

The procedure for clock synchronization mode adjustment is as follows:

Step 1 Plan and prepare the data for the synchronization source using time synchronization.

Step 2 Reconfigure the base station's clock working mode. Run the **SET CLKMODE** command with the **TASM**.***MODE*** parameter set to **FREE**.

Step 3 Remove the original clock link by running the **RMV IPCLKLINK** command.

Step 4 Deploy the synchronization source using time synchronization. Run the **SET CLKSYNCMODE** command with the **TASM**.***CLKSYNCMODE*** parameter set to **TIME**.

Step 5 Observe the activation of the synchronization source using time synchronization.

----End

NOTE

For details about the network plan, data preparation, activation, and activation verification of the synchronization source using time synchronization, see sections "Requirements" and "Operation and Maintenance" for the corresponding synchronization source.

3.2 Document Outline

This document consists of the following contents:

- Synchronization sources available to gNodeBs, as listed in **Table 3-2**
- Switching between and combination of multiple synchronization sources, as listed in **Table 3-3**
- Clock out-of-synchronization detection and intelligent clock fault joint diagnosis and self-healing, as listed in **Table 3-4**

Table 3-2 Reference clocks available to gNodeBs

Synchronization Source	Supported Synchronization Mode	Described In
GPS/RGPS/CGPS	Frequency synchronization and time synchronization	**4 Synchronization with GPS**
BeiDou	Frequency synchronization and time synchronization	**5 Synchronization with BeiDou**

Synchronization Source	Supported Synchronization Mode	Described In
Galileo	Frequency synchronization and time synchronization	**6 Synchronization with Galileo**
IEEE 1588V2	Frequency synchronization and time synchronization	**7 IEEE 1588V2 Clock Synchronization**
1PPS+TOD clock	Frequency synchronization and time synchronization	**8 1PPS+TOD Clock**

Table 3-3 Switching between and combination of multiple synchronization sources

Function	Description	Described In
Combined synchronization sources	gNodeBs support the combined synchronization sources of IEEE 1588V2 and synchronous Ethernet. The IEEE 1588V2 and synchronous Ethernet combination implements time synchronization. The use of combined synchronization sources enhances the robustness and holdover performance of time synchronization.	**9 Combined Synchronization Sources**
Synchronization source switching	When a base station is configured with multiple synchronization sources, switching between these synchronization sources is supported. If a configured synchronization source is unavailable, the base station can switch to another available synchronization source. This ensures the normal operation of the base station.	**10 Synchronization Source Switching**

Table 3-4 Clock out-of-synchronization detection and intelligent clock fault joint diagnosis and self-healing

Function	Description	Described In
Clock out-of-synchronization detection (low-frequency TDD)	NR TDD has strict requirements for clock synchronization. If a base station has lost synchronization, it will cause interference to other base stations. When the interference is severe, UEs may be unable to access the network, or receive a poor service experience. For example, service drops or handover failures are likely to occur. Clock out-of-synchronization detection helps quickly and accurately identify base stations that are not synchronized.	**11 Clock Out-of-Synchronization Detection (Low-Frequency TDD)**
Intelligent clock fault joint diagnosis and self-healing (low-frequency TDD)	For clock synchronization faults that cannot be identified by base stations, the intelligent clock fault joint diagnosis and self-healing system can perform further diagnosis and automatic isolation and holdover for faulty base stations through the following tasks: ● Network-wide synchronization deviation detection ● Multi-site joint clock fault diagnosis ● Self-isolation of clock faults ● Ultra-long holdover for sites with abnormal clock sources	**12 Intelligent Clock Fault Joint Diagnosis and Self-healing (Low-Frequency TDD)**

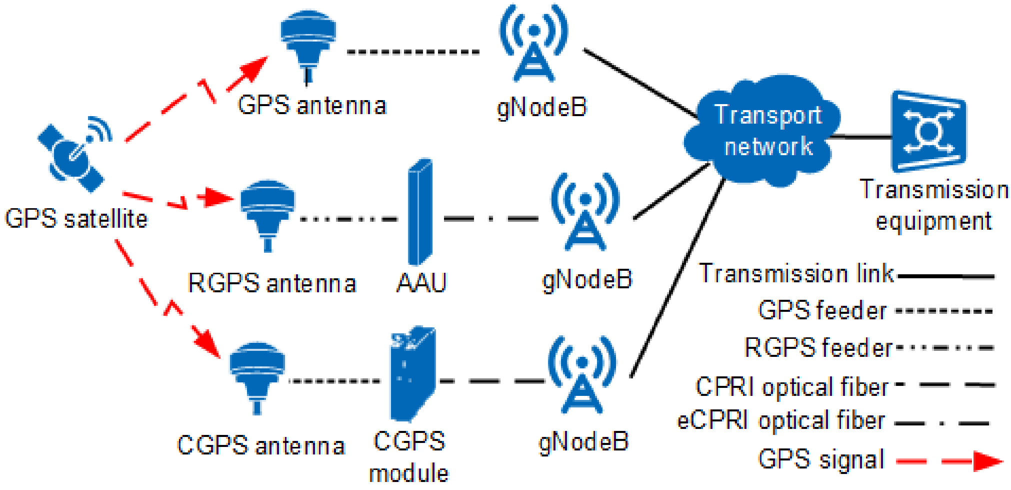
4 Synchronization with GPS

4.1 Principles

GPS is a global navigation system provided by the United States. It provides precise positioning, navigation, and timing services to objects around the world. The GPS synchronization source is highly accurate to the microsecond level, and supports both frequency synchronization and time synchronization. The GPS clock in this document can be a GPS clock, a remote global positioning system (RGPS) clock, or a CPRI GPS (CGPS) clock.

- If a gNodeB synchronizes with the GPS clock, it obtains synchronization signals from the GPS satellite synchronization system by connecting a board equipped with a GPS satellite card to the external GPS antenna system.
- If a gNodeB synchronizes with the RGPS clock, it obtains synchronization signals from the GPS satellite synchronization system by connecting the AAU to the external RGPS antenna system.
- If a gNodeB synchronizes with the CGPS clock, the gNodeB connects to an external CGPS antenna system to obtain synchronization signals from the GPS satellite synchronization system through a CGPS module. The CGPS module is connected to a baseband processing unit of the gNodeB through CPRI optical fibers.

Figure 4-1 GPS/RGPS/CGPS synchronization solution



As shown in **Figure 4-1**, the GPS/RGPS/CGPS antenna system receives GPS signals at 1575.42 MHz (L1C/A) and transmits them to the GPS satellite card. The GPS satellite card then processes the signals and transmits them to the main clock module of the main control board. To implement GPS synchronization, a gNodeB requires signals from at least four GPS satellites. (The number of tracked satellites is specified by the **GPS Satellites Traced** parameter in the **DSP GNSS** command output.)

NOTE

In the clock networking involving the cascading between the AAU and RGPS, GPS or BeiDou satellites are supported.

In normal cases, the GPS location does not have an offset. When the GPS becomes abnormal, an offset can be detected in the GPS location. When the offset exceeds a certain range, the base station reports ALM-26124 GNSS Clock Output Unavailable. This function is controlled by the **GNSS**.***POSCHECKSW*** parameter and is applicable only when the NE position is fixed.

While GPS synchronization supports both time synchronization and frequency synchronization, it requires additional investments in GPS equipment during base station deployment. For details on the GPS signal receiving conditions and engineering requirements for gNodeB sites, see *GNSS Satellite Antenna System Quick Installation Guide* in *3900 & 5900 Series Base Station Product Documentation*. For details on how to install, configure, commission, and maintain the RGPS system, see *RGPS Satellite Antenna System Quick Installation Guide* in *3900 & 5900 Series Base Station Product Documentation*. For details on how to install, configure, commission, and maintain the CGPS system, see *CGPS User Guide* in *3900 & 5900 Series Base Station Product Documentation*.

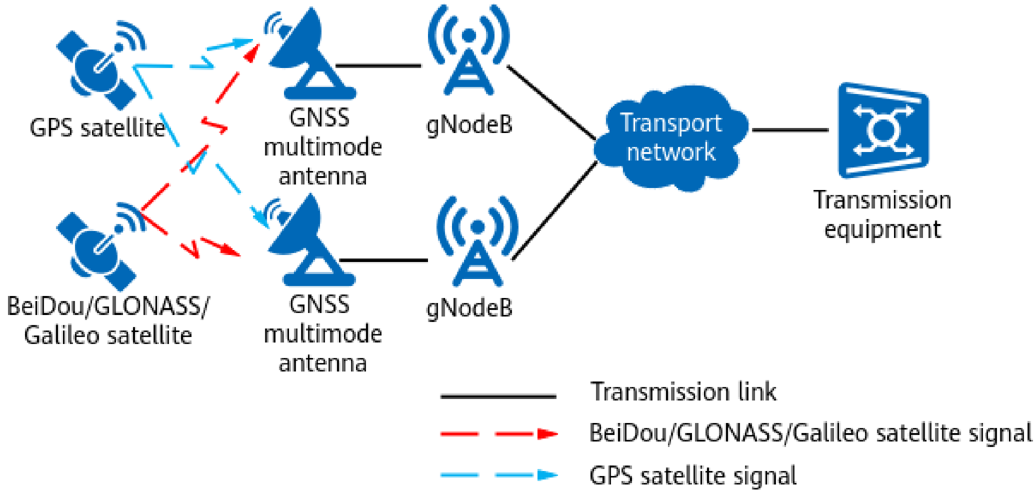
Multimode Satellite Cards

Currently, the BeiDou Navigation Satellite System (BDS), GLONASS Satellite Navigation System (referred to as GLONASS), or Galileo satellite navigation system (referred to as Galileo) can work with GPS to implement satellite synchronization when multimode satellite cards are configured to work in joint and concurrent searching mode. Multimode satellite cards support the simultaneous use of GPS with BDS, GLONASS, or Galileo.

Multimode satellite cards support joint and concurrent searching for the satellites of different satellite systems. In dual-satellite-system joint and concurrent searching mode, satellite systems of two different types work at the same time, and these satellite systems assist each other to meet clock synchronization requirements. The clock synchronization requirements can be met as long as at least five satellites are traced and the number of satellites of the same type is not less than four. For example, in GPS/BDS joint and concurrent searching mode, the corresponding base station can achieve clock synchronization as long as four GPS satellites and one BeiDou satellite are traced.

Base stations receive the reference time of the GNSS satellite system and obtain synchronization information to implement clock synchronization, which is irrelevant to the type of the satellite system in use. This is because the atomic clocks of satellites provide the same timing standard. For base stations, the configuration of multimode satellite cards in joint and concurrent searching mode only introduces the support for multiple reference clocks. This is the same as the case where multiple satellites of the same type are traced in single-satellite-system searching mode. **Figure 4-2** shows the networking solution of the joint and concurrent searching mode.

Figure 4-2 Networking solution of the joint and concurrent searching mode



In multi-satellite-system joint and concurrent searching mode, if one satellite system is faulty, base stations can work properly with other satellite systems, and base station services are not affected. If all satellite systems are faulty, base stations report alarms related to satellite cards.

Only the following combinations of satellite systems support the joint and concurrent searching mode:

- GPS/GLONASS
- BDS/GPS
- GPS/GALILEO
- GPS/GALILEO/GLONASS
- GPS/GALILEO/BDS
- GPS/GALILEO/BDS/GLONASS (supported only by the Book BBU)

The working mechanism of multimode satellite cards in single-satellite-system searching mode is the same as that when single-mode satellite cards are used.

NOTE

GLONASS is a global navigation satellite system provided by Russia, and has similar working principles and functions as GPS. To implement GLONASS synchronization, base stations must connect to the external GLONASS antenna system through boards equipped with the GLONASS satellite cards.

Unlike other navigation systems, GLONASS uses the local time of Russia. If only GLONASS is used, clock synchronization cannot be implemented between base stations. It is recommended that GLONASS be used together with GPS.

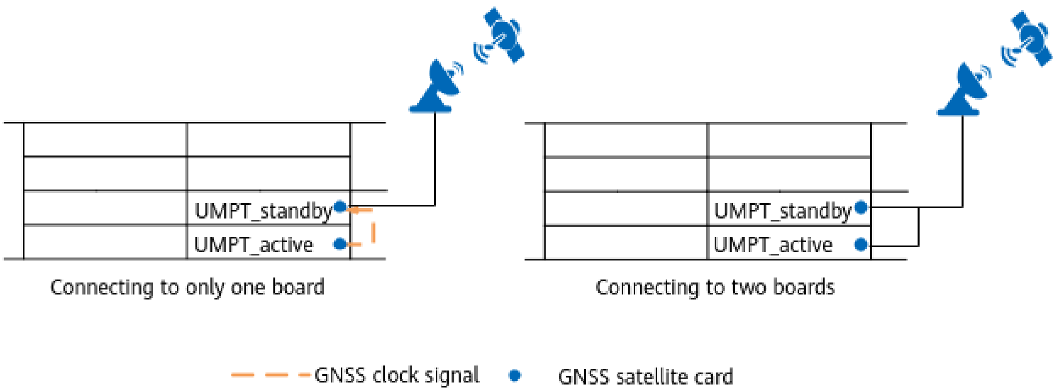
Synchronization Based on Signals from One to Three Satellites

In certain geographical environments, such as densely populated urban areas, the gNodeB may only be able to receive signals from one to three GPS satellites. In this case, synchronization can only be enabled based on signals from these satellites. This synchronization mode requires accurate longitude, latitude, and altitude configuration of the gNodeB. The GPS synchronization source supports synchronization based on signals from one to three satellites. The RGPS or CGPS synchronization source does not support synchronization based on signals from one to three satellites.

GNSS Backup When Main Control Boards Work in Active/Standby Mode

When the main control boards work in active/standby mode, the GNSS antenna system can be connected to only one or both main control boards, as shown in Figure 4-3. If it is connected to one board and the antenna access or board is faulty, the clock is unavailable and the reliability is reduced. If it is connected to two boards, the antenna access reliability can be improved. Therefore, connecting to two boards is recommended.

Figure 4-3 GNSS backup for main control boards in active/standby mode



GNSS clock source backup works in two modes: based on the board active/standby mode and based on reference source configuration slots.

- If the GNSS antenna system is connected to only one main control board, GNSS clock source backup can only work based on reference source configuration slots, and the clock source must be configured on the preceding main control board.

- If the GNSS antenna system is connected to both main control boards, the working modes differ as follows:
  - GNSS backup based on the board active/standby mode: The GNSS link can be configured only on one main control board, and this link takes effect on the active main control board. When the active main control board is faulty and a board switchover is triggered, the GNSS signal is automatically switched to the other main control board without manual cable connection adjustment.
  - GNSS backup based on reference source configuration slots: Two GNSS links can be configured respectively on the active and standby main control boards and one link is selected. If synchronization fails due to faults such as satellite card failures on one main control board, the GNSS link on the other main control board automatically takes effect without manual cable connection adjustment.

4.2 Network Analysis

4.2.1 Benefits

Wired networks do not affect synchronization with GPS/RGPS/CGPS. These synchronization sources are recommended when the wired network bandwidth is limited, or when wired networks frequently experience delay variation or packet loss.

4.2.2 Impacts

The impacts between this function and other functions are described in **4.3.2.3 Function Impacts**.

4.3 Requirements

4.3.1 Licenses

None

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- Synchronization with GPS
  - 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
  - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.
- Synchronization with RGPS

  3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with the BBU5900, BBU5910, or BBU5900A.
- Synchronization with CGPS
  - 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with the BBU5900, BBU5910, or BBU5900A.
  - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

- Synchronization with GPS: All NR-capable main control boards with GPS satellite cards and the USCU support this function. Only the UMPTha/UMPThb can receive GPS signals on multiple frequencies. For details, see the satellite card specifications for the main control boards and the USCU specifications in the BBU hardware description in *3900 & 5900 Series Base Station Product Documentation*.
- Synchronization with RGPS: All NR-capable main control boards support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*. The BookBBU5901 and BookBBU5901a do not support this function.
- Synchronization with CGPS: Baseband processing units that support the CPRI protocol must be configured. For details, see the BBU hardware description in *3900 & 5900 Series Base Station Product Documentation*. The BookBBU5901 and BookBBU5901a do not support this function.

RF Modules

- Non-RGPS synchronization source: All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
- The following RF modules support RGPS: AAU5639w/AAU5636w/AAU5831. For details, see the hardware description of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Others

Other requirements are as follows:

- Base stations must be configured with a GPS/RGPS/CGPS receive device to support the GPS/RGPS/CGPS clock.
- An external GPS/RGPS/CGPS antenna must be available at the site from which the GPS/RGPS/CGPS signals are received.
- In normal cases, at least four satellites are required for synchronization with the GPS/RGPS/CGPS clock. To trace as many satellites as possible, the external GPS/RGPS/CGPS antenna should be located in a sparsely populated area, or on the tallest buildings in a specified area. This synchronization source is not recommended for low spots in urban high-density building areas or mountainous areas.
- The GPS synchronization source is not recommended in rainy or foggy areas because the GPS signals are susceptible to bad weather.
- If an NR TDD base station synchronizes with the GPS clock using signals from only one to three GPS satellites, the longitude, latitude, and altitude information for the base station GPS device must be accurately configured. Otherwise, the synchronization accuracy is affected.
- The GNSS clock source cannot be configured on both the USCU and main control board.

The following are requirements of the CGPS synchronization source and suggestions for use:

- In separate-MPT GSM/NR SDR scenarios, if a GTMUc board is configured, NR does not support the CGPS synchronization source. Otherwise, if the clock mutual lock optimization switch (specified by **BTSOTHPARA**.***ClkMutualLockOptSw***) is turned off on the GTMUc, ALM-26270 Inter-System Communication Failure will be reported, and clock synchronization of other RATs will become abnormal.
- In inter-BBU SDR scenarios, the RAT configured with the CGPS synchronization source must be deployed on the primary BBU. Otherwise, ALM-26270 Inter-System Communication Failure will be reported and clock synchronization of other RATs will become abnormal.
- The CGPS synchronization source is mainly applied to scenarios where the GPS satellite card is installed remotely. The CGPS synchronization source does not support the function of base station positioning.
- The CGPS synchronization source supports only CPRI ports, and does not support IR or eCPRI ports.
- A single base station supports only two CGPS modules. Each CPRI link is connected to one CGPS module.

- The CGPS synchronization source does not support the CPRI chain cascading of AAUs+CGPS modules.
- The CGPS synchronization source does not support mutual backup between GPS, RGPS, and CGPS.
- The CGPS synchronization source does not support hardware upgrade of satellite cards.
- The CGPS synchronization source does not support the BeiDou satellite system.
- The BBU where the CGPS synchronization source is configured does not support inter-BBU distributed multiple-input multiple-output (D-MIMO) services.
- The CGPS synchronization source is not applicable to LampSite base stations.

The following are restrictions and requirements for the RGPS synchronization source:

- In separate-MPT GSM/NR SDR scenarios, if a GTMUc board is configured, NR does not support the RGPS synchronization source. When the clock mutual lock optimization switch (specified by **BTSOTHPARA**.***ClkMutualLockOptSw***) is turned off on the GTMUc, ALM-26270 Inter-System Communication Failure is reported, and the clock synchronization function of other RATs becomes abnormal.
- In inter-BBU SDR scenarios, the RAT configured with the RGPS synchronization source must be deployed on the primary BBU. Otherwise, ALM-26270 Inter-System Communication Failure will be reported and clock synchronization of other RATs will become abnormal.
- The RGPS synchronization source is mainly applied to scenarios where the GPS satellite card is installed remotely. The RGPS synchronization source does not support the function of base station positioning.
- A single base station supports a maximum of 18 RGPS modules. RGPS modules cannot be cascaded.
- The RGPS synchronization source does not support backup with the GPS or CGPS.
- The RGPS synchronization source does not support hardware upgrade of satellite cards.
- The BBU where the RGPS synchronization source is configured does not support inter-BBU distributed multiple-input multiple-output (D-MIMO) services.
- The RGPS synchronization source is not applicable to LampSite base stations.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-1 describes the parameters used for function activation.

Parameters in the **GNSS** MO can be configured using the **ADD GNSS** or **MOD GNSSPOS** command.

Parameters in the **TASM** MO can be configured using the **SET CLKMODE** or **SET CLKSYNCMODE** command.

Table 4-1 Parameters used for activating GNSS synchronization

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	**GNSS**.***CN***	Set these parameters based on the board to which the GNSS or RGPS feeder is connected. Before configuring a GNSS/RGPS clock link, ensure that the required board has been configured and no 1PPS+TOD clock link is configured on the board.
Subrack No.	**GNSS**.***SRN***	
Slot No.	**GNSS**.***SN***	
GNSS Clock No.	**GNSS**.***GN***	● If only one GNSS/RGPS/CGPS synchronization source is required, set this parameter to **0**. ● If two GNSS/RGPS/CGPS synchronization sources are required, set this parameter to **0** for one clock source and to **1** for the other.
Cable Length	**GNSS**.***CABLE_LEN***	If the feeder length cannot be measured, the difference between the value of this parameter and the actual length must be less than or equal to 20 m. Otherwise, the clock accuracy is affected.
GNSS Work Mode	**GNSS**.***MODE***	Set this parameter based on the type of the satellite card in use. If a board supports multiple types of satellite cards, it is recommended that this parameter be set to **GPS/GLONASS**, **BDS/GPS**, **GPS/GALILEO**, **GPS/GALILEO/GLONASS**, **GPS/GALILEO/BDS**, or **GPS/GALILEO/BDS/GLONASS**.
GNSS Primary Mode	**GNSS**.***PRIMARYMODE***	Set this parameter as planned. The value **NONE** is recommended.
Priority	**GNSS**.***PRI***	Set this parameter when two or more synchronization sources are configured. A smaller value of this parameter indicates a higher priority.

Parameter Name	Parameter ID	Setting Notes
Clock Working Mode	**TASM**.***MODE***	It is recommended that this parameter be set to **AUTO** for time synchronization and to **MANUAL** for frequency synchronization. NR TDD base stations support only time synchronization.
Selected Clock Source	**TASM**.***CLKSRC***	When the **TASM**.***MODE*** parameter is set to **MANUAL**, the default value is recommended.
Clock Source No.	**TASM**.***SRCNO***	When the **TASM**.***MODE*** parameter is set to **MANUAL**, it is recommended that this parameter be set to the link number specified when the clock link is created.
Clock Synchronization Mode	**TASM**.***CLKSYNCMODE***	Set this parameter as required.
Clock Source Backup Mode	**TASM**.***BKPMODE***	If the main control boards work in active/standby mode, you are advised to set this parameter to **CLKSRCBRDCFG**.
Head Cabinet No.	**RRUCHAIN**.***HCN***	Set these parameters based on hardware installation information.
Head Subrack No.	**RRUCHAIN**.***HSRN***	
Head Slot No.	**RRUCHAIN**.***HSN***	
Head Port No.	**RRUCHAIN**.***HPN***	
Chain No.	**RRUCHAIN**.***RCN***	Set this parameter to a unique value.
Topo Type	**RRUCHAIN**.***TT***	The CGPS clock supports only the star and chain topologies. In the star or chain topology, set this parameter to **CHAIN**. Each link can be configured with only one CGPS module.
Backup Mode	**RRUCHAIN**.***BM***	If the **RRUCHAIN**.***TT*** parameter is set to **CHAIN**, set this parameter to **COLD**.
Cabinet No.	**CXU**.***CN***	Set this parameter to **0** for the CGPS clock.
Subrack No.	**CXU**.***SRN***	Set this parameter to a value within the range of 60–254 for the CGPS clock.
Slot No.	**CXU**.***SN***	Set this parameter to **0** for the CGPS clock.

Parameter Name	Parameter ID	Setting Notes
Chain/Ring No.	**CXU**.***RCN***	Set this parameter to the value of the **RRUCHAIN**.***RCN*** parameter.
CXU Type	**CXU**.***CXUTYPE***	Set this parameter to **CGPS**.
CXU Position	**CXU**.***PS***	Set this parameter to **0**.
CXU Name	**CXU**.***CXUNAME***	Set this parameter to a user-defined name, which can consist of 0–99 characters. This parameter is optional.
Remote Flag	**CXU**.***REMOTEFLAG***	Set this parameter to **REMOTE**. If this parameter is not required, set this parameter to **UNDEFINED**.
User Label	**CXU**.***USERLABEL***	Set this parameter to user-defined CXU information, which consists of 0–255 characters.
Cabinet No.	**RETPORT**.***CN***	Set these parameters based on the board to which the RGPS feeder is connected.
Subrack No.	**RETPORT**.***SRN***	
Slot No.	**RETPORT**.***SN***	
Port No.	**RETPORT**.***PN***	Set this parameter to **RET_PORT**.
ALD Power Switch	**RETPORT**.***PWRSWITCH***	Set this parameter to **ON**.
Current Alarm Threshold Type	**RETPORT**.***THRESHOLDTYPE***	• If the RET port is connected only to the RGPS, set this parameter to **RGPS_ONLY**. • If the RET port is connected to both the RGPS and AISU, set this parameter to **RGPS_AISU**.
Way to Get Position	**GNSS**.***WPOS***	Set this parameter to **USER_CONFIG**.
Antenna Longitude	**GNSS**.***LONG***	Set these parameters based on the latitude, longitude, and other information of the GNSS/RGPS device.
Antenna Latitude	**GNSS**.***LAT***	
Antenna Altitude	**GNSS**.***ALT***	
Antenna Angle	**GNSS**.***AGL***	

4.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples for Synchronization with GPS

```
//Setting the clock synchronization mode (time synchronization is used as an example). NR TDD base stations must use time synchronization.
SET CLKSYNCMODE: CLKSYNCMODE=TIME;
//(Optional, required if the main control boards work in active/standby mode) Setting the clock source backup mode to CLKSRCBRDCFG
SET CLKSRCBKPMODE: BKPMODE=CLKSRCBRDCFG;
//Adding a GNSS synchronization source
ADD GNSS: GN=0, CN=0, SRN=0, SN=7, CABLE_LEN=50, MODE=GPS, PRI=4;
//(Optional, required if the main control boards work in active/standby mode) Adding a GNSS synchronization source for the standby main control board
ADD GNSS: GN=0, CN=0, SRN=0, SN=6, CABLE_LEN=50, MODE=GPS, PRI=4;
//Setting the clock working mode to AUTO
SET CLKMODE: MODE=AUTO;
//(Optional) Configuring the longitude, latitude, and altitude of the GPS device of a base station when the base station obtains signals from only one to three GPS satellites
MOD GNSSPOS: WPOS=USER_CONFIG, LONG=10, LAT=10, ALT=10;
```

Activation Command Examples for Synchronization with RGPS

```
//Setting the clock synchronization mode (time synchronization is used as an example). NR TDD base stations must use time synchronization.
SET CLKSYNCMODE: CLKSYNCMODE=TIME;
//Adding a synchronization source, which can be either GPS or BeiDou
//(Option 1) Adding a GPS synchronization source
ADD GNSS: GN=0, CN=0, SRN=61, SN=1, CABLE_LEN=50, MODE=GPS, PRI=4;
//(Option 2) Add a BDS synchronization source
ADD GNSS: GN=0, CN=0, SRN=61, SN=1, CABLE_LEN=50, MODE=BDS, PRI=4;
//Setting the clock working mode to AUTO
SET CLKMODE: MODE=AUTO;
//Using the RGPS 12 V power supply
MOD RETPORT: CN=0, SRN=61, SN=1, PN=RET_PORT, PWRSWITCH=ON, THRESHOLDTYPE=RGPS_ONLY;
```

Activation Command Examples for Synchronization with CGPS

```
//Adding an RRU chain or ring
ADD RRUCHAIN: RCN=1, TT=CHAIN, BM=COLD, HCN=0, HSRN=0, HSN=3, HPN=1;
//Adding a CGPS module with PS set to 0
ADD CXU:CN=0, SRN=61, SN=0, CXUTYPE=CGPS, RCN=1, PS=0, CXUNAME="CGPS", REMOTEFLAG=REMOTE, USERLABEL="CGPS1";
//Setting the clock synchronization mode (time synchronization is used as an example). NR TDD base stations must use time synchronization.
SET CLKSYNCMODE: CLKSYNCMODE=TIME;
//Adding a GNSS synchronization source
ADD GNSS: GN=0, CN=0, SRN=61, SN=0, CABLE_LEN=50, MODE=GPS, PRI=4;
//Setting the clock working mode to AUTO
SET CLKMODE: MODE=AUTO;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

After configuring a GNSS synchronization source, wait approximately 5 minutes. Then, run the **DSP CLKSTAT** command to check the system clock status. If the value of **Current Clock Source** is **GNSS Clock** and the value of **PLL Status** is **Locked**, this function has taken effect.

4.4.3 Network Monitoring

Synchronization with GNSS depends on the GNSS satellite tracing status, which can be monitored using the following counters.

Counter ID	Counter Name
1593835641	**VS.GNSS.SatellitesTraced.0**
1593835642	**VS.GNSS.SatellitesTraced.1**
1593835643	**VS.GNSS.SatellitesTraced.2**
1593835644	**VS.GNSS.SatellitesTraced.3**
1593835645	**VS.GNSS.SatellitesTraced.4**
1593835646	**VS.GNSS.SatellitesTraced.Mean**
1593835849	**VS.GNSS.GPSTraced.Mean**
1593835850	**VS.GNSS.BDSTraced.Mean**
1593835851	**VS.GNSS.GLONASSTraced.Mean**
1593835852	**VS.GNSS.GALILEOTraced.Mean**

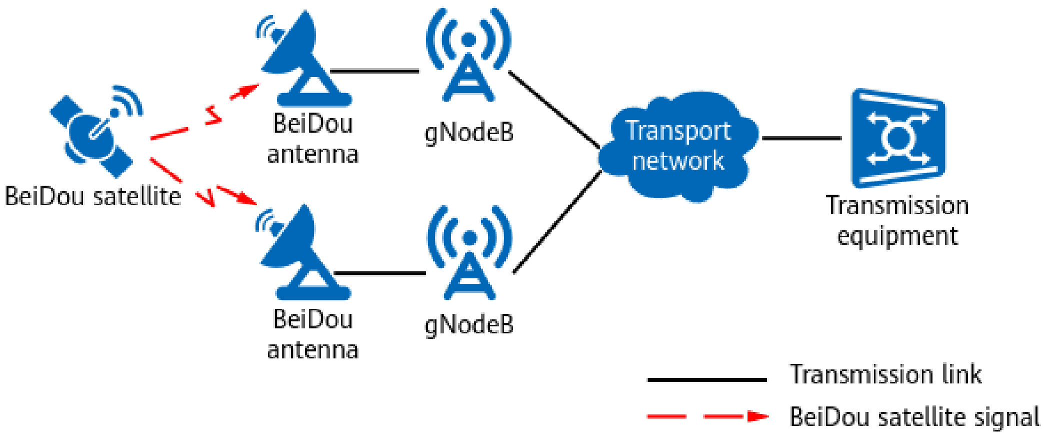
5 Synchronization with BeiDou

5.1 Principles

BDS is a global navigation satellite system developed by China. It is the third mature navigation satellite system following GPS and GLONASS. The working principles and functions of BDS are similar to those of GPS.

If the BeiDou clock is used for synchronization, gNodeBs can connect to the external BeiDou antenna system through boards equipped with BeiDou satellite cards, in order to obtain synchronization signals from the BeiDou satellite synchronization system.

Figure 5-1 BeiDou synchronization solution



As shown in **Figure 5-1**, the BeiDou antenna system receives BeiDou satellite signals and transmits them to the BeiDou satellite card. The BeiDou satellite card then transmits the signals to the main clock module of the main control board. To implement BeiDou synchronization, base stations require signals from at least four BeiDou satellites.

BeiDou provides frequency synchronization and time synchronization, but requires additional investments in BeiDou antennas and feeders during base station deployment. Base station sites must meet the conditions and engineering requirements of BeiDou signal reception. For details, see *GNSS Satellite Antenna System Quick Installation Guide* in *3900 & 5900 Series Base Station Product Documentation*.

Synchronization Based on Signals from One to Three Satellites

In certain geographical environments such as densely populated urban areas, gNodeBs may only be able to receive signals from one to three BeiDou satellites. In this case, BeiDou synchronization can be enabled based on signals from only one to three BeiDou satellites. This synchronization mode requires accurate longitude, latitude, and altitude configuration of gNodeBs.

BeiDou-Primary Mode

When the **GNSS**.***MODE*** parameter is set to a value indicating multi-satellite-system joint searching mode involving BDS (that is, **BDS/GPS**, **GPS/GALILEO/BDS**, or **GPS/GALILEO/BDS/GLONASS**), the **GNSS**.***PRIMARYMODE*** parameter can be set to **BDS** to enable the BeiDou-primary mode. In this mode, the base station uses only BeiDou satellites for synchronization if BeiDou satellite signals are available, and uses the other satellite systems for synchronization if BeiDou satellite signals are unavailable. When BeiDou satellite signals become available again, that is, the base station working in the multi-satellite-system joint searching mode can trace at least six BeiDou satellites within 24 hours and the SNR of BeiDou satellite signals is at least 35 dB, the base station automatically switches back to using only BeiDou satellites for synchronization. This condition is determined based on the operating period of BeiDou satellites.

5.2 Network Analysis

5.2.1 Benefits

Wired networks do not affect synchronization with BeiDou. This synchronization source is recommended when the wired network bandwidth is limited, or when wired networks frequently experience delay variations or packet losses.

5.2.2 Impacts

The impacts between this function and other functions are described in **5.3.2.3 Function Impacts**.

This function has no impact on the network.

5.3 Requirements

5.3.1 Licenses

None

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	GNSS Work Mode	**GNSS**.***MODE***	*Synchronization*	The **GNSS**.***PRIMARYMODE*** parameter can be set to **BDS** to enable the BeiDou-primary mode only if the **GNSS**.***MODE*** parameter is set to any of the following values: • **BDS/GPS** • **GPS/GALILEO/BDS** • **GPS/GALILEO/BDS/GLONASS**

5.3.2.2 Mutually Exclusive Functions

None

5.3.2.3 Function Impacts

None

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards that are configured with BeiDou satellite cards support synchronization with BeiDou. For details, see the satellite card specifications for the main control boards in the BBU hardware description in *3900 & 5900 Series Base Station Product Documentation*.

Only main control boards that are configured with multimode satellite cards support the BeiDou-primary mode. The USCU does not support this mode.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.3.4 Others

If synchronization with BeiDou is required, a BeiDou satellite antenna system must be configured.

The CGPS and RGPS do not support the BeiDou-primary mode.

The requirements described in **4.3.4 Others** for synchronization with GPS also apply.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

Table 5-1, **Table 5-2**, and **Table 5-3** describe the parameters used for function activation.

Parameters in the **GNSS** MO can be configured using the **ADD GNSS** or **MOD GNSSPOS** command.

Parameters in the **TASM** MO can be configured using the **SET CLKMODE** or **SET CLKSYNCMODE** command.

Table 5-1 Parameters in the **GNSS** MO

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	**GNSS**.***CN***	Set these parameters based on the board to which the BeiDou feeder is connected. Before configuring a BeiDou clock link, ensure that the required board has been configured and no 1PPS+TOD clock link is configured on the board.
Subrack No.	**GNSS**.***SRN***	
Slot No.	**GNSS**.***SN***	
GNSS Clock No.	**GNSS**.***GN***	● If only one BeiDou synchronization source is required, set this parameter to **0**. ● If two BeiDou synchronization sources are required, set this parameter to **0** for one synchronization source and to **1** for the other.
Cable Length	**GNSS**.***CABLE_LEN***	If the feeder length cannot be measured, the difference between the value of this parameter and the actual length must be less than or equal to 20 m. Otherwise, the clock accuracy is affected.
GNSS Work Mode	**GNSS**.***MODE***	Set this parameter based on the type of the satellite card in use. If the satellite card supports multiple modes, it is recommended that this parameter be set to multimode.
GNSS Primary Mode	**GNSS**.***PRIMARYMODE***	Set this parameter as planned. To enable the BeiDou-primary mode, set this parameter to **BDS**.
Priority	**GNSS**.***PRI***	Set this parameter when two or more synchronization sources are configured. A smaller value of this parameter indicates a higher priority.

Table 5-2 Parameters in the **TASM** MO

Parameter Name	Parameter ID	Setting Notes
Clock Working Mode	**TASM**.***MODE***	It is recommended that this parameter be set to **AUTO** for time synchronization and to **MANUAL** for frequency synchronization. NR TDD base stations support only time synchronization.

Parameter Name	Parameter ID	Setting Notes
Selected Clock Source	**TASM**.***CLKSRC***	When the **TASM**.***MODE*** parameter is set to **MANUAL**, the default value is recommended.
Clock Source No.	**TASM**.***SRCNO***	When the **TASM**.***MODE*** parameter is set to **MANUAL**, it is recommended that this parameter be set to the link number specified when the clock link is created.
Clock Synchronization Mode	**TASM**.***CLKSYNCMODE***	Set this parameter as required. For NR TDD base stations, set this parameter to **TIME**.

Table 5-3 Parameters for synchronization based on signals from one to three satellites

Parameter Name	Parameter ID	Setting Notes
Way to Get Position	**GNSS**.***WPOS***	Set this parameter to **USER_CONFIG**.
Antenna Longitude	**GNSS**.***LONG***	Set these parameters based on the longitude, latitude, and altitude of the base station BeiDou device.
Antenna Latitude	**GNSS**.***LAT***	
Antenna Altitude	**GNSS**.***ALT***	
Antenna Angle	**GNSS**.***AGL***	

5.4.1.2 Using MML Commands

Before using MML commands, refer to **5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

```
//Setting the clock synchronization mode (time synchronization is used as an example). NR TDD base stations must use time synchronization.
SET CLKSYNCMODE: CLKSYNCMODE=TIME;
//Adding a BeiDou synchronization source
ADD GNSS: GN=0, CN=0, SRN=0, SN=6, CABLE_LEN=50, MODE=BDS, PRI=4;
//(Optional) Enabling the BeiDou-primary mode
ADD GNSS: GN=0, CN=0, SRN=0, SN=6, CABLE_LEN=50, MODE=BDS/GPS, PRIMARYMODE=BDS, PRI=4;
//Setting the clock working mode
```

```
SET CLKMODE: MODE=AUTO;
//(Optional) Configuring the longitude, latitude, and altitude of the BeiDou device of a base station when the base station obtains signals from only one to three BeiDou satellites
MOD GNSSPOS: WPOS=USER_CONFIG, LONG=10, LAT=10, ALT=10;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

After configuring a BeiDou clock as the base station synchronization source, wait approximately 5 minutes. Run the **DSP CLKSTAT** command to check the system clock status. If the value of **Current Clock Source** is **GNSS Clock** and the value of **PLL Status** is **Locked**, BeiDou synchronization has been enabled.

Run the **DSP GNSS** command, and check the value of **Current GNSS Primary Mode**. If the value is **BDS**, the BeiDou-primary mode has taken effect.

5.4.3 Network Monitoring

For details, see **4.4.3 Network Monitoring**.

6 Synchronization with Galileo

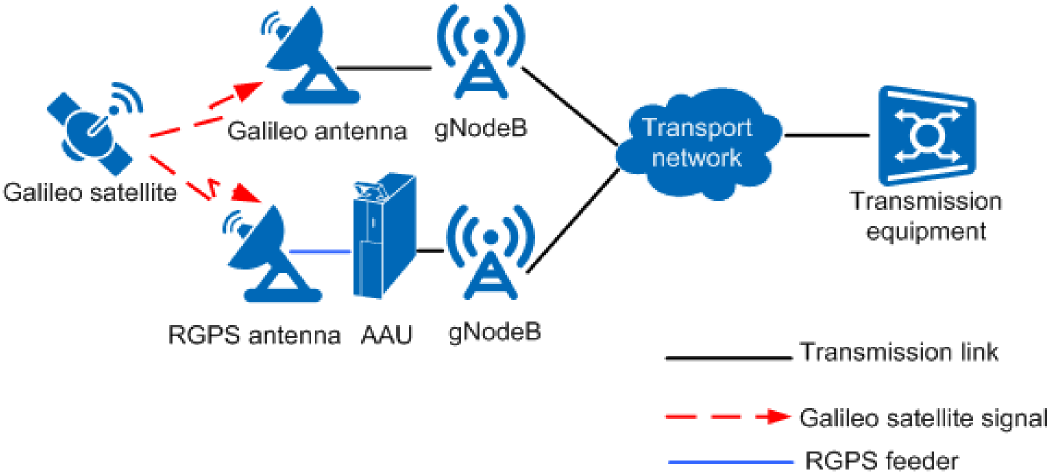
6.1 Principles

The Galileo is a global satellite navigation system developed by the European Union. This is the fourth fully mature system of its kind, following GPS developed in the United States, GLONASS in Russia, and BDS in China. The working principles and functions of Galileo are similar to those of GPS.

When synchronization with Galileo is implemented, the Galileo antenna system receives Galileo signals and transmits them to the Galileo satellite cards of the USCU/UMPT for processing. Alternatively, the RGPS antenna system receives Galileo signals, which are processed by the RGPS Galileo satellite cards. The processed signals are then transmitted to the main clock module of the main control board. For synchronization with Galileo, base stations require signals from at least four Galileo satellites.

Figure 6-1 shows the Galileo synchronization networking solution.

Figure 6-1 Galileo synchronization networking solution



Galileo provides both frequency and time synchronization, but requires additional investment in Galileo antennas and feeders during base station deployment. Base station locations must meet the conditions and engineering requirements of Galileo signal reception. The hardware and installation of the Galileo antenna system are similar to those of the GPS antenna system. For details, see *GNSS Satellite Antenna System Quick Installation Guide* in *3900 & 5900 Series Base Station Product Documentation*.

Synchronization Based on Signals from One to Three Satellites

In certain geographical environments, such as densely populated urban areas, base stations may only be able to receive signals from one to three Galileo satellites. In this case, synchronization can only be achieved based on signals from these Galileo satellites. This synchronization mode requires accurate longitude, latitude, and altitude configuration of base stations.

6.2 Network Analysis

6.2.1 Benefits

Wired networks do not affect synchronization with Galileo. This synchronization source is recommended when the wired network bandwidth is limited, or when wired networks frequently experience delay variations or packet losses.

6.2.2 Impacts

The impacts between this function and other functions are described in **6.3.2.3 Function Impacts**.

This function has no impact on the network.

6.3 Requirements

6.3.1 Licenses

None

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

None

6.3.2.2 Mutually Exclusive Functions

None

6.3.2.3 Function Impacts

None

6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

If synchronization with Galileo is not implemented through the RGPS antenna system, the gNodeB must be configured with the UMPT with the GNSS multimode satellite card or configured with the USCU. For details, see the satellite card specifications for the main control boards or the USCU specifications in BBU hardware description in *3900 & 5900 Series Base Station Product Documentation*.

If synchronization with Galileo is implemented through the RGPS antenna system, all NR-capable main control boards support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

For details about the Galileo multi-frequency capability of the UMPTha/UMPThb, see the satellite card specifications for the main control boards in the BBU hardware description in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

- If synchronization with Galileo is not implemented through the RGPS antenna system, all NR-capable RF modules support this function. For details about NR-capable RF modules, see the technical specifications of the related RF modules in *3900 & 5900 Series Base Station Product Documentation*.
- If synchronization with Galileo is implemented through the RGPS antenna system, the following RGPS-capable RF modules support this function: AAU5639w/AAU5636w/AAU5831. For details, see the hardware description of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

6.3.4 Others

- Synchronization with Galileo does not support the upgrade of satellite card firmware.

- Synchronization with Galileo requires the configuration of a Galileo antenna system.

The requirements described in **4.3.4 Others** for synchronization with GPS also apply.

6.4 Operation and Maintenance

6.4.1 Data Configuration

6.4.1.1 Data Preparation

Table 6-1, **Table 6-2**, and **Table 6-3** describe the parameters used for function activation.

Parameters in the **GNSS** MO can be configured using the **ADD GNSS** or **MOD GNSSPOS** command.

Parameters in the **TASM** MO can be configured using the **SET CLKMODE** or **SET CLKSYNCMODE** command.

Table 6-1 Parameters in the **GNSS** MO

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	**GNSS.*CN***	Set these parameters based on the board to which the Galileo feeder is connected. Before configuring a Galileo clock link, ensure that the required board has been configured and no 1PPS+TOD clock link is configured on the board.
Subrack No.	**GNSS.*SRN***	
Slot No.	**GNSS.*SN***	
GNSS Clock No.	**GNSS.*GN***	• If only one Galileo synchronization source is required, set this parameter to **0**. • If two Galileo synchronization sources are required, set this parameter to **0** for one synchronization source and to **1** for the other.
Cable Length	**GNSS.*CABLE_LEN***	If the feeder length cannot be measured, the difference between the value of this parameter and the actual length must be less than or equal to 20 m. Otherwise, the clock accuracy is affected.

Parameter Name	Parameter ID	Setting Notes
GNSS Work Mode	**GNSS**.***MODE***	This parameter can be set to **GALILEO**. If the satellite card supports multiple modes, it is recommended that this parameter be set to **GPS/GALILEO**.
GNSS Primary Mode	**GNSS**.***PRIMARYMODE***	Set this parameter as planned. The value **NONE** is recommended.
Priority	**GNSS**.***PRI***	Set this parameter when two or more synchronization sources are configured. A smaller value of this parameter indicates a higher priority.

Table 6-2 Parameters in the **TASM** MO

Parameter Name	Parameter ID	Setting Notes
Clock Working Mode	**TASM**.***MODE***	It is recommended that this parameter be set to **AUTO** for time synchronization and to **MANUAL** for frequency synchronization. NR TDD base stations support only time synchronization.
Selected Clock Source	**TASM**.***CLKSRC***	When the **TASM**.***MODE*** parameter is set to **MANUAL**, it is recommended that this parameter be set to **GNSS**.
Clock Source No.	**TASM**.***SRCNO***	When the **TASM**.***MODE*** parameter is set to **MANUAL**, it is recommended that this parameter be set to the link number specified when the clock link is created.
Clock Synchronization Mode	**TASM**.***CLKSYNCMODE***	Set this parameter as required. For NR TDD base stations, set this parameter to **TIME**.

Table 6-3 Parameters for synchronization based on signals from one to three satellites

Parameter Name	Parameter ID	Setting Notes
Way to Get Position	**GNSS**.***WPOS***	Set this parameter to **USER_CONFIG**.
Antenna Longitude	**GNSS**.***LONG***	Set these parameters based on the longitude, latitude, and altitude of the base station Galileo device.

Parameter Name	Parameter ID	Setting Notes
Antenna Latitude	**GNSS.*LAT***	
Antenna Altitude	**GNSS.*ALT***	
Antenna Angle	**GNSS.*AGL***	

6.4.1.2 Using MML Commands

Before using MML commands, refer to **6.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

```
//Setting the clock synchronization mode (time synchronization is used as an example). NR TDD base stations must use time synchronization.
SET CLKSYNCMODE: CLKSYNCMODE=TIME;
//Adding a GNSS synchronization source in the case that the main control board does not support Galileo
ADD GNSS: GN=0, CN=0, SRN=0, SN=1, CABLE_LEN=50, MODE=GALILEO, PRI=4;
//Adding a GNSS synchronization source in the case that the main control board supports Galileo
ADD GNSS: GN=0, CN=0, SRN=0, SN=7, CABLE_LEN=50, MODE=GALILEO, PRI=4;
//Setting the clock working mode
SET CLKMODE: MODE=AUTO;
//(Optional) Configuring the longitude, latitude, and altitude of the Galileo device of a base station when the base station obtains signals from only one to three Galileo satellites
MOD GNSSPOS: WPOS=USER_CONFIG, LONG=10, LAT=10, ALT=10;
```

6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.2 Activation Verification

After the Galileo synchronization source is configured, run the **DSP BRD** command to query the status of the USCU board. When the value of **Availability Status** is **Normal**, wait approximately 5 minutes. Then, run the **DSP CLKSTAT** command to check the system clock status. If the value of **Current Clock Source** is **GNSS Clock** and the value of **PLL Status** is **Locked**, this function has been activated.

6.4.3 Network Monitoring

For details, see **4.4.3 Network Monitoring**.

7 IEEE 1588V2 Clock Synchronization

7.1 Principles

IEEE 1588 messages are exchanged between primary and secondary equipment. To achieve frequency or time synchronization between the equipment, accurate time stamps are used to calculate the offset of time and frequency to the microsecond. IEEE 1588V2 synchronization is one of the clock solutions for IP networks and applies to the Ethernet transport network. IEEE 1588V2 supports both frequency synchronization and time synchronization.

gNodeBs support IEEE 1588V2 in layer 3 unicast and layer 2 multicast modes.

- In layer 3 unicast mode, IEEE 1588V2 packets are contained in IPv4/IPv6 User Datagram Protocol (UDP) packets.
- In layer 2 multicast mode, IEEE 1588V2 packets are contained in media access control (MAC) packets.

NOTE

- IEEE 1588V2 layer 3 unicast frequency synchronization and time synchronization are supported in IPv4 transmission.
- IEEE 1588V2 layer 3 unicast frequency synchronization and time synchronization are supported in IPv6 transmission.
- IEEE 1588V2 layer 2 multicast time synchronization and ITU-T G.8275.1 layer 2 multicast time synchronization are supported in both IPv4 and IPv6 transmission.
- IEEE 1588V2 does not support the configuration of multiple domains at the same time.
- A base station can be configured with a maximum of two IEEE 1588V2 reference clock sources.

Whether the gNodeB uses layer 3 unicast mode or layer 2 multicast mode is specified by the **IPCLKLNK**.***CNM*** parameter.

Table 7-1 lists the profile types complying with IEEE 1588V2. The IEEE 1588V2 clock is selected as the system clock of a gNodeB by setting the **IPCLKLNK**.***ICPT*** parameter to **PTP**.

Table 7-1 Profile types and networking modes of the IEEE 1588V2 synchronization

Profile Type	Networking Mode	Synchronization Mode
IEEE 1588V2	Layer 3 unicast	Time synchronization and frequency synchronization
1588V2_16.1	Layer 3 unicast	Time synchronization and frequency synchronization
IEEE 1588V2	Layer 2 multicast	Time synchronization
G.8275.2	Layer 3 unicast	Time synchronization
G.8275.1	Layer 2 multicast	Time synchronization
G.8265.1	Layer 3 unicast	Frequency Synchronization

When **IPCLKLNK**.***CNM*** is set to layer 3 unicast for the gNodeB, the VLAN configuration of 1588V2 packets is the same as that of the corresponding IP address.

When **IPCLKLNK**.***CNM*** is set to layer 2 multicast for the gNodeB, the VLAN mode and the corresponding VLAN ID can be configured for 1588V2 packets. Currently, three VLAN modes are supported: **UNTAGGED**, **TAGGED**, and **HYBRID**.

- In **UNTAGGED** mode, the 1588V2 packets without VLAN tags are supported. In this case, the transmitted 1588V2 packets do not carry VLAN tags, and strong verification is performed during packet transmission and reception (that is, the 1588V2 packets cannot contain VLAN tags).
- In **TAGGED** mode, 1588V2 packets with VLAN tags are supported. In this case, the VLAN ID needs to be configured. 1588V2 packets are transmitted based on the configured VLAN ID, and strong verification is performed during packet transmission and reception (that is, the 1588V2 packets must contain VLAN tags and the configured VLAN IDs).
- In **HYBRID** mode, VLAN tagged or untagged packets can be received. When VLAN tagged packets are received, the received VLANs are used for transmission. When VLAN untagged packets are received, the transmitted packets do not carry VLAN tags.

When the active and standby main control boards support transmission link backup, the 1588V2 clock source can be configured on the active main control link. When the transmission link of the 1588V2 clock source in use is interrupted or the main control board is faulty, the 1588V2 clock source attempts to switch to another transmission link for synchronization, not restricted by the configuration.

- Only IEEE 1588V2 layer 2 multicast time synchronization and ITU-T G.8275.1 layer 2 multicast time synchronization are supported.

7.1.1 Time Synchronization

The IEEE 1588V2 time synchronization solution requires that all intermediate transmission equipment on the data bearer network supports the Boundary Clock

(BC) function defined in IEEE 1588V2. You are advised to use the BC and layer 2 multicast mode networkwide. The layer 3 unicast mode also requires that all intermediate transmission equipment on the data bearer network supports the BC function defined in IEEE 1588V2.

IEEE 1588V2 time synchronization has the following deployment requirements:

- The IEEE 1588V2 BC function needs to be planned and enabled hop by hop on the transport network. The BC-related parameter settings must be consistent at the two ends on the transport network.
- The uplink and downlink transmission paths must be symmetric. If they are asymmetric, the delay difference must be compensated. Ensure that compensation is also performed for the primary and backup transmission paths.
- IEEE 1588V2 can be enabled only after acceptance testing by Huawei engineers. Ensure that the synchronization accuracy at the input ports of base stations is within the range of ±1100 ns.

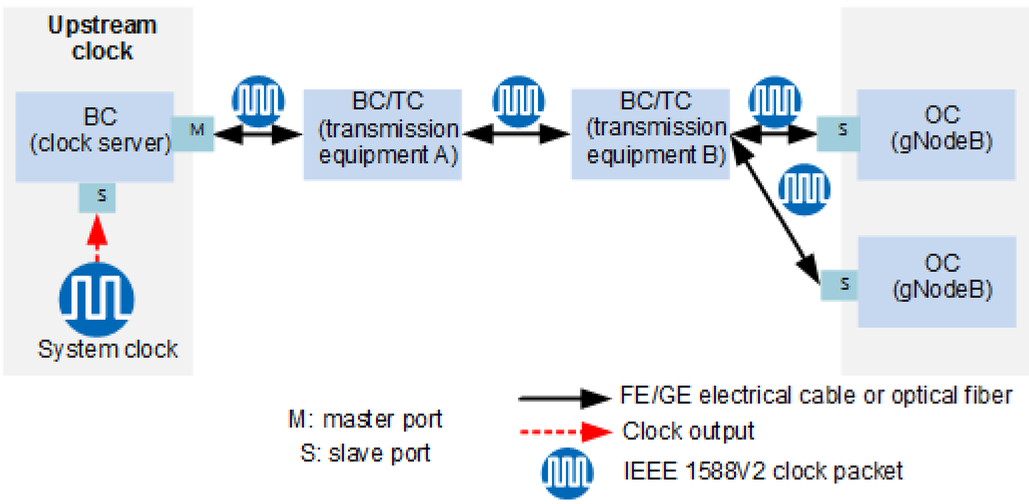
If the preceding requirements are not met but the single-reference-clock mode of IEEE 1588V2 is used on base stations to achieve time synchronization, base stations in a certain geographical area experience clock out-of-synchronization after a fault occurs. Multiple base stations may become out-of-synchronized and cause interference to neighboring base stations, or multiple base stations go out-of-service due to clock faults. It is difficult to demarcate such clock faults and takes a long time to rectify faults.

Therefore, the single-reference-clock mode of IEEE 1588V2 is not recommended for time synchronization on TDD sites where the deployment requirements are not met. The IEEE 1588V2 clock can work together with a GNSS clock as a standby clock source.

7.1.1.1 Network Clock Architecture

Figure 7-1 shows the network clock architecture defined in IEEE 1588V2. From the perspective of network clock architecture, the layer 3 unicast and layer 2 multicast modes are identical.

Figure 7-1 Network clock architecture defined in IEEE 1588V2



IEEE 1588V2 defines the following basic network components:

- Master port

  Used by the upstream equipment to distribute clock signals to the downstream equipment.

- Slave port

  Used by the downstream equipment to obtain clock signals from the upstream equipment.

- Ordinary clock (OC)

  OC is a clock with only one PTP communication port. The port can be used as the master port for distributing clock signals to the downstream equipment or as the slave port for obtaining clock signals from the upstream equipment. An OC restores the clock for frequency or time synchronization. The gNodeBs shown in **Figure 7-1** are used as OCs.

- Boundary clock (BC)

  BC is a clock with more than one PTP communication port. A complete BC has one slave port and one or more master ports. It restores the upstream clock (for example, GNSS clock) at the slave port by using the OC function and then distributes the clock signals through one or more master ports. The clock server shown in **Figure 7-1** is used as the BC.

  The BC is a point-to-point clock solution. The layer 2 multicast mode does not require negotiation, which facilitates interconnection between devices. If the BC solution is used for IEEE 1588V2 time synchronization, each BC node synchronizes with the upper-level equipment. When a synchronization issue occurs, it is easy to locate the issue.

- Transparent clock (TC)

  The TC supports the mechanism of link delay measurement of the IEEE 1588V2 clock and is located in the transmission equipment, as shown in **Figure 7-1**. Frequency synchronization and time synchronization can be implemented through TC after reconfiguration of all the transmission equipment on the path. If the TC is used for IEEE 1588V2 time synchronization, intermediate nodes compensate for only TC processing delay. When a synchronization issue occurs, it is difficult to locate the issue.

7.1.1.2 Delay Measurement

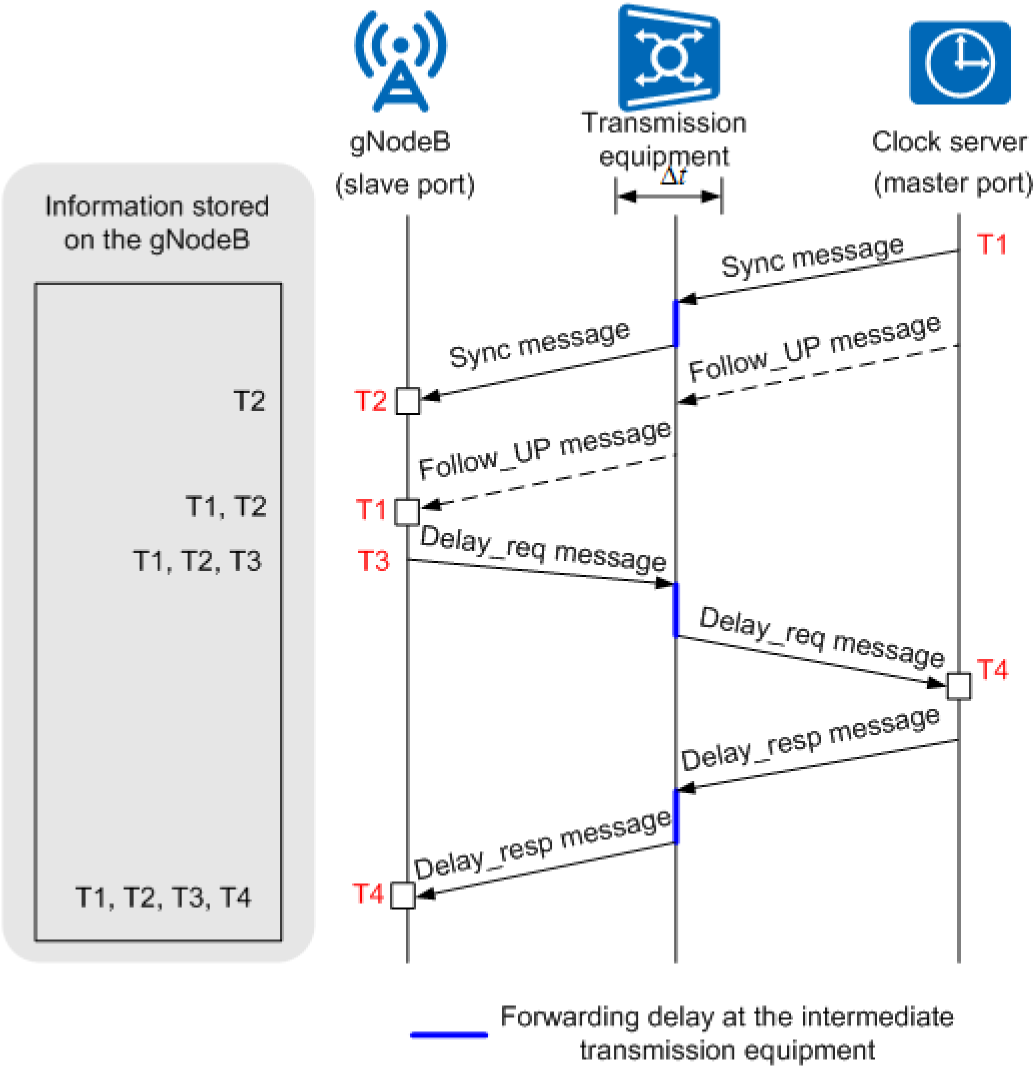
Delay is inevitable during the transmission of clock signals. Such delay needs to be measured and compensated for, in order to ensure clock accuracy. Delay measurement must be performed for all transmission interfaces of all intermediate transmission equipment, including layer 3 and layer 2 equipment. Delay compensation is performed based on the delay measurement results. The compensation value must be set again if routes or transmission paths change.

IEEE 1588V2 network clock delay measurement is classified into end-to-end (E2E) and point-to-point (P2P) delay measurement. The type of delay measurement is specified by the **IPCLKLNK**.***DELAYTYPE*** parameter.

E2E Delay Measurement

E2E delay measurement is used to calculate the processing delay at the intermediate transmission equipment between the upstream system clock and gNodeBs. **Figure 7-2** uses the E2E delay measurement between the clock server and a gNodeB as an example.

Figure 7-2 E2E delay measurement of the IEEE 1588V2 clock



The main messages in **Figure 7-2** are as follows:

- Sync message

  The Sync message contains standard time information, such as year, month, date, hour, minute, second, and nanosecond. gNodeBs record T2, the arrival time of the Sync message at gNodeBs. The time for sending or receiving the message needs to be measured and recorded at the underlying physical layer, or at the position close to the physical layer, to improve the clock accuracy.

  In the IEEE 1588 standard, the optional hardware assisted techniques are designed to improve the clock accuracy. If the Sync message is generated through the hardware assisted techniques, the message can also contain the timestamp T1, at which the message is sent. When an E2E TC (for example, transmission equipment in **Figure 7-1**) exists between the clock server and the gNodeB, the processing delay Δt on the transmission equipment is calculated and compensated.

- Follow_UP message
  - When IEEE 1588V2 uses the layer 3 unicast mode:

    If the delay of sending the Sync message is uncertain in the clock server, the clock server generates a Follow_UP message, which contains the timestamp T1. The Follow_UP message is optional.
  - When IEEE 1588V2 uses the layer 2 multicast mode:

    The Follow_UP message is not supported.
- Delay_resp message

  The delay of the Delay_resp message does not affect the value of T4. As a result, this message does not need to be processed in real time.

Principles for E2E delay measurement of the IEEE 1588V2 clock:

1. The clock server sends a Sync message to the gNodeB at the time point T1.
2. The gNodeB receives the Sync message at the time point T2.
3. The gNodeB sends a Delay_req message to the clock server at the time point T3.
4. The clock server receives the Delay_req message at the time point T4, and sends a Delay_resp message that contains T4 to the gNodeB.
5. The gNodeB stores the complete information about time points T1, T2, T3, and T4. The processing delay between the clock server and the gNodeB is calculated using the following formula:

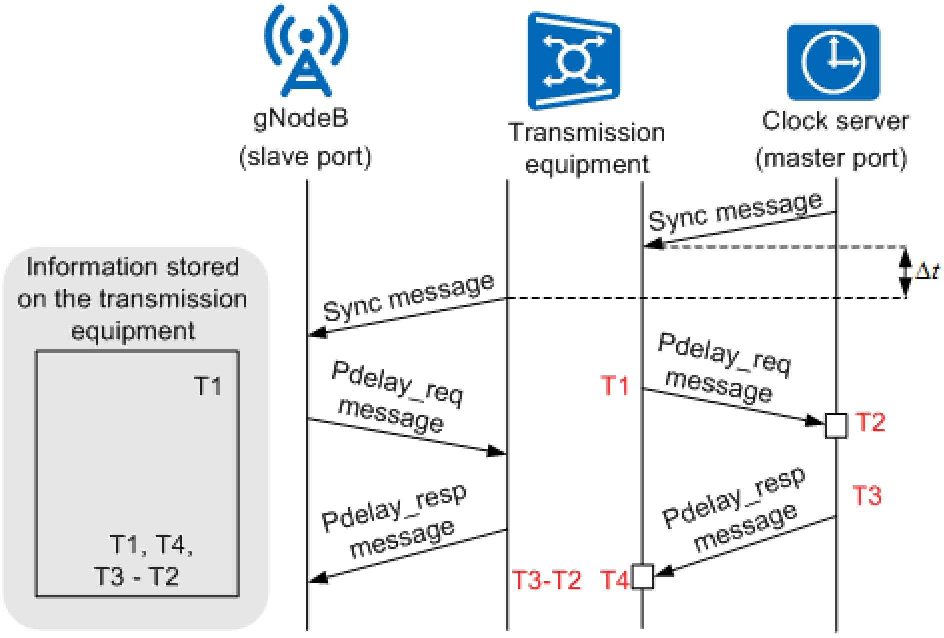
   Delay = [(T4 – T1) – (T3 – T2)]/2

P2P Delay Measurement

P2P delay measurement is used to calculate the link delay between adjacent TCs, between a TC and an OC, or between a TC and a BC. The link delay measurement between the transmission equipment and the base station is the same as that between the clock server and the transmission equipment.

Figure 7-3 uses the P2P delay measurement between the clock server and the transmission equipment as an example.

Figure 7-3 P2P delay measurement of the IEEE 1588V2 network clock



1. The clock server periodically sends a Sync message to the transmission equipment.
2. The transmission equipment calculates the processing delay Δt and compensates for the delay in the Sync message.
3. The transmission equipment generates a Pdelay_req message and sends the message to the clock server at T1.
4. The clock server receives the Pdelay_req message at T2, generates a Pdelay_resp message, and sends the message to the transmission equipment at T3. The Pdelay_resp message contains the deviation between T2 and T3 (the value of T3 minus T2).
5. The transmission equipment records the time the Pdelay_resp message is received as T4.
6. The transmission equipment stores the information about T1, T4, and the deviation between T2 and T3. The processing delay between the clock server and the transmission equipment is calculated as follows:

   Delay = [(T4 – T1) – (T3 – T2)]/2

Asymmetrical Delay Compensation

If the uplink and downlink physical paths between the master port and the slave port are symmetrical, the Sync message carries the absolute time, which equals the standard time plus the delay. If the uplink and downlink physical paths are asymmetrical, then asymmetrical delay compensation is required.

For asymmetrical delay compensation, the asymmetry of uplink and downlink physical paths (optical fibers, for example) results in the asymmetry of IEEE 1588V2 packets on uplink and downlink physical paths. As a result, the existing absolute time of the slave port is deviated, and a fixed deviation generated due to the asymmetry of uplink and downlink physical paths must be compensated on the slave port.

The delay deviation can be calculated and compensated in either of the following ways:

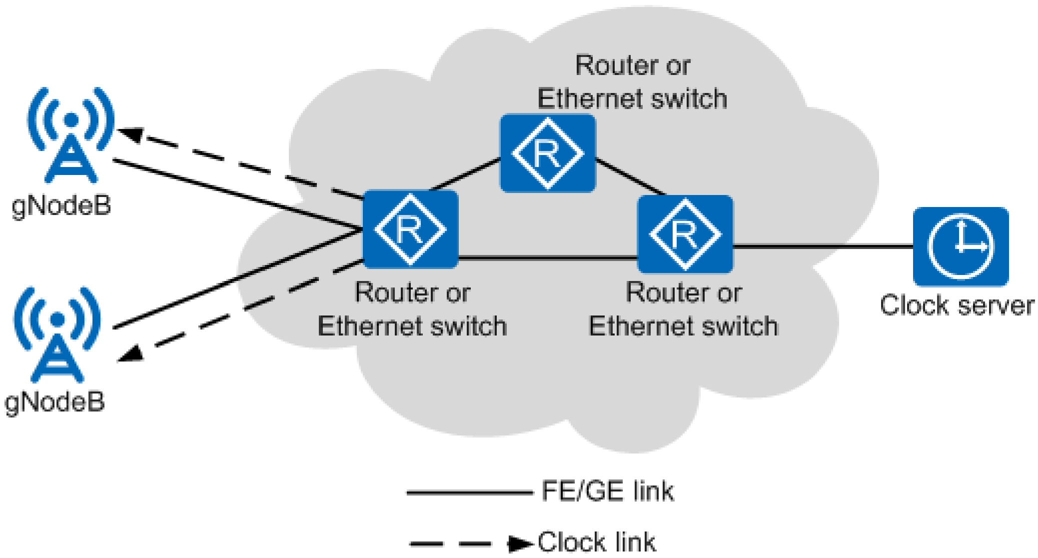
- Compare the IEEE 1588V2 clock and a standard clock such as the GNSS clock, and determine the deviation. Then, set the **IPCLKLNK**.***CMPST*** parameter to the deviation value for compensation.
- Test the difference between the transmission distances on the uplink and downlink physical paths, and determine the delay deviation. Then, set the **IPCLKLNK**.***CMPST*** parameter to the delay deviation for compensation.

7.1.1.3 Applications

Single-Reference-Clock Mode of IEEE 1588V2

Figure 7-4 shows the single-reference-clock mode of IEEE 1588V2.

Figure 7-4 Single-reference-clock mode of IEEE 1588V2

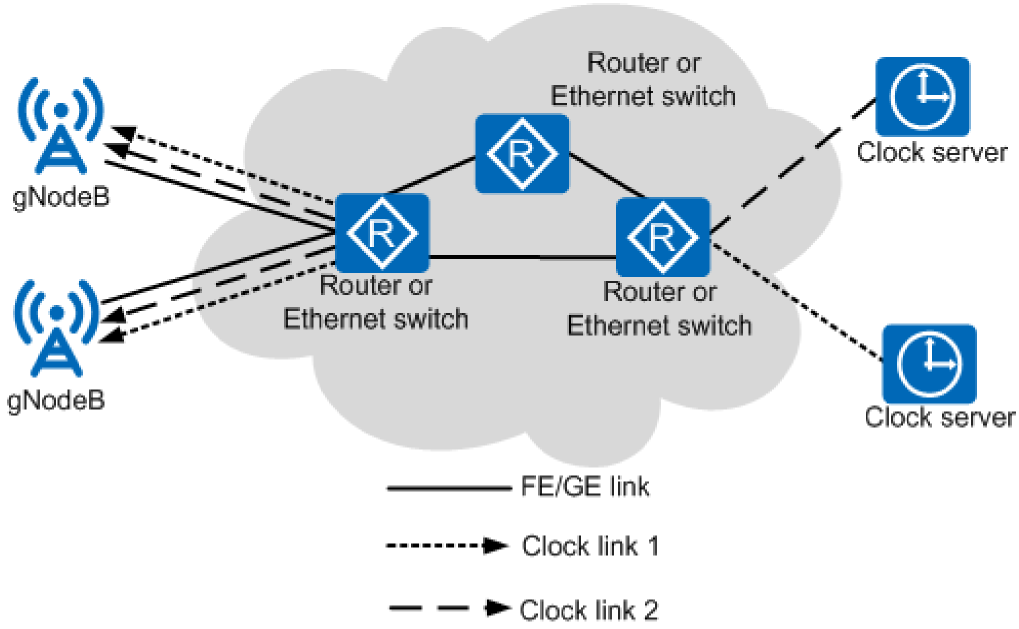


As shown in Figure 7-4, each dotted line depicts a clock link, which indicates an IEEE 1588V2 synchronization session. The clock server periodically sends a Sync message to the associated routers or Ethernet switches. Each gNodeB obtains the reference clock from the connected router or Ethernet switch. In this mode, each gNodeB has only one reference clock. If the reference clock becomes unavailable, the master clock of the gNodeB switches from locked mode, to holdover mode, and then to free running mode. The clock does not enter locked mode until the external reference clock becomes available again.

Dual-Reference-Clock Backup Mode of IEEE 1588V2

In this mode, two clock servers are deployed on the network to enhance the reliability of reference clocks. Each gNodeB is connected to these two clock servers through independent IEEE 1588V2 clock links. If the clock link of the master clock server is faulty, the gNodeB can still implement time synchronization through the backup clock server and backup clock link. Figure 7-5 shows the dual-reference-clock backup mode of IEEE 1588V2 for time synchronization.

Figure 7-5 Dual-reference-clock backup mode of IEEE 1588V2



In layer 2 multicast mode, the network-level master/backup solution is used, that is, master and backup clock servers are connected to the network. Each gNodeB can be configured with two IP clock links, as depicted by clock link 1 and clock link 2 in **Figure 7-5**. The two clock servers can be manually or automatically switched.

- In manual mode, the clock link to be used must be manually specified.
- In automatic mode, the gNodeB determines the clock server to be used based on the link status.
  - If one clock link works properly while the other does not, the clock server with the functional clock link is selected. During link switching, the gNodeB reports EVT-26269 Reference Clock Switchover.
  - If both clock links work properly, one clock link is selected based on the best master clock (BMC) algorithm. For details on the BMC algorithm, see section 9.3.2 "BMC algorithm" in IEEE_Std_1588-2008[1] or section 6.7.3 "Master selection process" in T-REC-G[1].8265.1-201010.
  - If neither clock link works properly, the synchronization source is abnormal, and ALM-26262 External Clock Reference Problem is reported.

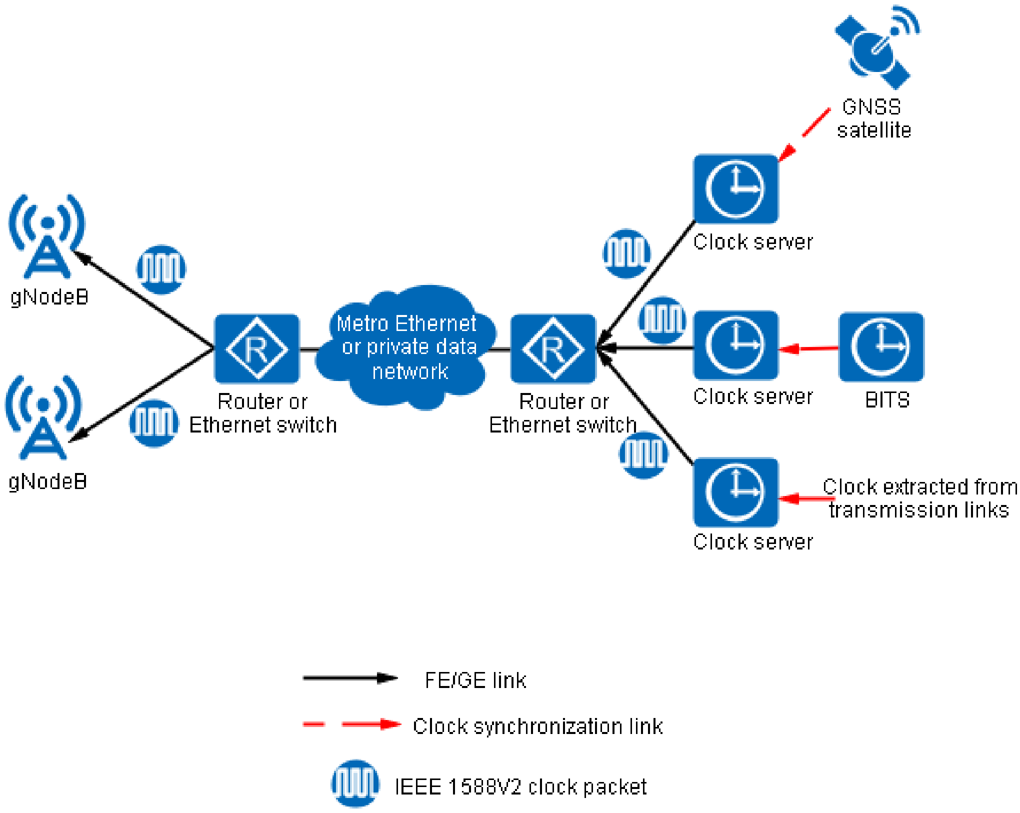
7.1.2 Frequency Synchronization

Layer 3 unicast mode is recommended when IEEE 1588V2 is used for frequency synchronization, as clock servers and base stations are connected through the transport network. In layer 3 unicast mode, the mechanism for packet transmission is simple and there are no special requirements for the transport network. IEEE 1588V2 frequency synchronization does not require delay measurement. When IEEE 1588V2 is used, only the source IP address at the local end is verified, and the destination IP address is not verified. The validity of the destination IP address is ensured by the clock server. If an invalid destination IP address is configured, this function does not take effect, and services are not affected.

7.1.2.1 Network Clock Architecture

IEEE 1588V2 frequency synchronization involves gNodeBs, clock servers, and intermediate transmission equipment between gNodeBs and clock servers. One or two clock servers constitute an independent clock domain, as shown in **Figure 7-6**. The configuration of two clock servers implements reference clock backup, improving reliability. The clock servers can use the GNSS clock or line clock as the reference clock.

Figure 7-6 IEEE 1588V2 frequency synchronization solution



The clock server sends clock packets to gNodeBs through the intermediate data bearer network. When IEEE 1588V2 frequency synchronization is used, the intermediate transmission equipment is not required to support the IEEE 1588V2 standard.

When a hub gNodeB is configured, the hub gNodeB routes the clock packets to leaf gNodeBs. A data forwarding device, such as router or Ethernet switch, is allowed between the hub gNodeB and leaf gNodeBs.

If the QoS of the network between gNodeBs and clock servers cannot be ensured, the clock servers and the hub gNodeB can be co-sited. In this case, the clock servers can directly provide clock signals for the hub gNodeB and leaf gNodeBs. This reduces the dependence on the quality of the bearer network.

gNodeBs using the IEEE 1588V2 clock synchronization support clock quality levels identified by clock classes that are specified in ITU-T G.8265.1. If the IEEE 1588V2 clock is used as the high-accuracy synchronization source for a gNodeB, the clock quality level cannot be lower than QL-SSU-B.

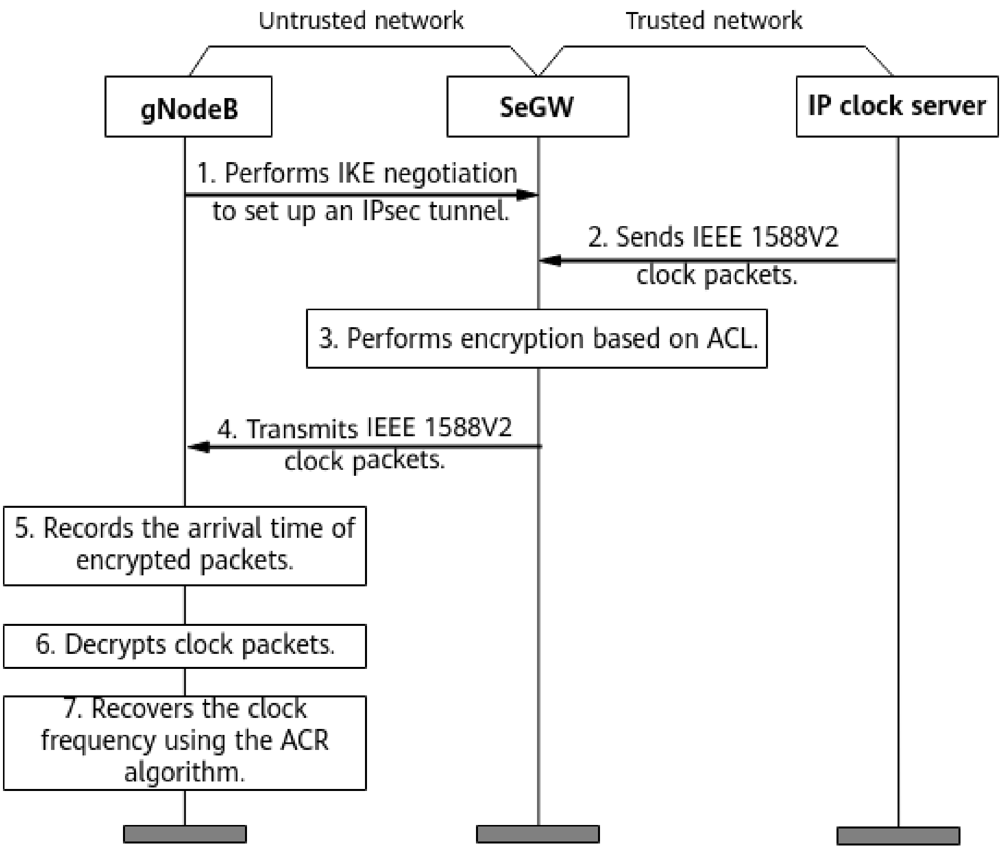
7.1.2.2 Encryption Process

Frequency synchronization can be achieved between a gNodeB and an IP clock server by using IEEE 1588V2 layer 3 unicast packets.

In a trusted transport network, the security of IEEE 1588V2 clock packets is ensured. Untrusted transport networks require higher-level security measures to protect the base station and IP clock server from being attacked by unauthorized packets. The base station encrypts clock packets by using IP Security (IPsec), and the security gateway (SeGW) uses IPsec to encrypt clock packets for the clock server. IEEE 1588V2 over IPsec supports only frequency synchronization.

The SeGW uses IPsec to encrypt all packets. The arrival time of all encrypted packets is recorded by the gNodeB. After decrypting the packets by using IPsec, the gNodeB identifies the IEEE 1588V2 clock packets based on UDP port numbers, and then recovers the clock frequency by using the adaptive clock recover (ACR) algorithm. **Figure 7-7** shows the detailed procedure.

Figure 7-7 IPsec encryption process



1. The gNodeB performs an Internet Key Exchange (IKE) negotiation to set up an IPsec tunnel with the SeGW.
2. The IP clock server sends the SeGW an IEEE 1588V2 clock packet with a timestamp attached.
3. The SeGW encrypts the IEEE 1588V2 clock packet through the established IPsec tunnel.

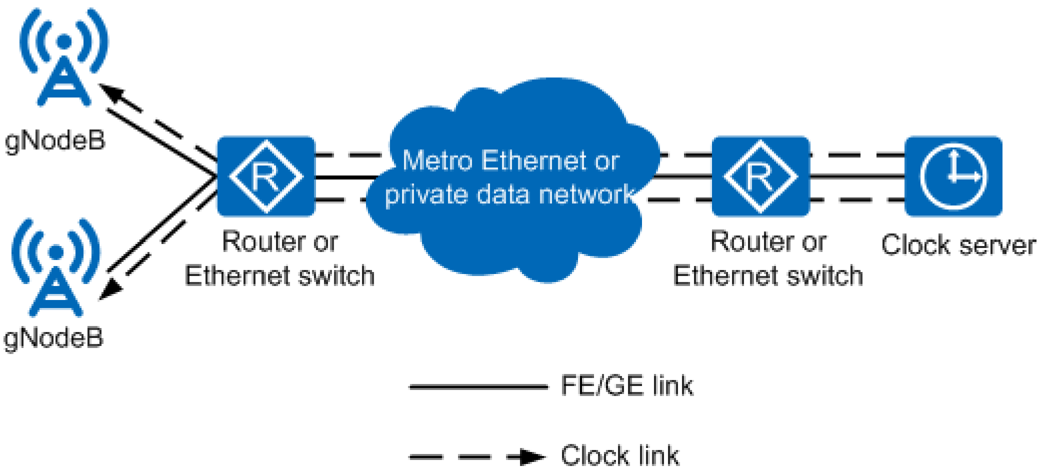
4. The SeGW transmits the encrypted clock packet to the gNodeB.
5. After receiving the encrypted clock packet, which cannot be identified by the gNodeB, the gNodeB records the arrival time and sends the timestamp to the upper layer together with the encrypted packet.
6. The gNodeB decrypts the clock packet by using IPsec and reads the time information from PTP packets.
7. The gNodeB recovers the clock frequency by using the ACR algorithm.

7.1.2.3 Applications

Single-Reference-Clock Mode of IEEE 1588V2

Figure 7-8 shows the single-reference-clock mode of IEEE 1588V2.

Figure 7-8 Single-reference-clock mode of IEEE 1588V2

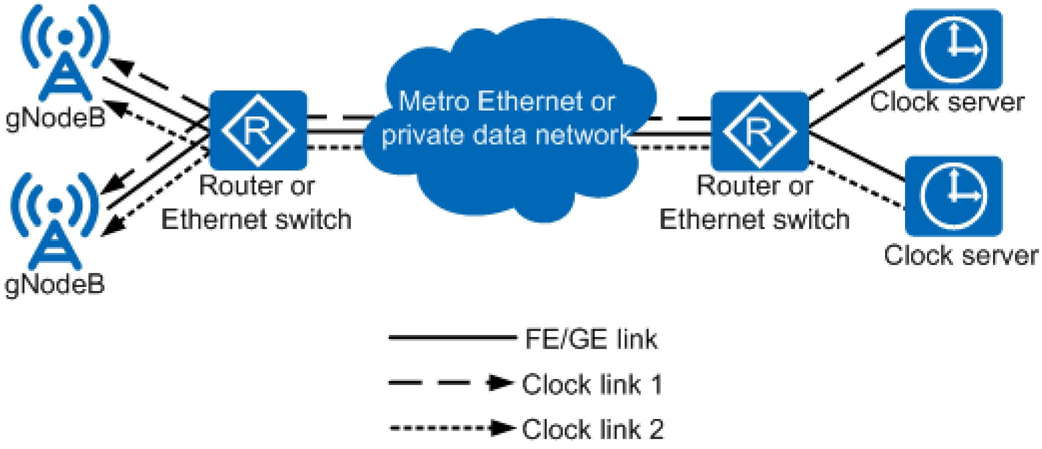


Each dotted line shown in Figure 7-8 depicts a clock link, which indicates an IEEE 1588V2 synchronization session. The clock server periodically sends a Sync message to the associated gNodeBs for clock synchronization. In this mode, each gNodeB has only one reference clock. If the reference clock becomes unavailable, the master clock of the gNodeB switches from locked mode, to holdover mode, and then to free running mode. The clock does not enter the locked mode until the external reference clock becomes available again.

Dual-Reference-Clock Backup Mode of IEEE 1588V2

In this mode, two clock servers are deployed on the network to enhance the reliability of reference clocks. Each gNodeB is connected to these two clock servers through independent IEEE 1588V2 clock links. If the clock link of the master clock server is disconnected, the gNodeB can still implement frequency synchronization through the backup clock server and backup clock link. Figure 7-9 shows the dual-reference-clock backup mode of IEEE 1588V2.

Figure 7-9 Dual-reference-clock backup mode of IEEE 1588V2



Each gNodeB can be configured with two IP clock links, as depicted by clock link 1 and clock link 2 in **Figure 7-9**. The two clock servers can be manually or automatically switched.

- In manual mode, the clock server to be used must be manually specified.
- In automatic mode, the gNodeB determines the clock server to be used based on the status of clock links.
  - If one clock link works properly while the other does not, the clock server with the functional clock link is selected. During link switching, the gNodeB reports EVT-26269 Reference Clock Switchover.
  - If both clock links work properly, one clock link is selected based on the BMC algorithm. For details on the BMC algorithm, see section 9.3.2 "BMC algorithm" in IEEE_Std_1588-2008[1] or section 6.7.3 "Master selection process" in T-REC-G[1].8265.1-201010.
  - If neither clock link works properly, the clock source is abnormal, and ALM-26262 External Clock Reference Problem is reported.

7.1.3 Clock Distribution

In base station cascading scenarios, IEEE 1588V2 clock distribution enables a lower-level base station to obtain IEEE 1588V2 clock signals from an upper-level base station and to distribute the clock signals to its lower-level base station. This eliminates the need of investing in the GNSS synchronization source for cascaded base stations.

IEEE 1588V2 clock distribution supports only the layer 2 multicast time synchronization mode. Therefore, the **TASM**.***CLKSYNCMODE*** parameter must be set to **TIME** for cascaded base stations.

IEEE 1588V2 clock distribution consists of the following two functions:

- IEEE 1588V2 master clock distribution

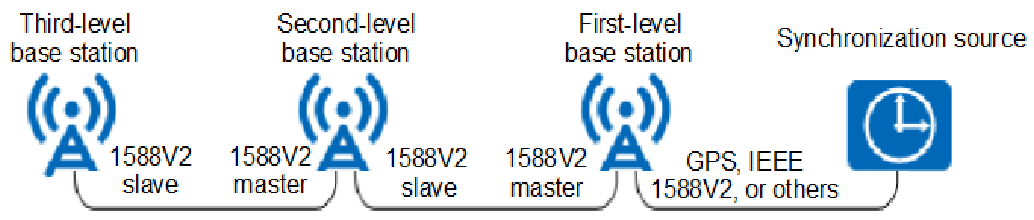
  After obtaining clock data from the IP network, GNSS, or an upper-level base station, the base station works as an IEEE 1588V2 master clock, and generates and sends IEEE 1588V2 clock packets to a lower-level base station.
- IEEE 1588V2 slave clock reception

A base station working as an IEEE 1588V2 slave clock receives IEEE 1588V2 clock packets from an upper-level base station (working as an IEEE 1588V2 master clock), and extracts time data from these packets.

IEEE 1588V2 clock distribution requires that the output port of an upper-level base station be connected to the input port of a lower-level base station. The PTP port for the IEEE 1588V2 master clock is set by using the **ADD PTPPORT** command.

To ensure clock synchronization precision, no more than three levels of base stations can be cascaded. In addition, the base station on each level must support IEEE 1588V2 master clock distribution and slave clock reception. The input synchronization source for first-level cascaded base stations can be GNSS, IEEE 1588V2, or others. **Figure 7-10** illustrates clock distribution based on IEEE 1588V2.

Figure 7-10 IEEE 1588V2 clock distribution



The first-level base station obtains the time synchronization source from the GNSS or IP network, and works as an IEEE 1588V2 master clock to send clock packets to the lower-level base station. Base stations in each subsequent level work as a slave clock to obtain synchronization source, and then as a master clock to send clock packets to lower-level base stations.

NOTE

If upper-level base stations are cascaded with lower-level base stations through intermediate transmission equipment, the intermediate transmission equipment must support the BC function defined in IEEE 1588V2. In addition, a maximum of three hops (including base stations and transmission equipment) can be configured between the root base station and the leaf base station.

7.1.4 Interworking Between IEEE 1588V2-compliant Equipment from Different Manufacturers

The IEEE 1588 protocol supports accurate clock synchronization in the industrial automation field, which allows for clock synchronization in distributed networks. The IEEE 1588 protocol also applies to the wide area network (WAN). The concept of "profile" was proposed in the IEEE 1588V2 protocol released in 2008. Manufacturers choose different subsets of IEEE 1588V2 (representing different profiles) to achieve clock synchronization based on different requirements. As a result, IEEE 1588V2-compliant equipment from different manufacturers cannot interwork with each other.

As an extension of the concept of "profile", the ITU proposes G.8265.1 and G.8275.1 for the telecommunications industry. G.8265.1 and G.8275.1 define interconnection standards for different vendors.

- ITU-T G.8265.1 applies to interworking between a base station and a clock server when IEEE 1588V2 layer 3 unicast frequency synchronization is implemented. ITU-T G.8265.1 is preferentially used in interworking between a base station and a third-party clock server.
- ITU-T G.8275.1 applies to interworking between a base station and transmission equipment when IEEE 1588V2 layer 2 multicast time synchronization is implemented.

IEEE 1588V2 16.1 applies to interworking between a base station and third-party clock servers when IEEE 1588V2 time synchronization is implemented. For details on IEEE 1588V2 16.1, see the optional section 16.1 in the IEEE 1588V2 standard.

Table 7-2 lists application scenarios of profile types complying with IEEE 1588V2 in interworking between IEEE 1588V2-compliant equipment from different manufacturers.

Table 7-2 Application scenarios of profile types complying with the IEEE 1588V2 standard

Profile Type	Synchronization Mode	Application Scenario	Supported Protocol Version
IEEE 1588V2 (Huawei proprietary) layer 3 unicast	Time synchronization and frequency synchronization	Interworking between gNodeBs and a clock server provided by Huawei	IEEE 1588-2008
IEEE 1588V2 16.1 layer 3 unicast	Time synchronization and frequency synchronization	Interworking between gNodeBs and a clock server provided by a third party	IEEE 1588-2008
ITU-T G.8275.2 layer 3 unicast	Time synchronization	• Interworking between gNodeBs and a clock server provided by Huawei • Interworking between gNodeBs and a clock server provided by a third party	ITU-T G.8275.2 ProfileVersion 1.0
ITU-T G.8265.1 layer 3 unicast	Frequency synchronization	• Interworking between gNodeBs and a clock server provided by Huawei • Interworking between gNodeBs and a clock server provided by a third party	ITU-T G.8265.1 ProfileVersion 1.0

Profile Type	Synchronization Mode	Application Scenario	Supported Protocol Version
IEEE 1588V2 layer 2 multicast	Time synchronization	• Interworking between gNodeBs and transmission equipment provided by Huawei • Interworking between gNodeBs and third-party transmission equipment	IEEE 1588-2008
ITU-T G.8275.1 layer 2 multicast	Time synchronization	• Interworking between gNodeBs and transmission equipment provided by Huawei • Interworking between gNodeBs and third-party transmission equipment	ITU-T G.8275.1 ProfileVersion 1.0

The **IPCLKLNK**.***PROFILETYPE*** parameter specifies a profile type used for interworking between IEEE 1588V2-compliant equipment from different manufacturers. The gNodeB and equipment interconnected with the gNodeB (for example, a clock server or transmission equipment) must use the same profile type for interworking.

When gNodeBs use IEEE 1588V2 time synchronization, set the **IPCLKLNK**.***PROFILETYPE*** parameter as follows:

- If a Huawei clock server is used, set this parameter to **1588V2**.
- If the gNodeB and transmission equipment are interconnected, set this parameter to **1588V2** or **G.8275.1**.

If a base station interworks with a third-party clock server, the **IPCLKLNK**.***ANNFREQ*** and **IPCLKLNK**.***NEGDURATION*** parameters can be set to specify the frequency at which the clock server sends Announce packets to the base station and the interval at which negotiations are performed between them, respectively. It is recommended that parameters **IPCLKLNK**.***ANNFREQ*** and **IPCLKLNK**.***NEGDURATION*** be set to their default values **1/2** and **300**, respectively. The two parameters can also be set to other values supported by the clock server that interworks with the base station.

When a base station interworks with third-party transmission equipment, the **IPCLKLNK**.***DSTMLTMACTYPE*** parameter specifies the destination multicast MAC type of G.8275.1 clock synchronization packets sent by the base station. The destination MAC type is used to match the MAC type of multicast packets that the transmission equipment receives. The base station can adaptively match the destination multicast MAC type of clock packets received from the transmission equipment. The **IPCLKLNK**.***DSTMLTMACTYPE*** parameter is used to match the clock packets for all types of transmission equipment.

When interworking with a Huawei clock server and the **IPCLKLNK**.***PROFILETYPE*** parameter is set to **1588V2**, a gNodeB supports the priority clock class function. To enable this function, run the **SET CLASSIDENTIFY** command with the **IPCLKLNK.*CLASSIDENTIFY*** parameter set to **ON**, and then run the **SET PRICLASS** command to set the priority clock class of the specified IP clock link. If this function is enabled, the gNodeB selects an available synchronization source with the highest priority clock class as the gNodeB synchronization source. If this function is disabled, the gNodeB does not select the synchronization source based on the priority clock class.

NOTE

The base station cannot work with a clock server that sends Announce packets at a frequency greater than 16 Hz or less than 1/64 Hz. If the base station is connected to such a clock server, it discards Announce packets, causing the clock source to become unavailable.

7.2 Network Analysis

7.2.1 Benefits

The IEEE 1588V2 clock is a cost-effective option for networks using IP transmission.

7.2.2 Impacts

The impacts between this function and other functions are described in **7.3.2.3 Function Impacts**.

This function has no impact on the network.

7.3 Requirements

7.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-010070	Network Synchronization	NR0S00STTN00	per gNodeB

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.2.1 Prerequisite Functions

None

7.3.2.2 Mutually Exclusive Functions

None

7.3.2.3 Function Impacts

None

7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

7.3.4 Networking

The networking requirements of the IEEE 1588V2 synchronization source are as follows:

- IEEE 1588V2 applies only to IP over FE/GE/xGE links. The port rate matches the board type, and IEEE 1588V2 is not affected by the port rate.
- When the dual-reference-clock backup mode of IEEE 1588V2 synchronization is used, the two clock links must use the same type of protocol and the same transmission mode. In addition, parameters specifying the clock topology, domain, compensation value, and delay type must be set to the same values for the two clock links.
- The IEEE 1588V2 clock is not recommended if the transport network frequently experiences delay variations or packet losses.
- IEEE 1588V2 clock distribution can be used only in gNodeB cascading scenarios.
- The layer 3 unicast mode is mutually exclusive with the layer 2 multicast mode.
  - If the layer 2 multicast mode is used, all intermediate equipment between gNodeBs and an IEEE 1588V2 server must support this mode.
  - If the layer 3 unicast mode is used:
    - When time synchronization is used, all intermediate transmission equipment on the data bearer network must support the BC or TC function defined in IEEE 1588V2.
    - When frequency synchronization is used, the mechanism for packet transmission is simple and there are no special requirements for the transport network.

If the IEEE 1588V2 time synchronization is used, the following extra networking requirements must be met:

- The frequency at which the upper-level reference clock transmits Sync packets must be greater than or equal to 0.5 packet per second. This ensures fast entrance to the locked mode and high clock accuracy.
- All intermediate equipment between gNodeBs and an IEEE 1588V2 clock server must support the BC or TC function defined in IEEE 1588V2.
- Delay measurement must be performed for all transmission interfaces of all intermediate transmission equipment including layer 3 and layer 2 routers. Delay compensation is performed based on the delay measurement results. The compensation value must be set again if routes or transmission paths change.
- When an IP clock is used in layer 3 unicast mode to implement time synchronization in IPv4 transmission, all transmission nodes on the E2E IP network must ignore the identification field in packets, as required by the RFC6864.

7.3.5 Others

None

7.4 Operation and Maintenance

7.4.1 Data Configuration

7.4.1.1 Data Preparation

Table 7-3 and Table 7-4 describe the parameters used for function activation.

Parameters in the **IPCLKLNK** MO can be configured using the **ADD IPCLKLINK** command or modified using the **MOD IPCLKLINK** command.

Parameters in the **TASM** MO can be configured using the **SET CLKMODE**, **SET CLKSYNCMODE**, or **SET IPCLKALGO** command.

Table 7-3 Parameters in the **IPCLKLNK** MO

Parameter Name	Parameter ID	Setting Notes
Link No.	**IPCLKLNK.*LN***	In single-reference-clock mode of IEEE 1588V2, set this parameter to **0**. In dual-reference-clock backup mode of IEEE 1588V2, two clock links are required. Set this parameter to **0** for one clock link and to **1** for the other. When configuring IEEE 1588V2 clock distribution for cascaded base stations, set this parameter to **2** or **3** for the IEEE 1588V2 master clock and to **0** or **1** for the IEEE 1588V2 slave clock.
Slot No.	**IPCLKLNK.*SN***	Set this parameter as required.
Connection Network Type	**IPCLKLNK.*CNTNETTYPE***	If the OC_MASTER link needs to take effect on the main control board, set this parameter to **BACKHAUL**. If the OC_MASTER link needs to take effect on a board of another type, set this parameter to **CLOUDBB**.
IP mode	**IPCLKLNK.*IPMODE***	Set this parameter to **IPV4** when an IPv4 address is used for an IP clock link. Set this parameter to **IPV6** when an IPv6 address is used for an IP clock link.

Parameter Name	Parameter ID	Setting Notes
Client IPv4	**IPCLKLNK**.***CIP***	Set this parameter to the IPv4 address used for the synchronization between the gNodeB and IP clock server. The IP address must be already configured through the **DEVIP**.***IP*** parameter (old model)/ **IPADDR4.*IP*** parameter (new model)[a]. Otherwise, this parameter is invalid.
Server IPv4	**IPCLKLNK**.***SIP***	Set this parameter to the IPv4 address of the IP clock server that provides clock signals. In dual-reference-clock backup mode of IEEE 1588V2, the two clock links must have different server IP addresses.
Client IPv6	**IPCLKLNK**.***CIPV6***	Set this parameter to the IPv6 address used for the synchronization between the gNodeB and IP clock server. The IP address must be already configured through the **IPADDR6**.***IPV6*** parameter. Otherwise, this parameter is invalid.
Server IPv6	**IPCLKLNK**.***SIPV6***	Set this parameter to the IPv6 address of the IP clock server that provides clock signals. In dual-reference-clock backup mode of IEEE 1588V2, the two clock links must have different server IPv6 addresses.
Domain	**IPCLKLNK**.***DM***	Set this parameter to the same value as the domain of the IP clock server in use.
Device Type[b]	**IPCLKLNK**.***DEVTYPE***	This parameter specifies the device type of an IP clock. If this parameter is set to **OC_MASTER**, the NE works as a master clock device to provide the IP clock, and you need to configure an external reference clock source for supplying reference clock signals. If this parameter is set to **OC_SLAVE**, the NE works as the slave clock device that is synchronized with the upstream clock. Set this parameter as required.
Priority	**IPCLKLNK**.***PRI***	When two or more synchronization sources are used, a smaller value of this parameter indicates a higher priority. In dual-reference-clock backup mode of IEEE 1588V2, the two clock links must have the same priority.
Protocol Type	**IPCLKLNK**.***ICPT***	Set this parameter to **PTP**.
Clock Net Mode[c]	**IPCLKLNK**.***CNM***	Set this parameter as required.
Compensation	**IPCLKLNK**.***CMPST***	Set this parameter as required in time synchronization mode.

Parameter Name	Parameter ID	Setting Notes
Delay Type	**IPCLKLNK**.***DELAYTYPE***	Set this parameter as required in time synchronization mode.
Profile Type	**IPCLKLNK**.***PROFILETYPE***	When an NR TDD base station is interconnected with a clock server: ● Set this parameter to **1588V2** if a Huawei clock server is used. ● Set this parameter to **1588V2_16.1** if a third-party clock server is used. When an NR TDD base station is interconnected with a transmission device, set this parameter to **G.8275.1**.
Clock Source Specified or Not	**IPCLKLNK**.***MACMODE***	Set this parameter as required.
Destination Multicast MAC Address Type	**IPCLKLNK**.***DSTMLTMACTYPE***	Set this parameter when the profile type is set to **G.8275.1**. This parameter is used to match the type of multicast MAC addresses in packets that can be received by transmission equipment.
Master Priority	**IPCLKLNK.*MASTERPRIO***	This parameter is valid only when **IPCLKLNK**.***CNM*** is set to **UNICAST** and **IPCLKLNK**.***PROFILETYPE*** is set to **G.8265.1**.
VRF Index	**IPCLKLNK.*VRFIDX***	Set this parameter to the index of the virtual routing and forwarding (VRF) to which the client IPv4 address or client IPv6 address belongs.
PTP VLAN Mode	**IPCLKLNK**.***PTPVLANMODE***	If **IPCLKLNK**.***CNM*** is set to **L2_MULTICAST**, the VLAN configuration must be the same as that of the transmission device. If the transmission device is not configured with a VLAN, the PTP VLAN mode must be set to **UNTAGGED**. If the transmission device is configured with a VLAN, the PTP VLAN mode must be set to **TAGGED**. If the VLAN configuration of the transmission device cannot be determined, the PTP VLAN mode must be set to **HYBRID**.
Continuous Sending Sync Packets Switch	**IPCLKLNK**.***CSSPSW***	Set this parameter as required.

Parameter Name	Parameter ID	Setting Notes
a: When the **GTRANSPARA**.***TRANSCFGMODE*** parameter is set to **OLD**, the old model is used. When this parameter is set to **NEW**, the new model is used. b: If the **IPCLKLNK**.***DEVTYPE*** parameter is set to **OC_SLAVE** for a clock link, you can run the **MOD IPCLKLINK** command to modify the parameters for a single clock link, such as **CLKSRVDESCRI**, **AUTOCMPSTSW**, **CMPSTINTERVAL**, **CMPST**, **PRI**, and **CSSPSW**. However, in dual-reference-clock backup mode of IEEE 1588V2, the parameter configurations of the two PTP clock links must be the same, and therefore the parameters of a single clock link cannot be directly modified. In this case, you are advised to delete the two clock links by running the **RMV IPCLKLINK** command and then modify the parameters of the two clock links by running the **ADD IPCLKLINK** command to restore the two clock links. c: In dual-reference-clock backup mode of IEEE 1588V2, the two PTP clock links must be configured with the same clock networking mode, compensation value, and delay type. In addition, the two clock links must be configured on the same board. For details, see the online help of the **ADD IPCLKLINK** command.		

Table 7-4 Parameters in the **TASM** MO

Parameter Name	Parameter ID	Setting Notes
Clock Working Mode	**TASM**.***MODE***	It is recommended that this parameter be set to **AUTO** for time synchronization and to **MANUAL** for frequency synchronization. NR TDD base stations support only time synchronization.
Selected Clock Source	**TASM**.***CLKSRC***	When the **TASM**.***MODE*** parameter is set to **MANUAL**, it is recommended that this parameter be set to **IPCLK**.
Clock Source No.	**TASM**.***SRCNO***	When the **TASM**.***MODE*** parameter is set to **MANUAL**, it is recommended that this parameter be set to the link number specified when the clock link is created.
Clock Synchronization Mode	**TASM**.***CLKSYNCMODE***	Set this parameter as required. For NR TDD base stations, set this parameter to **TIME**. If this parameter is set to **TIME**, the **TASM**.***SYNMODE*** parameter must be set to **OFF**.
IP Clock Abnormality Filter Switch	**TASM**.***IPCLKABNFILTERSW***	Set this parameter to **ON** if attempts to filter out IP clock link abnormalities are required before alarm reporting. With this setting, when the system detects a link abnormality that occurs on an IP clock serving as the standby reference clock source, ALM-26263 IP Clock Link Failure will be reported only if this abnormality is not filtered out.

> **NOTE**
>
> The IP clock link can be configured on the main control board and UTRPc of a Huawei base station. Before configuring an IP clock link, ensure that either board has been configured.

7.4.1.2 Using MML Commands

Before using MML commands, refer to **7.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

```
//Setting the clock synchronization mode of the base station
SET CLKSYNCMODE: CLKSYNCMODE=TIME;
//Adding an IP clock link in either layer 2 multicast mode or layer 3 unicast mode
//Adding an IP clock link in layer 2 multicast mode. (IP clock links cannot be added in layer 3 unicast mode
in this case.)
ADD IPCLKLINK:LN=0, ICPT=PTP, SN=6, CNM=L2_MULTICAST, DELAYTYPE=E2E, MACMODE=NO,
PROFILETYPE=1588V2, PTPVLANMODE=UNTAGGED;
//Adding an IP clock link in layer 3 unicast mode. (IP clock links cannot be added in layer 2 multicast mode
in this case.)
ADD IPCLKLINK: LN=0,ICPT=PTP, SN=6, CNM=UNICAST, IPMODE=IPV4, CIP="172.18.11.10",
SIP="172.20.16.10", DELAYTYPE=E2E, PROFILETYPE=1588V2, VRFIDX=0;
//Setting the clock working mode
SET CLKMODE:MODE=AUTO;
//Configuring the 1588V2 master clock and slave clock for clock distribution in cascading scenarios
//Configuring the 1588V2 master clock
ADD IPCLKLINK: LN=2, ICPT=PTP, CNM=L2_MULTICAST, CNTNETTYPE=BACKHAUL, PROFILETYPE=1588V2;
ADD PTPPORT: CN=0, SRN=0, SN=6, PN=0;
//Configuring the 1588V2 slave clock
ADD IPCLKLINK: LN=0, ICPT=PTP, SN=6, CNM= L2_MULTICAST, DELAYTYPE=E2E, MACMODE=NO,
PROFILETYPE=1588V2, PTPVLANMODE=UNTAGGED;
//(Optional) Running the MOD IPCLKLINK command to modify the parameters of an IP clock link when
IPCLKLNK.DEVTYPE is set to OC_SLAVE (This operation is not supported if two OC_SLAVE links are
configured.)
MOD IPCLKLINK: LN=0, AUTOCMPSTSW=ON, CMPSTINTERVAL=300, CMPST=0, PRI=3,
CLKSRVDESCRI="xxx", CSSPSW=ON;
```

7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

> **NOTE**
>
> Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.4.2 Activation Verification

After configuring an IEEE 1588V2 synchronization source for a gNodeB, wait approximately 5 minutes (in the case of IEEE 1588V2 time synchronization) or 20 to 30 minutes (in the case of IEEE 1588V2 frequency synchronization) when the transmission quality is good. Then, run the **DSP CLKSTAT** command to check the system clock status. If the value of **Current Clock Source** is **IP Clock** and the value of **PLL Status** is **Locked**, this function has taken effect.

After an IP clock link is set up, you can run the **DSP IPCLKLINK** command to query the status of the active link and clock synchronization information. The query results include the following information:

- **Grandmaster Quality Class**: Indicates the status of the IP clock server. **Table 7-5** lists the values and their meanings. (IEEE 1588V2 and IEEE 1588V2_16.1 are used as examples.) You can run the **DSP IPCLKCLASSREC** command to query the historical status of **Grandmaster Quality Class**.

Table 7-5 Values and meanings of Grandmaster Quality Class

Value	Meaning
6, 13	The IP clock server locks an external synchronization source and works properly. It provides the optimum level of clock quality to base stations.
7, 14	The external synchronization source of the IP clock server is lost and the IP clock server works in holdover state. The quality level of the clock provided to base stations is degraded but the clock is still available.
52	The IP clock server is faulty and the clock is unavailable. Base stations cannot trace the clock of the IP clock server.

- **Grandmaster Clock ID**: Indicates the ID of the clock server that is directly connected to base stations. You can identify the base stations that are connected to the same clock server based on the clock server ID.
- **Step from Grandmaster**: Indicates the number of nodes supporting IEEE 1588 between the IP clock server and base stations.

7.4.3 Network Monitoring

IEEE 1588V2 synchronization packets are transmitted over transmission links. The transmission status can be monitored using the following counters.

Counter ID	Counter Name
1593835843	**VS.IPCLK.PDV.OFFSET.MAX**
1593835844	**VS.IPCLK.PDV.OFFSET.MIN**
1593835845	**VS.IPCLK.PDV.OFFSET.AVG**
1593835846	**VS.IPCLK.PDV.FPP.SUM**
1593835847	**VS.IPCLK.PDV.FLOOR.MAX**
1593835848	**VS.IPCLK.PDV.FLOOR.MIN**
1593835916	**VS.IPCLK.PDV.TWD.SUM**

8 1PPS+TOD Clock

8.1 Principles

1PPS+TOD clock synchronization enables base stations to achieve synchronization using the 1PPS+TOD serial port signals from external transmission equipment or an external clock device. When the 1PPS+TOD clock is used, the loss of satellite signals can be detected, and the TOD pulse per second (PPS) state of the level-2 time synchronization equipment in the holdover state can also be detected. 1PPS signals are used for clock synchronization. TOD signals are used to transfer the UTC, reference clock type, and working status. The 1PPS+TOD clock supports both frequency and time synchronization.

In 1PPS+TOD clock synchronization, gNodeBs use the 1PPS+TOD reference clock provided by the RGPS clock source of an upper-level BBU or by transmission equipment. The 1PPS+TOD ports on gNodeBs use the RS422 level.

The 1PPS+TOD clock synchronization supports both time and frequency synchronization, and provides accurate UTC. However, 1PPS+TOD signals are transmitted through cables with a short transmission distance. Therefore, the transmission equipment or clock devices that provide the 1PPS+TOD clock must be deployed near base stations, which increases costs.

8.2 Network Analysis

8.2.1 Benefits

The 1PPS+TOD clock supports both time and frequency synchronization, and provides accurate UTC.

8.2.2 Impacts

The impacts between this function and other functions are described in **8.3.2.3 Function Impacts**.

The 1PPS+TOD signals are transmitted through cables with a short transmission distance. Therefore, the transmission equipment or clock devices that provide the 1PPS+TOD clock must be deployed near gNodeBs, which increases costs.

8.3 Requirements

8.3.1 Licenses

None

8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.3.2.1 Prerequisite Functions

None

8.3.2.2 Mutually Exclusive Functions

None

8.3.2.3 Function Impacts

None

8.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

Only the UMPT and USCU support this function.

The BookBBU5901 and BookBBU5901a do not support this function.

RF Modules

No requirements

8.3.4 Networking

None

8.3.5 Others

None

8.4 Operation and Maintenance

8.4.1 Data Configuration

8.4.1.1 Data Preparation

Before configuring a 1PPS+TOD clock link, ensure that the required board has been configured and no GNSS or BITS clock link is configured on the board.

Table 8-1 describes the parameters used for function activation.

Table 8-1 Parameters in the **TOD** MO

Parameter Name	Parameter ID	Setting Notes
TOD Clock No.	**TOD**.***TN***	It is recommended that this parameter be set to the number of the TOD port on the USCU or the UMPT. TOD clock links cannot be configured on both the USCU and UMPT.
TOD Type	**TOD**.***TYPE***	Set this parameter to **IN** when a gNodeB receives clock signals through a TOD port. Set this parameter to **OUT** when a gNodeB transmits clock signals through a TOD port to another gNodeB.
Cabinet No.	**TOD**.***CN***	Set this parameter based on the board to which the TOD feeder is connected. Before configuring a TOD clock link, ensure that the board to which the TOD feeder is connected has been configured.
Subrack No.	**TOD**.***SRN***	Set this parameter based on the board to which the TOD feeder is connected. Before configuring a TOD clock link, ensure that the board to which the TOD feeder is connected has been configured.
Slot No.	**TOD**.***SN***	Set this parameter based on the board to which the TOD feeder is connected. Before configuring a TOD clock link, ensure that the board to which the TOD feeder is connected has been configured.

Parameter Name	Parameter ID	Setting Notes
Cable Length	**TOD**.*CABLE_LEN*	Set this parameter based on the actual length of the TOD feeder. If the feeder length cannot be measured, the difference between the value of this parameter and the actual length must not exceed 20 m. If the length difference is greater than 20 m, the clock accuracy is affected.
Priority	**TOD**.*PRI*	When two or more clock sources are used, a smaller parameter value indicates a higher priority.

Table 8-2 lists the parameters for configuring the clock working mode, specified reference clock, reference clock number, and clock synchronization mode.

Parameters in the **TASM** MO can be configured using the **SET CLKMODE** or **SET CLKSYNCMODE** command.

Table 8-2 Parameters in the **TASM** MO

Parameter Name	Parameter ID	Setting Notes
Clock Working Mode	**TASM**.*MODE*	It is recommended that this parameter be set to **AUTO** for time synchronization and to **MANUAL** for frequency synchronization. TDD must use time synchronization.
Selected Clock Source	**TASM**.*CLKSRC*	Set this parameter to **TOD**.
Clock Source No.	**TASM**.*SRCNO*	Set this parameter to the link number specified when the clock link is created.
Clock Synchronization Mode	**TASM**.*CLKSYNCMODE*	Set this parameter as required. TDD base stations support only time synchronization.

8.4.1.2 Using MML Commands

Before using MML commands, refer to **8.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

```
//Adding a TOD clock source
ADD TOD: TN=0,TYPE=IN,SN=0;
//Setting the clock synchronization mode
SET CLKSYNCMODE: CLKSYNCMODE=TIME;
//Setting the clock working mode
SET CLKMODE: MODE=AUTO;
```

8.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.4.2 Activation Verification

TDD:

Step 1 After the base station is configured with the 1PPS+TOD clock and starts successfully, wait approximately 30 minutes.

Step 2 Run the **DSP CLKSTAT** command to check the system clock status. If the value of **Current Clock Source** is **TOD Clock** and the value of **PLL Status** is **Locked**, this function has taken effect.

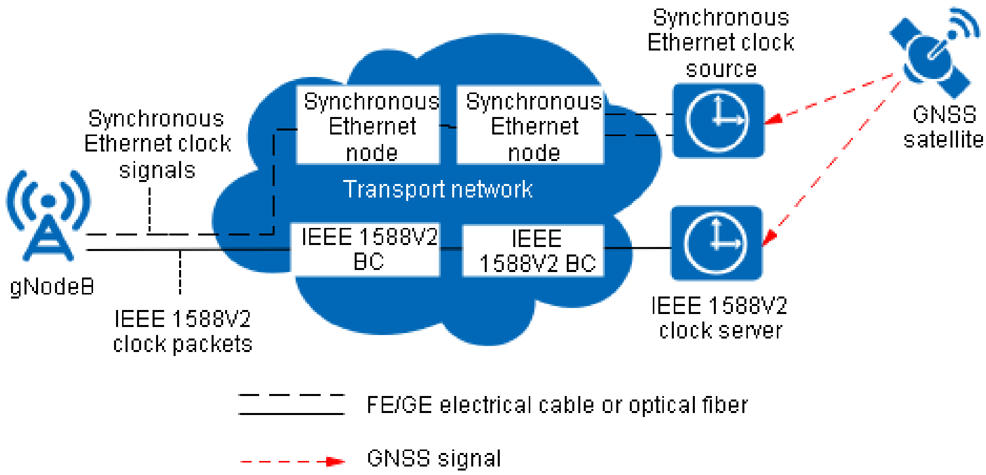
----End

9 Combined Synchronization Sources

9.1 Principles

gNodeBs support the combined synchronization sources of IEEE 1588V2 and synchronous Ethernet. This combination of IEEE 1588V2 and synchronous Ethernet implements synchronization for the base station, with the IEEE 1588V2 clock providing time synchronization and the synchronous Ethernet clock providing frequency synchronization. As synchronous Ethernet has a high frequency stability, the use of these combined synchronization sources enhances time synchronization robustness and improves time holdover performance. If both IEEE 1588V2 and synchronous Ethernet clocks work properly, the IEEE 1588V2 clock is used to perform phase calibration and the synchronous Ethernet clock is used to maintain the phase. If the IEEE 1588V2 clock becomes faulty while the synchronous Ethernet clock works properly, the synchronous Ethernet clock with high frequency stability can be used to maintain the phase for less than or equal to 24 hours. If the synchronous Ethernet clock becomes faulty, services may be affected. Figure 9-1 shows an example of the combination of IEEE 1588V2 and synchronous Ethernet. Combined synchronization sources are supported when the **TASM**.***CLKSRC*** parameter is set to **SYNCETH+IPCLK**.

Figure 9-1 Example combination of IEEE 1588V2 and synchronous Ethernet



9.2 Network Analysis

9.2.1 Benefits

The use of combined synchronization sources enhances time synchronization robustness and improves time holdover performance.

9.2.2 Impacts

The impacts between this function and other functions are described in **9.3.2.3 Function Impacts**.

This function has no impact on the network.

9.3 Requirements

9.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-010070	Network Synchronization	NR0S00STTN00	per gNodeB

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

9.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

9.3.2.1 Prerequisite Functions

None

9.3.2.2 Mutually Exclusive Functions

None

9.3.2.3 Function Impacts

None

9.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

9.3.4 Networking

The networking requirements of the IEEE 1588V2 and synchronous Ethernet combination are as follows:

- The IEEE 1588V2 server and synchronous Ethernet must extract clock signals from the same synchronization source.
- Only IEEE 1588V2 in layer 2 multicast mode or G.8275.1 can be configured as a synchronization source in synchronization source combination. The IEEE 1588V2 in layer 3 unicast mode cannot be configured as the IEEE 1588V2 in the combined synchronization sources. For details about other networking requirements of the IEEE 1588V2 synchronization source, see **7.3.4 Networking**.
- The networking requirements of synchronous Ethernet are as follows:
  - To implement synchronous Ethernet, all the intermediate transmission equipment, such as hubs and LAN switches, must be capable of transparent transmission or regeneration of clock signals at the physical layer.
  - Before configuring a synchronous Ethernet clock link, ensure that the required board has been configured.
  - It is recommended that SSM be enabled for devices in end-to-end synchronous Ethernet networking to improve the reliability of the IEEE 1588V2 and synchronous Ethernet combination.
  - The frequency of the synchronous Ethernet synchronization source must be normal and stable. Otherwise, there will be a phase offset between

the IEEE 1588V2 and synchronous Ethernet synchronization sources. As a result, the base station will report ALM-26262 External Clock Reference Problem with "Specific Problem" being "Inconsistent Clock References". This affects the reliability of the IEEE 1588V2 and synchronous Ethernet combination.

9.3.5 Others

None

9.4 Operation and Maintenance

9.4.1 Data Configuration

9.4.1.1 Data Preparation

- The parameters related to IP clock link addition are described in **7.4.1.1 Data Preparation**.
- The parameters related to synchronous Ethernet addition are described in **Table 9-1** and **Table 9-2**.

  Parameters in the **TASM** MO can be configured using the **SET CLKMODE** or **SET CLKSYNCMODE** command.

Table 9-1 Parameters in the **SYNCETH** MO

Parameter Name	Parameter ID	Setting Notes
SyncEth Clock No.	**SYNCETH.*LN***	Set this parameter to **0**.
Port No.	**SYNCETH.*PN***	Set this parameter to the number of the port where a synchronous Ethernet clock link is configured. Ensure that the port and the synchronous Ethernet synchronization source are connected to the same network.
SSM Selection	**SYNCETH.*SSM***	Set this parameter to the same value as that for the synchronous Ethernet clock server.
Priority	**SYNCETH.*PRI***	When two or more synchronization sources are used, a smaller value of this parameter indicates a higher priority. If only the synchronous Ethernet clock is used, use the default value of this parameter.

Table 9-2 Parameters in the **TASM** MO

Parameter Name	Parameter ID	Setting Notes
Clock Working Mode	**TASM**.***MODE***	Set this parameter to **MANUAL**.
Selected Clock Source	**TASM**.***CLKSRC***	Set this parameter to **SYNCETH**.
Clock Source No.	**TASM**.***SRCNO***	Set this parameter to the link number specified when the clock link was created.
Clock Synchronization Mode	**TASM**.***CLKSYNCMODE***	Set this parameter to **FREQ** as the synchronization source supports only frequency synchronization.

- **Table 9-3** describes the parameters for adding combined synchronization sources.

Table 9-3 Parameters in the **TASM** MO

Parameter Name	Parameter ID	Setting Notes
Clock Working Mode	**TASM**.***MODE***	Set this parameter to **MANUAL**.
Selected Clock Source	**TASM**.***CLKSRC***	Set this parameter to **SYNCETH+IPCLK**.
Clock Source No.	**TASM**.***SRCNO***	Set this parameter to the link number specified when the clock link was created.
Clock Synchronization Mode	**TASM**.***CLKSYNCMODE***	Set this parameter to **TIME**.

9.4.1.2 Using MML Commands

Before using MML commands, refer to **9.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Example of the IEEE 1588V2 and Synchronous Ethernet Combination

```
//Setting the clock synchronization mode of the base station
SET CLKSYNCMODE: CLKSYNCMODE=TIME;
//Configuring an IP clock as the synchronization source
ADD IPCLKLINK: LN=0, ICPT=PTP, SN=7, CNM=L2_MULTICAST, DM=0, DELAYTYPE=E2E, PRI=4,
MACMODE=NO, PROFILETYPE=1588V2, PTPVLANMODE=UNTAGGED;
//Configuring synchronous Ethernet as the synchronization source
ADD SYNCETH: LN=0, SN=7, PN=0;
//Setting the clock working mode
SET CLKMODE: MODE=MANUAL, CLKSRC=SYNCETH+IPCLK;
```

9.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

9.4.2 Activation Verification

Combination of IEEE 1588V2 and Synchronous Ethernet

After configuring the IEEE 1588V2 and synchronous Ethernet combination as the base station synchronization source, wait approximately 5 minutes in scenarios where the transmission quality is good. Then, run the **DSP CLKSTAT** command to query the synchronization source status. If the value of **Current Clock Source** is **SyncEth Clock+IP Clock** and the value of **PLL Status** is **Locked**, this function has taken effect.

9.4.3 Network Monitoring

None

10 Synchronization Source Switching

10.1 Principles

A base station configured with multiple synchronization sources can switch among these synchronization sources. For example, if a configured synchronization source is unavailable, the base station can switch to another available synchronization source for its normal operation. Base stations support both manual and automatic synchronization source switching. **Table 10-1** lists the supported synchronization source types.

Table 10-1 Synchronization source switching types supported by base stations

Switching Type	Time Synchronization	Frequency Synchronization
Manual switching	• GNSS • IEEE 1588V2 • 1PPS+TOD clock • IEEE 1588V2 and synchronous Ethernet combination	• GNSS • IEEE 1588V2 • 1PPS+TOD clock

Switching Type	Time Synchronization	Frequency Synchronization
Automatic switching	● GNSS and IEEE 1588V2 ● GNSS and the "IEEE 1588V2 and synchronous Ethernet" combination ● IEEE 1588V2 and 1PPS+TOD ● IEEE 1588V2 and the peer clock	● GNSS and IEEE 1588V2
Notes: 1. In the case of automatic switching between GNSS and the "IEEE 1588V2 and synchronous Ethernet" combination in time synchronization, the GNSS synchronization source does not include RGPS or CGPS. 2. In automatic switching, the IEEE 1588V2 and synchronous Ethernet combination can be configured by running the **ADD HYBRIDCLK** command.		

10.1.1 Automatic Synchronization Source Selection and Switching

Basic Principles

When the **TASM**.***MODE*** parameter is set to **AUTO**, base stations operate in automatic synchronization source selection and switching mode. Base stations observe the following rules when performing automatic synchronization source selection and switching:

- Initial synchronization source selection
  - In the case of a soft reset, the base station preferentially selects the synchronization source used before the reset. If the synchronization source is abnormal, the base station waits for about 1 minute, and then selects the first available synchronization source.
  - In the case of a power-off reset:

    1. If the synchronization sources have different priorities, the base station preferentially selects the synchronization source with the highest priority. If this synchronization source is abnormal, the base station waits for about 1 minute, and then selects the first available synchronization source.

    2. If the synchronization sources have the same priority, the base station selects the first available one.

- Synchronization source selection during operation
  - The base station automatically selects an available synchronization source.
  - If multiple synchronization sources are available, the base station selects the synchronization source with the highest priority as the current synchronization source.
  - After the synchronization source is selected, the base station does not perform synchronization source switching (except automatic switchback) unless the current synchronization source is faulty.
    - If a low-priority synchronization source is selected, the base station does not switch to a high-priority synchronization source after the high-priority synchronization source becomes available.
    - If the active synchronization source becomes abnormal (due to insufficient GNSS satellites searched, excessive synchronization source frequency offsets or phase deviations, clock link exceptions, or other causes) and the fault persists, the base station automatically switches to the standby synchronization source with the highest priority. The original active synchronization source functions as the standby synchronization source upon recovery.
- After the maximum holdover duration of the base station elapses:
  - If the priorities of the configured synchronization sources are different, the synchronization sources are selected in descending order of priority.
  - If the priorities of the configured synchronization sources are the same, the synchronization source that was last locked is preferentially selected.

NOTE

When the IEEE 1588V2 synchronization source is configured in hybrid synchronization mode, automatic switching among synchronization sources is not supported.

Assume that automatic synchronization source selection is enabled and the standby synchronization source is abnormal (for example, the phase deviation is too large). If the active synchronization source becomes faulty and the maximum holdover duration of the base station elapses, the base station clock may forcibly trace the standby synchronization source, causing clock out-of-synchronization.

In inter-base-station or inter-BBU clock synchronization scenarios, the system takes a long time to enter the stable state after it starts up, as negotiation between boards is required. As a result, it is possible that the synchronization source used before the reset cannot be selected within 1 minute.

Switching upon IEEE 1588V2 Degradation

The GNSS clock has higher accuracy and reliability than the IEEE 1588V2 clock. In the case that GNSS and IEEE 1588V2 are configured as the synchronization sources, when the quality of the IEEE 1588V2 clock deteriorates and no **Inconsistent Clock References** problem exists between the two clocks, then the base station can automatically switch to the GNSS clock.

- When the priority of IEEE 1588V2 is not lower than that of GNSS in time synchronization, switching upon IEEE 1588V2 degradation can be enabled by setting **IPCLKLNK**.***CLASSENHHOSW*** to **ON**.

- The **DSP CLKSRC** command can be used to query the reference clock state. When the IEEE 1588V2 clock source deteriorates, the value of the corresponding item is **Available but Degraded**.
- If the synchronization source type of the base station is GNSS+dual IEEE 1588V2 clock synchronization, the base station automatically switches to the GNSS only when both IEEE 1588V2 clock links deteriorate.

10.1.2 Automatic Switchback

To improve the reliability and accuracy of the clock source for a base station, automatic switchback is supported in clock source combination scenarios. If the **TASM**.***SWITCHPOLICY*** parameter (specifying the reference clock source switchback policy) is set to **SYSTEMSELECT**, automatic switchback takes effect based on the configured synchronization source priorities. If this parameter is set to another value, automatic switchback takes effect based on the clock source sequence listed in the value, and the configured priorities of related clock sources do not take effect. In clock source combination scenarios, automatic switchback requires that the phase deviation between the two reference clock sources is small. If it is too large, ALM-26262 External Clock Reference Problem will be reported with **Specific Problem** being **Inconsistent Clock References**, and the base station will continue to use the original reference clock source.

Reference Clock Source Switchback Policy Set to SYSTEMSELECT

If the reference clock source switchback policy is set to **SYSTEMSELECT**, automatic switchback works in different clock source combination scenarios as follows:

- In dual-reference-clock mode of GNSS+IEEE 1588V2 or GNSS+the "IEEE 1588V2 and synchronous Ethernet" combination in time synchronization, when the GNSS clock is configured with a higher priority than the IEEE 1588V2 clock or the IEEE 1588V2 and synchronous Ethernet combination:
  - With IEEE 1588V2 or the IEEE 1588V2 and synchronous Ethernet combination being used as the current synchronization source, base stations automatically switch back to the GNSS clock if it is available, regardless of whether the IEEE 1588V2 clock or the IEEE 1588V2 and synchronous Ethernet combination is normal.
  - With GNSS being used as the current synchronization source: (1) Base stations automatically switch to the IEEE 1588V2 clock or the IEEE 1588V2 and synchronous Ethernet combination if the GNSS clock becomes faulty; (2) Base stations automatically switch back to the GNSS clock if it recovers, regardless of whether the IEEE 1588V2 clock or the IEEE 1588V2 and synchronous Ethernet combination is normal.
- In other scenarios, automatic switchback to the reference clock source is not supported.

Reference Clock Source Switchback Policy Set to IPCLK+GNSS

If the reference clock source switchback policy is set to **IPCLK+GNSS**, automatic switchback works in different clock source combination scenarios as follows:

- In dual-reference-clock mode of GNSS+IEEE 1588V2 in time synchronization:

  - With GNSS being used as the current synchronization source, base stations automatically switch back to the IEEE 1588V2 clock if it is available, regardless of whether the GNSS clock is normal.
  - With IEEE 1588V2 being used as the current synchronization source: (1) Base stations automatically switch to the GNSS clock if the IEEE 1588V2 clock becomes faulty; (2) Base stations automatically switch back to the IEEE 1588V2 clock if it recovers, regardless of whether the GNSS clock is normal.
- In other scenarios, automatic switchback to the reference clock source is not supported.

Reference Clock Source Switchback Policy Set to IPCLK+TOD

If the reference clock source switchback policy is set to **IPCLK+TOD**, automatic switchback works in different clock source combination scenarios as follows:

- In dual-reference-clock mode of IEEE 1588V2+"1PPS+TOD" in time synchronization:
  - With 1PPS+TOD being used as the current synchronization source, base stations automatically switch back to the IEEE 1588V2 clock if it is available, regardless of whether the 1PPS+TOD clock is normal.
  - With IEEE 1588V2 being used as the current synchronization source: (1) Base stations automatically switch to the 1PPS+TOD clock if the IEEE 1588V2 clock becomes faulty; (2) Base stations automatically switch back to the IEEE 1588V2 clock if it recovers, regardless of whether the 1PPS +TOD clock is normal.
- In other scenarios, automatic switchback to the reference clock source is not supported.

NOTE

If the reference clock source switchback policy is set to **IPCLK+TOD**, other types of synchronization sources cannot be configured, but another TOD synchronization source of the output type can be configured.

Reference Clock Source Switchback Policy Set to IPCLK+PEERCLK

If the reference clock source switchback policy is set to **IPCLK+PEERCLK**, automatic switchback works in different clock source combination scenarios as follows:

- In dual-reference-clock mode of IEEE 1588V2+peer clock in time synchronization:
  - With the peer clock being used as the current synchronization source, base stations automatically switch back to the IEEE 1588V2 clock if it is available, regardless of whether the peer clock is normal.
  - With IEEE 1588V2 being used as the current synchronization source: (1) Base stations automatically switch to the peer clock if the IEEE 1588V2 clock becomes faulty; (2) Base stations automatically switch back to the IEEE 1588V2 clock if it recovers, regardless of whether the peer clock is normal.
- In other scenarios, automatic switchback to the reference clock source is not supported.

NOTE

The preceding mechanism does not differentiate the type of the peer clock.

If the reference clock source switchback policy is set to **IPCLK+PEERCLK**, other types of synchronization sources cannot be configured, but another TOD synchronization source of the output type can be configured.

10.1.3 Manual Synchronization Source Selection and Switching

When the **TASM**.***MODE*** parameter is set to **MANUAL**, base stations operate in manual synchronization source selection and switching mode. Manual synchronization source selection and switching observe the following rules:

- Manual synchronization source switching can be directly performed between two synchronization sources without phase deviations.
- If there is a phase deviation between the synchronization source in use and the target synchronization source, and the phase deviation is caused by the target synchronization source, correct the phase deviation of the target synchronization source, and then manually switch the synchronization sources.
- If there is a phase deviation between the synchronization source in use and the target synchronization source, and the phase deviation is caused by the synchronization source in use, set the clock working mode to free running first, and then manually switch the synchronization sources.

A base station in manual synchronization source switching mode will fail to obtain clock signals if the link to the manually selected synchronization source becomes faulty.

- If the synchronization source is locked before it is lost, the base station clock will work in holdover mode after the synchronization source is lost.
- If the link to the synchronization source is not restored for a long period of time, the base station clock finally enters free running mode.

10.2 Network Analysis

10.2.1 Benefits

If the configured synchronization source is faulty, the base station can switch to another available synchronization source. This ensures the normal operation of the base station.

10.2.2 Impacts

The impacts between this function and other functions are described in **10.3.2.3 Function Impacts**.

This function has no impact on the network.

10.3 Requirements

10.3.1 Licenses

When a reference source is configured as the active clock source, the corresponding license is required. For details, see the license requirements of the reference source. The active clock source cannot be used if the corresponding license is unavailable.

When the reference source is configured as the standby clock source, no license is required.

NOTE

To check whether the corresponding license is available to the active clock source, run the **DSP CLKSRC** command. If the value of **License Authorized** is **Forbid** for the active clock source in the command output, no license is available for this clock source. In this case, you need to apply for the corresponding license.

10.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

10.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Synchronization with GNSS	None	*Synchronization*	None
Low-frequency TDD High-frequency TDD	IEEE 1588V2 clock synchronization	None	*Synchronization*	None

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	IEEE 1588V2 and synchronous Ethernet combination	None	*Synchronization*	None

10.3.2.2 Mutually Exclusive Functions

None

10.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Switching upon IEEE 1588V2 degradation	**IPCLKLNK.*CLASSENHHOSW***	*Synchronization*	Switching upon IEEE 1588V2 degradation does not take effect when the function of automatic switchback in IEEE 1588V2+GNSS clock source scenarios is in effect.

10.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

Board requirements vary depending on the synchronization source to be used. For details, see board requirements of each synchronization source.

For switching upon IEEE 1588V2 degradation:

- The GNSS synchronization source can only be the BBU GPS.
- Only UMPT series boards support this function.

RF Modules

RF module requirements vary depending on the synchronization source to be used. For details, see RF module requirements of each synchronization source.

10.3.4 Networking

- Before the clock working mode is set, ensure that a synchronization source is available and that the networking mode matches the synchronization source.
- Manual mode is used in either of the following situations:
  - A synchronization source needs to be specified, or there is only one synchronization source available. The specified synchronization source must have been configured. In manual mode, base stations cannot automatically select another synchronization source. If both the active and standby clock sources of the selected synchronization source become unavailable, the base station clock finally works in free running mode.
  - In a separate-MPT multimode base station, each RAT supports only one synchronization source type.
- In automatic switching between GNSS and the IEEE 1588V2 and synchronous Ethernet combination, only one IEEE 1588V2 clock link can be configured.

10.3.5 Others

None

10.4 Operation and Maintenance

10.4.1 Data Configuration

10.4.1.1 Data Preparation

Parameters in the **TASM** MO can be configured using the **SET CLKMODE**, **SET CLKSYNCMODE**, or **SET CLKSWITCHPOLICY** command.

Automatic Switching/Automatic Switchback

- If the **TASM**.***SWITCHPOLICY*** parameter is set to **SYSTEMSELECT**, the parameter configurations for automatic switching between synchronization sources are similar, except for the automatic switching from GNSS to the IEEE 1588V2 and synchronous Ethernet combination. This section uses the parameter configuration for automatic switching from GNSS to IEEE 1588V2 as an example.

- Detailed instructions on how to configure the GNSS synchronization source and specify the synchronization source priority are described in **4.4.1.1 Data Preparation**.
- Detailed instructions on how to configure the IEEE 1588V2 synchronization source and specify the synchronization source priority are described in **7.4.1.1 Data Preparation**.
- The parameters in the **TASM** MO related to automatic switching are described in **Table 10-2**.

Table 10-2 Parameters in the **TASM** MO

Parameter Name	Parameter ID	Setting Notes
Clock Working Mode	**TASM**.***MODE***	Set this parameter to **AUTO**.
Clock Synchronization Mode	**TASM**.***CLKSYNCMODE***	Set this parameter as required.
Reference Clock Source Switchback Policy	**TASM**.***SWITCHPOLICY***	Retain the default value.

- If the **TASM**.***SWITCHPOLICY*** parameter is set to **SYSTEMSELECT**, the parameter configuration for automatic switching from GNSS to the IEEE 1588V2 and synchronous Ethernet combination is as follows:
  - Detailed instructions on how to configure the GNSS synchronization source and specify the synchronization source priority are described in **4.4.1.1 Data Preparation**.
  - The IEEE 1588V2 and synchronous Ethernet combination is configured, and the frequency source (synchronous Ethernet) and time source (IEEE 1588V2 synchronization source) for the synchronization source combination are specified. The priority of synchronization source combination is the same as that of the time source. **9.4.1.1 Data Preparation** and **Table 10-3** describe the related parameters.

Table 10-3 Parameters in the **HYBRIDCLK** MO

Parameter Name	Parameter ID	Setting Notes
Hybrid Clock Number	**HYBRIDCLK**.***LN***	Retain the default value.
Hybrid Clock Type	**HYBRIDCLK**.***HYBRIDCLKTYPE***	Retain the default value.

Parameter Name	Parameter ID	Setting Notes
Time Synchronization Clock Reference Source Type	**HYBRIDCLK**.***TCLKSRC***	Retain the default value.
Time Synchronization Clock Reference Source Number	**HYBRIDCLK**.***TSRCNO***	Retain the default value.
Frequency Synchronization Clock Reference Source Type	**HYBRIDCLK**.***FCLKSRC***	Retain the default value.
Frequency Synchronization Clock Reference Source Number	**HYBRIDCLK**.***FSRCNO***	Retain the default value.

– The parameters in the **TASM** MO related to automatic switching are described in **Table 10-4**.

Table 10-4 Parameters in the **TASM** MO

Parameter Name	Parameter ID	Setting Notes
Clock Working Mode	**TASM**.***MODE***	Set this parameter to **AUTO**.
Clock Synchronization Mode	**TASM**.***CLKSYNCMODE***	Set this parameter to **TIME**.
Reference Clock Source Switchback Policy	**TASM**.***SWITCHPOLICY***	Retain the default value.

- If the **TASM**.***SWITCHPOLICY*** parameter is set to **IPCLK+GNSS**:
  – Detailed instructions on how to configure the IEEE 1588V2 synchronization source are described in **7.4.1.1 Data Preparation**.

– Detailed instructions on how to configure the GNSS synchronization source are described in **4.4.1.1 Data Preparation**.
– The parameters in the **TASM** MO related to automatic switching are described in **Table 10-5**.

Table 10-5 Parameters in the **TASM** MO

Parameter Name	Parameter ID	Setting Notes
Clock Working Mode	**TASM.*MODE***	Set this parameter to **AUTO**.
Clock Synchronization Mode	**TASM.*CLKSYNCMODE***	Set this parameter to **TIME**.
Reference Clock Source Switchback Policy	**TASM.*SWITCHPOLICY***	Set this parameter to **IPCLK+GNSS**.

- If the **TASM.*SWITCHPOLICY*** parameter is set to **IPCLK+TOD**:
  – Detailed instructions on how to configure the IEEE 1588V2 synchronization source are described in **7.4.1.1 Data Preparation**.
  – Detailed instructions on how to configure the 1PPS+TOD synchronization source are described in **8.4.1.1 Data Preparation**.
  – The parameters in the **TASM** MO related to automatic switching are described in **Table 10-6**.

Table 10-6 Parameters in the **TASM** MO

Parameter Name	Parameter ID	Setting Notes
Clock Working Mode	**TASM.*MODE***	Set this parameter to **AUTO**.
Clock Synchronization Mode	**TASM.*CLKSYNCMODE***	Set this parameter to **TIME**.
Reference Clock Source Switchback Policy	**TASM.*SWITCHPOLICY***	Set this parameter to **IPCLK+TOD**.

- If the **TASM.*SWITCHPOLICY*** parameter is set to **IPCLK+PEERCLK**:
  – Detailed instructions on how to configure the IEEE 1588V2 synchronization source are described in **7.4.1.1 Data Preparation**.
  – Detailed instructions on how to configure the peer clock are described in the section "eGBTS/NodeB/eNodeB/gNodeB/Co-MPT Base Station

Sharing Clock Source Provided by Another RAT" of "Sharing Clock Source Provided by Another RAT" in *Common Clock*.

– The parameters in the **TASM** MO related to automatic switching are described in **Table 10-7**.

Table 10-7 Parameters in the **TASM** MO

Parameter Name	Parameter ID	Setting Notes
Clock Working Mode	**TASM**.***MODE***	Set this parameter to **AUTO**.
Clock Synchronization Mode	**TASM**.***CLKSYNCMODE***	Set this parameter to **TIME**.
Peer Reference Clock Backup Switch	**TASM**.***PEERCLKBKUPSW***	Set this parameter to **ON**.
Reference Clock Source Switchback Policy	**TASM**.***SWITCHPOLICY***	Set this parameter to **IPCLK+PEERCLK**.

Manual Switching

The parameter configurations for manual switching between synchronization sources are similar. This section uses the manual switching from GNSS to the IEEE 1588V2 and synchronous Ethernet combination as an example.

- Detailed instructions on how to configure the GNSS synchronization source are described in **4.4.1.1 Data Preparation**.
- The parameters used for configuring the IEEE 1588V2 and synchronous Ethernet combination are described in **9.4.1.1 Data Preparation** and **Table 10-8**.

Table 10-8 Parameters in the **TASM** MO

Parameter Name	Parameter ID	Setting Notes
Clock Working Mode	**TASM**.***MODE***	Set this parameter to **MANUAL**.
Selected Clock Source	**TASM**.***CLKSRC***	Set this parameter based on the synchronization source in use. For example, set this parameter to **GNSS** if the GNSS synchronization source is used.

Parameter Name	Parameter ID	Setting Notes
Clock Source No.	**TASM.*SRCNO***	Set this parameter to the link number specified when the clock link was created.
Clock Synchronization Mode	**TASM.*CLKSYNCMODE***	Set this parameter as required.

10.4.1.2 Using MML Commands

Before using MML commands, refer to **10.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Example of Automatic Switching Between the GNSS and IEEE 1588V2 Synchronization Sources

- If the **TASM.*SWITCHPOLICY*** parameter is set to its default value, the priority of the GNSS synchronization source needs to be higher than that of the IEEE 1588V2 synchronization source.
```
//Setting the clock synchronization mode of the base station
SET CLKSYNCMODE: CLKSYNCMODE=TIME;
//Adding a GNSS clock link
ADD GNSS: GN=0, CN=0, SRN=0, SN=7, CABLE_LEN=50, MODE=GPS, PRI=1;
//Adding an IP clock link
ADD IPCLKLINK: LN=0, ICPT=PTP, SN=7, CNM=L2_MULTICAST, DELAYTYPE=E2E, PRI=2,
MACMODE=NO, PROFILETYPE=1588V2, PTPVLANMODE=UNTAGGED;
//Setting the clock working mode
SET CLKMODE: MODE=AUTO;
```
- If the **TASM.*SWITCHPOLICY*** parameter is set to **IPCLK+GNSS**, the priorities of the IEEE 1588V2 and GNSS synchronization sources do not need to be specified.
```
//Setting the clock synchronization mode of the base station
SET CLKSYNCMODE: CLKSYNCMODE=TIME;
//Adding an IP clock link
ADD IPCLKLINK: LN=0, ICPT=PTP, SN=7, CNM=L2_MULTICAST, DELAYTYPE=E2E, MACMODE=NO,
PROFILETYPE=1588V2, PTPVLANMODE=UNTAGGED;
//Adding a GNSS clock link
ADD GNSS: GN=0, CN=0, SRN=0, SN=7, CABLE_LEN=50, MODE=GPS;
//Setting the clock working mode
SET CLKMODE: MODE=AUTO;
//Setting the reference clock source switchback policy
SET CLKSWITCHPOLICY: SWITCHPOLICY=IPCLK+GNSS;
```


Activation Command Example of Automatic Switching Between the IEEE 1588V2 and 1PPS+TOD Synchronization Sources

Setting the reference clock source switchback policy to **IPCLK+TOD**

```
//Setting the clock synchronization mode of the base station
SET CLKSYNCMODE: CLKSYNCMODE=TIME;
//Adding an IP clock link
ADD IPCLKLINK: LN=0, ICPT=PTP, SN=7, CNM=L2_MULTICAST, DELAYTYPE=E2E, MACMODE=NO,
PROFILETYPE=1588V2, PTPVLANMODE=UNTAGGED;
//Adding a 1PPS+TOD clock link
ADD TOD: TN=0, TYPE=IN, SN=7;
//Setting the clock working mode
SET CLKMODE: MODE=AUTO;
//Setting the reference clock source switchback policy
SET CLKSWITCHPOLICY: SWITCHPOLICY=IPCLK+TOD;
```

Activation Command Example of Automatic Switching Between the IEEE 1588V2 Synchronization Source and Peer Clock

```
//Setting the clock synchronization mode of the base station
SET CLKSYNCMODE: CLKSYNCMODE=TIME;
//Adding an IP clock link
ADD IPCLKLINK: LN=0, ICPT=PTP, SN=7, CNM=L2_MULTICAST, DELAYTYPE=E2E, MACMODE=NO,
PROFILETYPE=1588V2, PTPVLANMODE=UNTAGGED;
//Adding a peer reference clock link
ADD PEERCLK: PN=0, PNT=BBU;
//Setting the clock working mode
SET CLKMODE: MODE=AUTO, PEERCLKBKUPSW=ON;
//Setting the reference clock source switchback policy
SET CLKSWITCHPOLICY: SWITCHPOLICY=IPCLK+PEERCLK;
```

Activation Command Example of Automatic Switching Between GNSS and the "IEEE 1588V2 and Synchronous Ethernet" Combination

```
//Setting the clock synchronization mode of the base station
SET CLKSYNCMODE: CLKSYNCMODE=TIME;
//Adding a GNSS clock link
ADD GNSS: GN=0, CN=0, SRN=0, SN=7, CABLE_LEN=50, MODE=GPS, PRI=1;
//Adding an IP clock link
ADD IPCLKLINK: LN=0, ICPT=PTP, SN=7, CNM=L2_MULTICAST, DELAYTYPE=E2E, PRI=2, MACMODE=NO,
PROFILETYPE=1588V2, PTPVLANMODE=UNTAGGED;
//Adding a synchronous Ethernet clock link
ADD SYNCETH: LN=0, SN=7, PN=0;
//Adding a hybrid reference clock source
ADD HYBRIDCLK: HYBRIDCLKTYPE=TF;
//Setting the clock working mode
SET CLKMODE: MODE=AUTO;
```

Activation Command Example of Manual Switching Between GNSS and the "IEEE 1588V2 and Synchronous Ethernet" Combination

```
//Setting the clock synchronization mode of the base station
SET CLKSYNCMODE: CLKSYNCMODE=TIME;
//Adding a GNSS clock link
ADD GNSS: GN=0, CN=0, SRN=0, SN=7, CABLE_LEN=50, MODE=GPS, PRI=1;
//Adding an IP clock link
ADD IPCLKLINK: LN=0, ICPT=PTP, SN=7, CNM=L2_MULTICAST, DELAYTYPE=E2E, PRI=2, MACMODE=NO,
PROFILETYPE=1588V2, PTPVLANMODE=UNTAGGED;
//Adding a synchronous Ethernet clock link
ADD SYNCETH: LN=0, SN=7, PN=0;
//Setting the clock working mode
SET CLKMODE: MODE=MANUAL, CLKSRC=GNSS, SRCNO=0;
//(Optional) Adding the synchronization source combination when synchronization source switching mode
is to be changed from manual to automatic
```



```
ADD HYBRIDCLK: HYBRIDCLKTYPE=TF;
//(Optional) Manually setting the clock working mode when the GNSS synchronization source is unavailable
SET CLKMODE: MODE=MANUAL, CLKSRC=SYNCETH+IPCLK;
```

Example for Configuring Automatic Switching to GNSS upon IEEE 1588V2 Deterioration

```
//Setting the clock synchronization mode of the base station
SET CLKSYNCMODE: CLKSYNCMODE=TIME;
//Adding a GNSS clock link
ADD GNSS: GN=0, CN=0, SRN=0, SN=7, CABLE_LEN=50, MODE=GPS, PRI=2;
//Adding an IP clock link
ADD IPCLKLINK: LN=0, ICPT=PTP, SN=7, CNM=L2_MULTICAST, DELAYTYPE=E2E, PRI=1, MACMODE=NO,
PROFILETYPE=1588V2, PTPVLANMODE=UNTAGGED;
//Setting the clock working mode
SET CLKMODE: MODE=AUTO;
//Setting the priority clock class
SET PRICLASS: CLASS0=6, CLASS1=13, CLASS2=7, CLASS3=14;
//Setting the priority clock switch
SET CLASSIDENTIFY: LN=0, CLASSIDENTIFY=ON, CLASSENHHOSW=ON;
```

10.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

10.4.2 Activation Verification

The following uses MML commands as an example for activation verification.

Automatic Switching Between the GNSS and IEEE 1588V2 Synchronization Sources

- If the **TASM**.***SWITCHPOLICY*** parameter is set to its default value:
  a. Run the **DSP CLKSRC** command to query the status of the active and standby clock sources. If the value of **Clock Source Type** is **GNSS Clock**, the value of **Clock Source State** is **Available**, and the value of **Clock Source Active State** is **Activated** for one clock source, and if the value of **Clock Source Type** is **IP Clock**, the value of **Clock Source State** is **Available**, and the value of **Clock Source Active State** is **Deactivated** for the other clock source, the current active synchronization source is GNSS and the standby synchronization source is IEEE 1588V2.
  b. Run the **DSP CLKSTAT** command to query the synchronization source status. If the value of **Current Clock Source** is **GNSS Clock** and the value of **PLL Status** is **Locked**, the active GNSS clock source is working properly.
  c. When the GNSS clock becomes unavailable due to reasons such as a GNSS satellite card fault or a GNSS antenna open-circuit, automatic switching to the IEEE 1588V2 synchronization source is performed. Wait for about 15 minutes (the wait time varies depending on the synchronization source after the switchover), and run the **DSP CLKSTAT** command to query the synchronization source status. If the value of

Current Clock Source is **IP Clock** and that of **PLL Status** is **Locked**, automatic switching has succeeded.

d. After the GNSS synchronization source recovers, automatic switchback to the GNSS synchronization source is performed. Wait for about 15 minutes (the wait time varies depending on the synchronization source after the switchover), and run the **DSP CLKSTAT** command to query the synchronization source status. If the value of **Current Clock Source** is **GNSS Clock** and that of **PLL Status** is **Locked**, automatic switchback has succeeded.

- If the **TASM.*SWITCHPOLICY*** parameter is set to **IPCLK+GNSS**:
  a. Run the **DSP CLKSRC** command to query the status of the active and standby clock sources. If the value of **Clock Source Type** is **IP Clock**, the value of **Clock Source State** is **Available**, and the value of **Clock Source Active State** is **Activated** for one clock source, and if the value of **Clock Source Type** is **GNSS Clock**, the value of **Clock Source State** is **Available**, and the value of **Clock Source Active State** is **Deactivated** for the other clock source, the current active synchronization source is IEEE 1588V2 and the standby synchronization source is GNSS.
  b. Run the **DSP CLKSTAT** command to query the synchronization source status. If the value of **Current Clock Source** is **IP Clock** and the value of **PLL Status** is **Locked**, the active IEEE 1588V2 clock source is working properly.
  c. When the IEEE 1588V2 synchronization source becomes unavailable, automatic switching to the GNSS synchronization source is performed. Wait for about 15 minutes (the wait time varies depending on the synchronization source after the switchover), and run the **DSP CLKSTAT** command to query the synchronization source status. If the value of **Current Clock Source** is **GNSS Clock** and that of **PLL Status** is **Locked**, automatic switching has succeeded.
  d. After the IEEE 1588V2 synchronization source recovers, automatic switchback to the IEEE 1588V2 synchronization source is performed. Wait for about 15 minutes (the wait time varies depending on the synchronization source after the switchover), and run the **DSP CLKSTAT** command to query the synchronization source status. If the value of **Current Clock Source** is **IP Clock** and that of **PLL Status** is **Locked**, automatic switchback has succeeded.

Automatic Switching Between GNSS and the IEEE 1588V2 and Synchronous Ethernet Combination

1. Run the **DSP CLKSRC** command to query the status of the active and standby clock sources. If the value of **Clock Source Type** is **GNSS Clock**, the value of **Clock Source State** is **Available**, and the value of **Clock Source Active State** is **Activated** for one clock source, and if the value of **Clock Source Type** is **Hybrid Clock**, the value of **Clock Source State** is **Available**, and the value of **Clock Source Active State** is **Deactivated** for the other clock source, the current active synchronization source is GNSS and the standby synchronization source is IEEE 1588V2 and synchronous Ethernet combination.
2. Run the **DSP CLKSTAT** command to query the synchronization source status. If the value of **Current Clock Source** is **GNSS Clock** and the value of **PLL Status** is **Locked**, the active GNSS clock source is working properly.

3. When the GNSS clock becomes unavailable due to reasons such as a GNSS satellite card fault or a GNSS antenna open-circuit, automatic switching to the IEEE 1588V2 and synchronous Ethernet combination is performed. Wait for about 15 minutes (the wait time varies depending on the synchronization source after the switchover), and run the **DSP CLKSTAT** command to query the synchronization source status. If the value of **Current Clock Source** is **Hybrid Clock** and that of **PLL Status** is **Locked**, automatic switching has succeeded.
4. After the GNSS synchronization source recovers, automatic switchback to the GNSS synchronization source is performed. Wait for about 15 minutes (the wait time varies depending on the synchronization source after the switchover), and run the **DSP CLKSTAT** command to query the synchronization source status. If the value of **Current Clock Source** is **GNSS Clock** and that of **PLL Status** is **Locked**, automatic switchback has succeeded.

Manual Switching Between GNSS and the IEEE 1588V2 and Synchronous Ethernet Combination

1. Run the **DSP CLKSRC** command to query the status of the active and standby clock sources. If the value of **Clock Source Type** is **GNSS Clock**, the value of **Clock Source State** is **Available**, and the value of **Clock Source Active State** is **Activated** for one clock source, and if the value of **Clock Source Type** is **Hybrid Clock**, the value of **Clock Source State** is **Available**, and the value of **Clock Source Active State** is **Deactivated** for the other clock source, the current active synchronization source is GNSS and the standby synchronization source is IEEE 1588V2 and synchronous Ethernet combination.
2. Run the **DSP CLKSTAT** command to query the synchronization source status. If the value of **Current Clock Source** is **GNSS Clock** and the value of **PLL Status** is **Locked**, the active GNSS clock source is working properly.
3. When the GNSS clock becomes unavailable due to reasons such as a GNSS satellite card fault or a GNSS antenna open-circuit, manual switching to the IEEE 1588V2 and synchronous Ethernet combination is required. Wait for about 15 minutes (the wait time varies depending on the synchronization source after the switchover), and run the **DSP CLKSTAT** command to query the synchronization source status. If the value of **Current Clock Source** is **Hybrid Clock** and that of **PLL Status** is **Locked**, this function has been successfully activated.

Automatic Switching to GNSS upon IEEE 1588V2 Deterioration

1. Run the **DSP CLKSRC** command to query the status of the active and standby clock sources. If the value of **Clock Source Type** is **IP Clock**, the value of **Clock Source State** is **Available**, and the value of **Clock Source Active State** is **Activated** for one clock source, and if the value of **Clock Source Type** is **GNSS Clock**, the value of **Clock Source State** is **Available**, and the value of **Clock Source Active State** is **Deactivated** for the other clock source, the current active synchronization source is IEEE 1588V2 and the standby synchronization source is GNSS.
2. Run the **DSP CLKSTAT** command to query the synchronization source status. If the value of **Current Clock Source** is **IP Clock** and the value of **PLL Status** is **Locked**, the active IEEE 1588V2 clock source is working properly.

3. If the quality level of the 1588V2 clock deteriorates, the time synchronization source is automatically switched to the GNSS clock and EVT-26269 Reference Clock Switchover is reported. Wait for about 15 minutes (the wait time varies depending on the synchronization source after the switchover), and run the **DSP CLKSTAT** command to query the synchronization source status. If the value of **Current Clock Source** is **GNSS Clock** and that of **PLL Status** is **Locked**, run the **DSP CLKSRC** command to query the reference clock source status. If the value of **Clock Source State** for the clock source with **Clock Source Type** being **IP Clock** is **Available but Degraded**, this function has been activated.

Automatic Switching Between the IEEE 1588V2 and 1PPS+TOD Synchronization Sources

1. Run the **DSP CLKSRC** command to query the status of the active and standby clock sources. If the value of **Clock Source Type** is **IP Clock**, the value of **Clock Source State** is **Available**, and the value of **Clock Source Active State** is **Activated** for one clock source, and if the value of **Clock Source Type** is **TOD Clock**, the value of **Clock Source State** is **Available**, and the value of **Clock Source Active State** is **Deactivated** for the other clock source, the current active synchronization source is IEEE 1588V2 and the standby synchronization source is 1PPS+TOD.
2. Run the **DSP CLKSTAT** command to query the synchronization source status. If the value of **Current Clock Source** is **IP Clock** and the value of **PLL Status** is **Locked**, the active IEEE 1588V2 clock source is working properly.
3. When the IEEE 1588V2 synchronization source becomes unavailable, automatic switching to the 1PPS+TOD synchronization source is performed. Wait for about 15 minutes (the wait time varies depending on the synchronization source after the switchover), and run the **DSP CLKSTAT** command to query the synchronization source status. If the value of **Current Clock Source** is **TOD Clock** and that of **PLL Status** is **Locked**, automatic switching has succeeded.
4. After the IEEE 1588V2 synchronization source recovers, automatic switchback to the IEEE 1588V2 synchronization source is performed. Wait for about 15 minutes (the wait time varies depending on the synchronization source after the switchover), and run the **DSP CLKSTAT** command to query the synchronization source status. If the value of **Current Clock Source** is **IP Clock** and that of **PLL Status** is **Locked**, automatic switchback has succeeded.

Automatic Switching Between the IEEE 1588V2 Synchronization Source and Peer Clock

1. Run the **DSP CLKSRC** command to query the status of the active and standby clock sources. If the value of **Clock Source Type** is **IP Clock**, the value of **Clock Source State** is **Available**, and the value of **Clock Source Active State** is **Activated** for one clock source, and if the value of **Clock Source Type** is **Peer Clock**, the value of **Clock Source State** is **Available**, and the value of **Clock Source Active State** is **Deactivated** for the other clock source, the current active synchronization source is IEEE 1588V2 and the standby synchronization source is the peer clock.

2. Run the **DSP CLKSTAT** command to query the synchronization source status. If the value of **Current Clock Source** is **IP Clock** and the value of **PLL Status** is **Locked**, the active IEEE 1588V2 clock source is working properly.
3. When the IEEE 1588V2 synchronization source becomes unavailable, automatic switching to the peer clock is performed. Wait for about 15 minutes (the wait time varies depending on the synchronization source after the switchover), and run the **DSP CLKSTAT** command to query the synchronization source status. If the value of **Current Clock Source** is **Peer Clock** and that of **PLL Status** is **Locked**, automatic switching has succeeded.
4. After the IEEE 1588V2 synchronization source recovers, automatic switchback to the IEEE 1588V2 synchronization source is performed. Wait for about 15 minutes (the wait time varies depending on the synchronization source after the switchover), and run the **DSP CLKSTAT** command to query the synchronization source status. If the value of **Current Clock Source** is **IP Clock** and that of **PLL Status** is **Locked**, automatic switchback has succeeded.

NOTE

The **DSP CLKSRC** command can be used to query the availability and activation status of clock sources. An activated clock source is valid only when it is available. Otherwise, it cannot be locked. An available clock source only indicates that the physical signals of the clock source are normal or the network connection is normal. When the clock source is activated, the NE can synchronize its signals with the signals of the clock source, but this does not mean that the NE will lock the clock source after the synchronization.

In the case of synchronization source switching, the **PLL Status** displayed in the **DSP CLKSTAT** command output may be the status before synchronization source switching. You are advised to query the status about 15 minutes after synchronization source switching is completed. (The waiting time depends on the selected synchronization source and may be longer in some cases.)

10.4.3 Network Monitoring

None

11 Clock Out-of-Synchronization Detection (Low-Frequency TDD)

11.1 Principles

NR TDD has strict requirements for clock synchronization. If a base station has lost synchronization, it may cause interference to other base stations. When the interference is severe, UEs may be unable to access the network, or receive a poor service experience. For example, service drops or handover failures are likely to occur. Clock out-of-synchronization detection helps quickly and accurately identify base stations that are not synchronized.

Clock out-of-synchronization detection can be used to identify out-of-synchronization base stations using the **11.1.1 Clock Out-of-Synchronization Detection 1.0** or **11.1.2 Clock Out-of-Synchronization Detection 2.0** solution.

- Clock out-of-synchronization detection 1.0 can be used when a GNSS clock source is configured and can take effect only when there are two or fewer faulty base stations in an area.
- Clock out-of-synchronization detection 2.0 can be used when an IEEE 1588V2 clock source is configured or when multiple BBUs are deployed in the same equipment room and can take effect when more than two base stations are faulty in an area.

After clock out-of-synchronization detection is complete:

- If the **CLK_PROCESS_SW** option of the **gNodeBParam**.***ClkOutofsyncDetSwitch*** parameter is selected, then:
  - The out-of-synchronization base stations automatically deactivate all their NR TDD cells and report ALM-29840 gNodeB Out of Service. The out-of-synchronization base stations will not deactivate their NR TDD cells if more than five base stations are detected to be out of synchronization. This is because a single MAE allows at most five out-of-synchronization base stations to deactivate their NR TDD cells within a day.
  - Deselect this option for synchronized base stations that were incorrectly considered as out-of-synchronization base stations, so as to reactivate all deactivated NR TDD cells of these base stations.

- If the **CLK_PROCESS_SW** option of the **gNodeBParam**.***ClkOutofsyncDetSwitch*** parameter is deselected, the out-of-synchronization base stations will not automatically deactivate their NR TDD cells, but will report ALM-29840 gNodeB Out of Service.

11.1.1 Clock Out-of-Synchronization Detection 1.0

Clock out-of-synchronization detection 1.0 can be used when a GNSS clock source is configured and can take effect only when there are two or fewer faulty base stations in an area. This solution takes effect when both of the following conditions are met: (1) On the base station, the **CLK_DETECT_SW** option of the **gNodeBParam**.***ClkOutofsyncDetSwitch*** parameter is selected. (2) On the MAE-Access, the **Clock out-of-sync check** switch is turned on, and the **Enhanced out-of-synchronization detection** switch is turned off.

This solution involves the following procedures: network interference monitoring, synchronization status detection between base stations, and fault determination and alarm reporting for base stations.

- Network interference monitoring

  If the uplink interference strength on a base station frequency is greater than the value of **gNodeBParam**.***ClkOutofsyncIntrfRptThld***, the base station will report an interference event for this frequency to the MAE.

  When the uplink interference strength on a base station frequency is greater than the value of **gNodeBParam**.***ClkOutofsyncIntrfRptThld*** continuously, the base station will repeatedly report interference events. The initial reporting interval is specified by the **gNodeBParam**.***ClkOutOfSyncCIntrfRptIntvl*** parameter, and then the reporting interval will be gradually prolonged. When the reporting interval exceeds 8 hours, the repeated reporting of interference events related to the frequency is stopped.
- Synchronization status detection between base stations

  Synchronization sequence detection and silence detection are used to determine the status of the to-be-detected base stations.
  - Synchronization sequence detection

    During synchronization sequence detection, a to-be-detected base station sends synchronization sequences, and its intra-frequency neighboring base stations check for the synchronization sequences at the same frame locations. Only the intra-frequency neighboring base stations that are synchronized with the to-be-detected base station can detect the synchronization sequences.

    If the synchronization sequences can be detected by more than one intra-frequency neighboring base station, the to-be-detected base station is operating properly. Otherwise, the to-be-detected base station may be unsynchronized and it requires silence detection.
  - Silence detection

    During silence detection, a to-be-detected base station stops transmitting downlink signals, including traffic signals and reference signals, on some frames according to certain rules. If the interference on more than two intra-frequency neighboring base stations disappears during the period when the downlink signal transmission stops, and reappears after the downlink signal transmission restarts, it can be determined that the to-be-detected base station is interfering with these intra-frequency neighboring base stations and therefore it is an out-of-synchronization base station. Otherwise, the to-be-detected base station is not considered as an out-of-synchronization base station.

Silence detection works as follows: It becomes invalid when the strength of interference on the neighboring base stations is less than the value of **gNodeBParam**.***ClkOutofsyncIntrfRptThld*** minus 10. The to-be-detected base station is considered as unsynchronized if the interference strength change for the neighboring base stations exceeds the value of **gNodeBParam**.***SilDetIntrfDiffThld*** during the detection.

- Fault determination and alarm reporting for base stations

  The MAE can determine the out-of-synchronization base stations after network interference monitoring and synchronization status detection between base stations.

  If the **Send out-of-sync alarm** switch on the MAE is turned on, out-of-synchronization base stations will report ALM-29830 gNodeB Out of Clock Synchronization.

Clock out-of-synchronization detection 1.0 also supports re-detection during off-peak hours. With this function, base stations that have reported ALM-29830 gNodeB Out of Clock Synchronization are re-detected during a period of time when clock out-of-synchronization detection is not performed. If a base station is considered synchronized during re-detection, the alarm will be automatically cleared. Re-detection during off-peak hours is not performed on base stations that have automatically deactivated their served out-of-synchronization cells.

11.1.2 Clock Out-of-Synchronization Detection 2.0

With the development of NR TDD networks, a large number of base stations use the IEEE 1588V2 clock source, or the number of cases are large that multiple BBUs are deployed in the same equipment room. These changes increase the probability that multiple base stations are out of synchronization at the same time. Clock out-of-synchronization detection 2.0 is introduced as clock out-of-synchronization detection 1.0 cannot detect more than two out-of-synchronization base stations at a time.

Clock out-of-synchronization detection 2.0 can be used when an IEEE 1588V2 clock source is configured or when multiple BBUs are deployed in the same equipment room and can take effect when more than two base stations are faulty in an area. For cells not covering subways or high-speed railways, this solution takes effect when both of the following conditions are met: (1) On the base station, the **CLK_DETECT_SW** option of the **gNodeBParam**.***ClkOutofsyncDetSwitch*** parameter is selected. (2) On the MAE-Access, the **Clock out-of-sync check** and **Enhanced out-of-synchronization detection** switches are both turned on. For cells covering subways or high-speed railways, this solution (except synchronization sequence detection, as this function is not supported) takes effect when both of the following conditions are met: (1) On the base station, the **CLK_DETECT_IN_COMPLEX_SW** option of the **gNodeBParam**.***ClkOutofsyncDetSwitch*** parameter is selected. (2) On the MAE-Access, the **Clock out-of-sync check** and **Enhanced out-of-synchronization detection** switches are both turned on.

This solution involves the following procedures: network interference monitoring, synchronization status detection between base stations, and fault determination and alarm reporting for base stations.

- Network interference monitoring

  The principles are the same as those of network interference monitoring in clock out-of-synchronization detection 1.0. For details, see **11.1.1 Clock Out-of-Synchronization Detection 1.0**.

- Synchronization status detection between base stations

  Synchronization sequence detection and out-of-synchronization sequence detection are used to determine whether base stations are synchronized.

  – Synchronization sequence detection

    During synchronization sequence detection, a to-be-detected base station sends synchronization sequences, and its intra-frequency neighboring base stations check for the synchronization sequences at the same frame locations. Only the intra-frequency neighboring base stations that are synchronized with the to-be-detected base station can detect the synchronization sequences.

    If an intra-frequency neighboring base station detects the synchronization sequences, then this base station and the to-be-detected base station are considered to be synchronized with each other.

  – Out-of-synchronization sequence detection

    During out-of-synchronization sequence detection, a to-be-detected base station sends out-of-synchronization sequences in downlink slots, and its intra-frequency neighboring base stations check for the out-of-synchronization sequences in uplink slots.

    If an intra-frequency neighboring base station detects the out-of-synchronization sequences, then this base station and the to-be-detected base station are considered to be out of synchronization with each other.

    The **gNBClkResMode**.***OutOfSynSeqThld*** parameter can be reconfigured to reduce the false detection rate of out-of-synchronization sequences on the network.

- Fault determination and alarm reporting for base stations

  Based on the results of synchronization sequence detection and out-of-synchronization sequence detection between base stations, the MAE performs the following operations:

  – Groups synchronized base stations into synchronization groups.

  – Selects a synchronization group as the reference group.

  – If base station A is out of synchronization with a base station in the reference group, all base stations in the synchronization group of base station A are considered to be out of synchronization.

  If the **Send out-of-sync alarm** switch on the MAE is turned on, out-of-synchronization base stations will report ALM-29830 gNodeB Out of Clock Synchronization.

NOTICE

If the **CLK_DETECT_SW** option of the **gNodeBParam**.***ClkOutofsyncDetSwitch*** parameter is selected for the base station serving a cell that covers subways or high-speed railways, RF modules in the base station may be damaged.

Clock out-of-synchronization faults can be detected for such a base station only if both of the following conditions are met: (1) The **CLK_DETECT_IN_COMPLEX_SW** option of the **gNodeBParam**.***ClkOutofsyncDetSwitch*** parameter is selected for the base station. (2) Neighbor relationships are configured between the preceding cell and an intra-frequency low-speed cell.

11.2 Network Analysis

11.2.1 Benefits

Clock out-of-synchronization detection 1.0 can be used when a GNSS clock source is configured. This solution can detect out-of-synchronization base stations in either of the following scenarios:

- Only one base station is unsynchronized in the area where clock out-of-synchronization detection is enabled.
- Two or more base stations are unsynchronized in the area where clock out-of-synchronization detection is enabled, and there is a long distance between these base stations.

Clock out-of-synchronization detection 2.0 can be used when an IEEE 1588V2 clock source is configured or when multiple BBUs are deployed in the same equipment room. The maximum number of faulty base stations that can be detected is limited by the alarm reporting threshold on the MAE-Access.

Clock out-of-synchronization detection does not directly produce network performance gains. However, it can quickly identify unsynchronized base stations and help eliminate interference sources, thereby reducing the impact of interference on network performance.

11.2.2 Impacts

The impacts between this function and other functions are described in **11.3.2.3 Function Impacts**.

During clock out-of-synchronization detection, a base station suspected to be unsynchronized stops transmitting uplink and downlink data, which may cause a few paging messages to be discarded due to timeout. As a result, the traffic volume in its served cells slightly decreases.

11.3 Requirements

11.3.1 Licenses

None

11.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

11.3.2.1 Prerequisite Functions

None

11.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Clock out-of-synchronization detection in complex network scenarios	**CLK_DETECT_IN_COMPLEX_SW** option of the **gNodeBParam**.***ClkOutofsyncDetSwitch*** parameter	*Synchronization*	The switch controlling clock out-of-synchronization detection and the switch controlling clock out-of-synchronization detection in complex network scenarios cannot be both turned on.

11.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Remote interference management	**NRDUCellAlgoSwitch**.***RimAlgoSwitch***	*Remote Interference Management (Low-Frequency TDD)*	• A cell sending silence sequences will stop sending atmospheric duct characteristic sequences. • A cell sending out-of-synchronization sequences will stop sending atmospheric duct characteristic sequences. • A cell detecting out-of-synchronization sequences will stop detecting atmospheric duct characteristic sequences.
Low-frequency TDD	Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*	Clock out-of-synchronization detection does not take effect in hyper cells.
Low-frequency TDD	Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** and **NRDUCellMultiTrp**.***DataTransMode*** set to **BASIC_MODE** or **DPS_MODE**	*Cell Combination*	Clock out-of-synchronization detection does not take effect in combined cells.
Low-frequency TDD	Distributed massive MIMO	**DM_MIMO_SERVICE_SWITCH** option of the **NRDUCellAlgoSwitch**.***DmMimoSwitch*** parameter	*Distributed Antenna Solutions (Low-Frequency TDD)*	Clock out-of-synchronization detection does not take effect in distributed massive MIMO cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Network-wide synchronization deviation detection	**GAP_BASED_SYNC_DIFF_DET_SW** option of the **gNodeBParam**.***SyncDiffDetectSwitch*** parameter	*Synchronization*	When a resource conflict occurs between the synchronization sequences of clock out-of-synchronization detection 1.0 and sequences of network-wide synchronization deviation detection, the resources are allocated in a first-come first-served manner. This increases the missing detection rate and false detection rate of clock out-of-synchronization detection 1.0.
Low-frequency TDD	Channel calibration	**NRDUCellAlgoSwitch**.***ChnCalibSwitch***	None	Clock out-of-synchronization detection is mutually exclusive with channel calibration. These two functions cannot be enabled at the same time. If operators have enabled one function and then attempt to enable the other, a message is displayed indicating that these two functions conflict with each other.
Low-frequency TDD	RF module deep dormancy	**RRU**.***DORMANCYSW*** set to **ON** or **ON_DEEP_SUPER**	*Multi-RAT Coordinated Energy Saving*	During the period when RF module deep dormancy is in effect, clock out-of-synchronization may fail to be detected.
Low-frequency TDD	RF module super dormancy	**RRU**.***DORMANCYSW*** set to **ON_SUPER** or **ON_DEEP_SUPER**	*Multi-RAT Coordinated Energy Saving*	During the period when RF module super dormancy is in effect, clock out-of-synchronization may fail to be detected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intelligent carrier shutdown	**INTRA_GNB_MULTI_CARR_SD_SW** or **INTER_GNB_MULTI_CARR_SD_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	During the period when intelligent carrier shutdown is in effect, clock out-of-synchronization may fail to be detected.
Low-frequency TDD	Uplink symbol power saving	**UL_SYMBOL_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	When clock out-of-synchronization detection takes effect, uplink symbol power saving offers lower energy saving gains.
Low-frequency TDD	Fast carrier shutdown	Capacity-layer cells: • **INTRA_GNB_MULTI_CARR_SD_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter (intra-base-station multi-carrier scenarios) or **INTER_GNB_MULTI_CARR_SD_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter (inter-base-station multi-carrier scenarios) • **FAST_CARRIER_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingExtSwitch*** parameter Basic-layer cells: **FAST_CARR_SHUT_SERV_ASSUR_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingExtS-witch*** parameter	*Energy Conservation and Emission Reduction*	During the period when fast carrier shutdown is in effect, clock out-of-synchronization may fail to be detected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Interfering cell detection	**NRDUCellTimeIntrf**.***CrossLinkIntrfCfg***	*Standards Compliance*	During the silence detection in clock out-of-synchronization detection, cross-link sequence transmission is stopped. During the out-of-synchronization sequence detection in clock out-of-synchronization detection, cross-link sequence detection is stopped.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Extended Cell Range (TDD)	**NRDUCell**.***CellRadius*** set to a value greater than 14500 m and less than or equal to 60000 m	*Extended Cell Range*	If the switch for clock out-of-synchronization detection and the switch for time-domain interference avoidance in certain scenarios (controlled by the corresponding option of the **gNodeBAlgo**.***TimeIntrfAvoidSw*** parameter) are turned on for cells with Extended Cell Range enabled, the positions for sending and detecting synchronization sequences will be adjusted. As a result, synchronization sequence detection will fail between the base station serving these cells and any neighboring base station with the switch for time-domain interference avoidance in certain scenarios turned off. Therefore, the switch setting for time-domain interference avoidance must be the same for all cells with Extended Cell Range enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Extended Cell Range of 90 km (TDD)	**NRDUCell**.***CellRadius*** set to a value greater than 60000 m and less than or equal to 90000 m	*Extended Cell Range*	After Extended Cell Range of 90 km (TDD) is enabled in cells with the slot structure being SS518, if the switch for clock out-of-synchronization detection is turned on and the scenario-specific option in the **gNodeBAlgo**.***TimeIntrfAvoidSw*** parameter is selected, the positions for sending and detecting synchronization sequences will be adjusted in these cells. As a result, synchronization sequence detection will fail between the base station serving these cells and any neighboring base station with the scenario-specific option deselected. Therefore, the setting of the scenario-specific option **SS518_4GP_TIME_INTRF_AVOID_SW** must be the same for all cells with the slot structure being SS518.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Proactive avoidance of remote interference	**SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	*Remote Interference Management (Low-Frequency TDD)*	When proactive avoidance of remote interference is enabled: If SRSs are transmitted using the last symbol of the uplink slot, the RBs specific to SRS transmission are used for clock out-of-synchronization detection; if SRSs are not transmitted using the last symbol of the uplink slot, the idle RBs for the PUSCH are used for clock out-of-synchronization detection.
Low-frequency TDD	Proactive avoidance of remote interference	**SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	*Remote Interference Management (Low-Frequency TDD)*	When proactive avoidance of remote interference is enabled, the measured value of **N.UL.Last.Symbol13.Pwr** may be greater than expected, affecting the interference detection results of remote interference management and clock out-of-synchronization detection.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PUSCH scheduling based on SRS interference avoidance	**SRS_INTRF_AVOID_PUSCH_SCH_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	None	When PUSCH scheduling based on SRS interference avoidance is enabled, the measured value of **N.GAP.LastSymbol.Pwr** may be greater than expected, affecting the interference detection results of remote interference management and clock out-of-synchronization detection.
Low-frequency TDD	SRS remote interference avoidance	**SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	None	When both SRS remote interference avoidance and SRS interference coordination based on self-contained slots are enabled: If SRSs are transmitted using the last symbol of the uplink slot, the RBs specific to SRS transmission are used for clock out-of-synchronization detection; if SRSs are not transmitted using the last symbol of the uplink slot, the idle RBs for the PUSCH are used for clock out-of-synchronization detection. The measured SRS interference noise strength may be higher than expected when the signal to interference plus noise ratio (SINR) is high. In this case, more alarms indicating clock out-of-synchronization may be reported.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Co-MM	**INTRA_GNB_SU_COMM_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter	*Co-MM (Low-Frequency TDD)*	Clock out-of-synchronization detection may affect the downlink user-perceived rates of Co-MM UEs. It is recommended that the **INTRF_STAT_ALGO_ENH_SW** option of the **gNBClkResMode**.***ClkOutOfSyncAlgoSw*** parameter be selected to reduce the number of times clock out-of-synchronization detection is performed in the case of non-continuous external interference.
Low-frequency TDD	Single DCI	**DL_SINGLE_DCI_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoSwitch*** parameter	*Multi-Frequency Smart Aggregation*	Clock out-of-synchronization detection does not take effect for UEs for which single DCI is in effect.
Low-frequency TDD	RF module fast deep dormancy	**FAST_DEEP_DORM_SW** option of the **RRUEXTINFO**.***RRUDOROPTFEATSW*** parameter	*Multi-RAT Coordinated Energy Saving*	During the dormancy period, clock out-of-synchronization may fail to be detected.

11.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with the BBU5900, BBU5910, or BBU5900A.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

The BookBBU5901 and BookBBU5901a do not support this function.

RF Modules

All NR TDD-capable 4T4R/8T8R/32T32R/64T64R low-frequency RF modules, except for certain ones, support clock out-of-synchronization detection.

The following NR TDD-capable 4T4R/8T8R/32T32R/64T64R low-frequency RF modules do not support clock out-of-synchronization detection:

- RRU5254
- RRU5258
- RRU5836E

Cells

Clock out-of-synchronization detection does not take effect in 4T4R cells served by two combined 2T2R RF modules. For details about combined-RRU cells, see *Cell Management*.

11.3.4 Networking

- Before enabling clock out-of-synchronization detection, ensure that the time of the MAE is synchronized with that of the base station. You can log in to the MAE, run the **DSP TIME** command to query the time of any base station, and compare it with the time of the MAE. The time difference must be less than or equal to 30 seconds. Otherwise, sequence detection may time out, leading to incorrect detection results of suspected out-of-synchronization sites.
- It is recommended that clock out-of-synchronization detection be enabled on the entire network. For a base station to be enabled with this function, there is at least one cell configured with a minimum of five valid intra-frequency neighboring cells. These intra-frequency neighboring cells must be served by different base stations. The distance between the base station to be enabled with this function and its intra-frequency neighboring base stations must be less than 5 km. If the base station is not configured with intra-frequency neighboring cells or the distance between intra-frequency neighboring base stations exceeds 5 km, missing or incorrect detection may occur.
- Only the following slot structures in 4:1, 8:2, and 7:3 slot configuration support sequence detection: SS2, SS3, SS4, SS5, SS6, SS18, SS52, SS53, SS54, SS55, SS56, SS518, SS82, SS83, SS84, SS85, SS86, SS818, SS102, SS103, SS104, SS105, SS106, and SS1017.
- Clock out-of-synchronization detection does not apply when a base station serves both synchronized and out-of-synchronization cells.
- Clock out-of-synchronization detection does not apply when out-of-synchronization base stations are not configured with intra-frequency neighboring cells.

- Clock out-of-synchronization detection does not apply when the uplink interference of out-of-synchronization base stations and neighboring intra-frequency base stations does not reach the interference reporting threshold.
- If the slot structures, slot configurations, or bandwidths configured for intra-frequency cells of neighboring base stations are inconsistent, the detection results of suspected out-of-synchronization sites may be incorrect and missing detection may occur.
- When second-value hopping exists between the to-be-detected base stations and their neighboring base stations, the actual positions where data is transmitted and received differ from the indicated ones. Consequently, the base stations with second-value hopping are incorrectly considered as synchronized base stations in sequence detection and silence detection, causing missing detection of suspected out-of-synchronization base stations or synchronization faults.
- The enhanced clock out-of-synchronization detection algorithm requires that the base station version be V100R018C10 or later. If the **Enhanced out-of-synchronization detection** switch is turned on on the OSS but the base station version is earlier than V100R018C10, the out-of-synchronization fault of the base station cannot be detected.
- If the faulty base station is located at the boundary of two or more OSSs, missing or incorrect detection of clock out-of-synchronization faults may occur because the number of neighboring base stations of the faulty base station in the same OSS is insufficient.
- External interference may cause incorrect or missing out-of-synchronization detection.

11.3.5 Others

None

11.4 Operation and Maintenance

11.4.1 Data Configuration

11.4.1.1 Data Preparation

Table 11-1 describes the parameters used for function activation.

Table 11-1 Parameters used for function activation

Parameter Name	Parameter ID	Setting Notes
Clock Out-of-sync Detect Switch	**gNodeBParam**.***ClkOutofsyncDetSwitch***	By default, the **CLK_DETECT_SW** option is selected. However, it must be deselected for base stations serving cells that cover subways or high-speed railways. For such base stations, select the **CLK_DETECT_IN_COMPLEX_SW** option and deselect the **CLK_DETECT_SW** option. It is recommended that the **CLK_PROCESS_SW** option of this parameter use its default value.
Clock Out-of-sync Intrf Report Thld	**gNodeBParam**.***ClkOutofsyncIntrfRptThld***	The default value is recommended. ● A smaller value of this parameter results in a higher probability of triggering clock out-of-synchronization detection. ● A larger value of this parameter results in a lower probability of triggering clock out-of-synchronization detection.
Clock Out-of-Sync C-Intrf Report Interval	**gNodeBParam**.***ClkOutOfSyncCIntrfRptIntvl***	The default value is recommended.
Silence Detect Intrf Difference Thld	**gNodeBParam**.***SilDetIntrfDiffThld***	The default value is recommended. ● A smaller value of this parameter results in a higher probability that a base station is considered to be out-of-synchronization, and results in a possibility of false detection. ● A larger value of this parameter results in a lower probability that a base station is considered to be out-of-synchronization, and results in a possibility of missing detection.

11.4.1.2 Using MML Commands

Before using MML commands, refer to **11.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After

Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Clock out-of-synchronization detection takes effect only when the related switches are enabled on both the base station and OSS sides. For details about configurations on the OSS side, see **11.4.1.4 Using the MAE-Access**.

Enabling clock out-of-synchronization detection in scenarios other than subways and high-speed railways

```
//Enabling clock out-of-synchronization detection
MOD GNODEBPARAM: ClkOutofsyncDetSwitch=CLK_DETECT_SW-1;
//Setting the threshold for reporting the clock out-of-synchronization interference value
MOD GNODEBPARAM: ClkOutofsyncIntrfRptThld=-90, ClkOutOfSyncCIntrfRptIntvl=MIN_30;
//Setting the interference difference threshold for silence detection in clock out-of-synchronization detection
MOD GNODEBPARAM: SilDetIntrfDiffThld=15;
//(Optional) Disabling automatic deactivation of NR TDD cells served by an identified out-of-synchronization base station
MOD GNODEBPARAM: ClkOutofsyncDetSwitch=CLK_PROCESS_SW-0;
```

Enabling clock out-of-synchronization detection in subway or high-speed railway scenarios

```
//Enabling clock out-of-synchronization detection
MOD GNODEBPARAM: ClkOutofsyncDetSwitch=CLK_DETECT_IN_COMPLEX_SW-1;
//Setting the threshold for reporting the clock out-of-synchronization interference value
MOD GNODEBPARAM: ClkOutofsyncIntrfRptThld=-90, ClkOutOfSyncCIntrfRptIntvl=MIN_30;
//Setting the interference difference threshold for silence detection in clock out-of-synchronization detection
MOD GNODEBPARAM: SilDetIntrfDiffThld=15;
//(Optional) Disabling automatic deactivation of NR TDD cells served by an identified out-of-synchronization base station
MOD GNODEBPARAM: ClkOutofsyncDetSwitch=CLK_PROCESS_SW-0;
```

Deactivation Command Examples

Clock out-of-synchronization detection only needs to be deactivated on the base station side.

```
//Disabling clock out-of-synchronization detection
MOD GNODEBPARAM: ClkOutofsyncDetSwitch=CLK_DETECT_SW-0;
```

11.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

11.4.1.4 Using the MAE-Access

To configure clock out-of-synchronization detection on the MAE-Access, choose **SON** > **Clock Out-of-Sync Check** > **NR Clock Out-of-Sync Check** > **Parameter Settings**. **Table 11-2** shows the recommended parameter settings.

Table 11-2 Recommended values of parameters related to clock out-of-synchronization detection on the OSS

Parameter Name	Recommended Value
Clock out-of-sync check	On
Send out-of-sync alarm	On
Enhanced out-of-synchronization detection	On[a]
Alarm reporting threshold	5
a: Before turning on the enhanced out-of-synchronization detection switch, check the version of the base station on the OSS. It is recommended that the base station be upgraded to V100R018C10 or later before turning on this switch.	

11.4.2 Activation Verification

Automatic Detection

After clock out-of-synchronization detection is activated, wait for five minutes. Then, log in to the MAE-Access, and choose **SON** > **Clock Out-of-Sync Check** > **NR Clock Out-of-Sync Check**. If neighboring base stations suffer from clock out-of-synchronization interference, wait for a maximum of 60 minutes for the OSS to automatically report the detection results.

- After the clock out-of-synchronization cause is confirmed and the issue is rectified, manually remove the out-of-synchronization or suspected base stations from the detection results on the OSS. The alarm on these base stations will be automatically cleared.
- If more than two base stations are suspected of clock out-of-synchronization in sequence detection, silence detection will not be performed. Consequently, it cannot be determined if these are out-of-synchronization base stations.
- If the base station that has reported ALM-29830 gNodeB Out of Clock Synchronization is disconnected from the MAE-Access and the out-of-synchronization issue is solved, obtain the list of out-of-synchronization base stations or suspected base stations after the base station is connected to the MAE-Access, and delete the entries related to the base station to clear the alarm. You are not advised to mask the out-of-synchronization alarm on the base station. Otherwise, this alarm may fail to be reported later.

Manual Detection

NOTE

- When manual sequence detection and manual silence detection fall into an automatic detection period, the automatic detection is delayed to avoid a conflict with the manual detection. When manual detection is performed, the lists of out-of-synchronization base stations are not delivered to the base stations, and no alarms are reported on the base station side.
- In manual sequence detection and silence detection, only low-band TDD frequencies are selected for detection to eliminate invalid detection and reduce the detection time.

Manual synchronization sequence detection

Log in to the MAE-Access, and choose **SON** > **Clock Out-of-Sync Check** > **NR Clock Out-of-Sync Check** > **Sequence Check**. Perform manual synchronization sequence detection on a target base station. The detection results will be displayed within 20 minutes.

The detection results are displayed in two panes:

- The upper pane displays the detection results of the target base station.
- The lower pane displays the detection results of the target base station and its neighboring base stations operating on the same frequency.

When manual synchronization sequence detection is performed on only one base station, the lower pane contains the detection results of up to ten base stations. Observe the detection results of all base stations. If the results of all base stations indicate successful detection, clock out-of-synchronization detection has taken effect.

The following exceptions may occur during synchronization sequence detection:

- If the detection results include less than ten base stations, fewer than nine neighboring intra-frequency base stations meet the detection requirements. Check the neighboring cells configured for the target base station.
- If the detection results of some base stations indicate function deactivation, check the base stations for which this function is not activated.
- If the detection results of a base station indicate detection timeout, NTP time synchronization is not achieved between the MAE-Access and the base station. Contact Huawei engineers.

Manual enhanced out-of-synchronization detection

Log in to the MAE-Access, and choose **SON** > **Clock Out-of-Sync Check** > **NR Clock Out-of-Sync Check** > **Out-of-Sync List**. Select the target suspected base station to be observed. After a maximum of 65 minutes, the result of the enhanced out-of-synchronization detection is reported.

- If the target base station is synchronized, the detection result indicates synchronization.
- If the target base station is out-of-synchronization, the detection result indicates out-of-synchronization.

Manual silence detection

Log in to the MAE-Access, and choose **SON** > **Clock Out-of-Sync Check** > **NR Clock Out-of-Sync Check** > **Silence Detection**. Perform manual silence detection on a target suspected base station. The detection results will be displayed within 20 minutes.

- If the target base station is not out-of-synchronization, the silence detection results indicate synchronization.
- If the target base station is considered to be out-of-synchronization, the silence detection results indicate out-of-synchronization.

11.4.3 Network Monitoring

None

12 Intelligent Clock Fault Joint Diagnosis and Self-healing (Low-Frequency TDD)

NR TDD is a strict clock synchronization system. Base stations can be synchronized with each other through external GNSS or 1588V2 reference sources.

The synchronization system of base stations cannot completely identify the phase deviation of reference sources and the deviation introduced internally. As a result, if the fault source is not isolated in time when the deviation exceeds the threshold and the deviation is introduced to the air interface, a large-scale interference storm will occur. One out-of-synchronization base station can cause service deterioration of dozens of surrounding base stations. As a result, UEs cannot access the network or the service experience is poor. Specifically, service drops and handover failures are likely to occur.

In addition, if the synchronization system of base stations detects a fault, the reference source of the faulty base station is isolated. In this case, the faulty base station uses its internal clock for clock holdover, but only for a maximum of 24 hours. After the 24 hours, the cells served by the faulty base station are out of service.

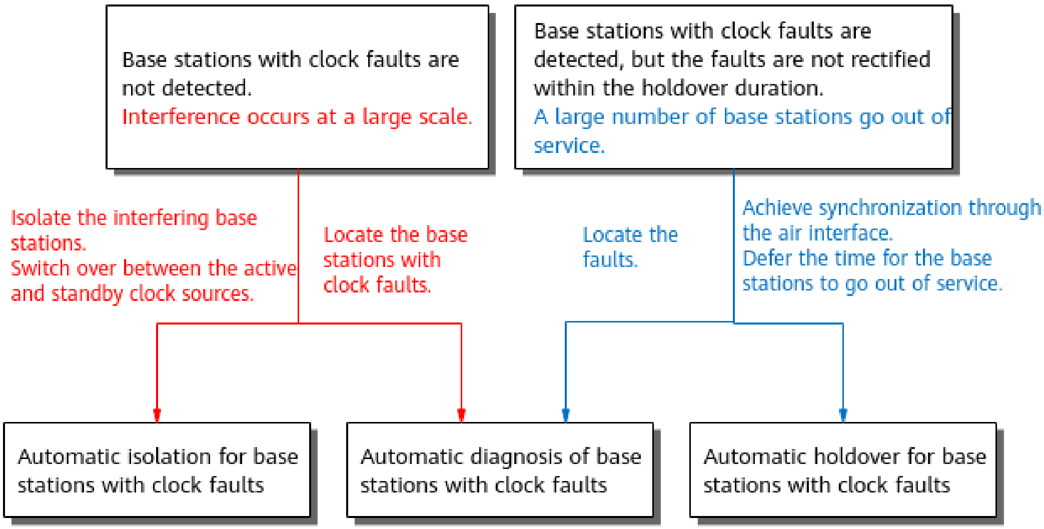
For clock synchronization faults that cannot be identified by base stations, the intelligent clock fault joint diagnosis and self-healing system shown in **Figure 12-1** can implement the following operations:

- Automatic diagnosis of base stations with clock faults: Base stations with out-of-synchronization interference can be identified based on air interface measurement and their clock faults. In addition, root cause locating information can be provided for base stations that are about to go out of service due to clock exceptions.
- Automatic isolation for base stations with clock faults: For an identified base station which interferes with surrounding base stations due to its excessive phase deviation or the out-of-synchronization state, if a standby clock source is available, the base station can switch to the standby clock source. If no standby clock source is available and the base station clock is in the holdover state, the base station clock forcibly traces the clock source. If no standby clock source is available and the base station clock is tracing the clock source,

the TDD cells go out of service to avoid interference to surrounding base stations.

- Automatic holdover for base stations with clock faults: For a base station that has a faulty clock source and enters the holdover state, the clock sources of surrounding synchronized base stations are measured through the air interface for synchronization reference to prolong the holdover duration of the base station. This prevents a large number of base stations from going out of service.

Figure 12-1 Intelligent clock fault joint diagnosis and self-healing system



Intelligent clock fault joint diagnosis and self-healing consists of the following functions:

- Network-wide synchronization deviation detection
- Multi-site joint clock fault diagnosis
- Self-isolation of clock faults
- Ultra-long holdover for sites with abnormal clock sources

NOTE

For the function of ultra-long holdover for sites with abnormal clock sources, it is recommended that the **gNodeBParam**.***GnssFailLfHoldoverDuration*** parameter be set to its default value **HOLDOVER_24_HOURS** if users need to rectify the clock fault of the base station through the air interface. If this parameter is set to another value, there is a low probability that cells served by this base station go out of service at an earlier time.

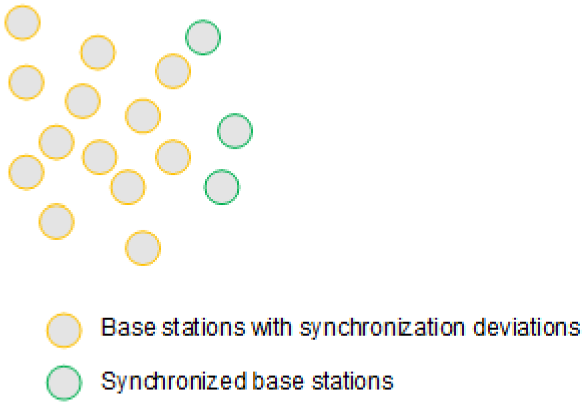
12.1 Network-wide Synchronization Deviation Detection

12.1.1 Principles

12.1.1.1 Air-Interface-based Network-wide Synchronization Deviation Detection

When multiple base stations have synchronization issues at the same time, the information about a single base station or base stations in a small area may be insufficient for quickly determining the base stations with synchronization issues. As shown in **Figure 12-2**, most of the base stations in a small area have synchronization deviations, and only few of them are synchronized. Under these circumstances, it is difficult to determine which type of base stations have synchronization issues.

Figure 12-2 Synchronization deviation



To accurately identify base stations with synchronization deviations, sufficient reference base stations are required for comparison. This function allows the inter-site synchronization deviation detection among all gNodeBs managed by the same OSS. The inter-site synchronization deviation data on the entire network is intelligently analyzed on the OSS, and the few base stations with synchronization deviations are identified by using the majority of base stations as reference base stations. Inter-site synchronization deviation is quantitatively identified through sequence detection. If base stations properly receive the characteristic sequences from each other, the bidirectional air interface delay between the base stations can be measured to calculate the deviation.

Air-interface-based network-wide synchronization deviation detection consists of the following three parts:

1. Centralized control of network-wide sequence detection on the OSS

   The sequence detection of base stations on the entire network is centrally controlled to ensure consistency and validity of the sequence detection. The control measures include specifying the NR-ARFCN, slot configuration, slot structure, detection period, detection sequence, as well as detection start time, and determining the to-be-detected base stations and their neighboring base stations.

2. Periodic execution of sequence detection by base stations

   The OSS controls the sequence, period, and NR-ARFCN of the to-be-detected base stations. These base stations periodically perform sequence detection and report the detection results to the OSS.

3. Centralized analysis of the network-wide detection data on the OSS

The OSS performs cluster analysis of the inter-site sequence detection results reported by the to-be-detected base stations on the entire network, determines the inter-site synchronization deviation and clock status of these base stations, and displays the results on the GUI. The OSS displays the list of sites with an excessive synchronization deviation and the list of suspected out-of-synchronization sites.

- Sites with an excessive synchronization deviation: base stations with a synchronization deviation greater than 3 μs from reference base stations
- Suspected out-of-synchronization sites: base stations that meet either of the following conditions:
  - They can receive the characteristic sequence from reference base stations in the previous period, but cannot in the current period.
  - They cannot receive the characteristic sequence from reference base stations in the current period, and are considered as suspected out-of-synchronization base stations in the previous period.

Air-interface-based network-wide synchronization deviation detection is implemented by the OSS and base stations on the entire network. On the base station side, this function is controlled by the **GAP_BASED_SYNC_DIFF_DET_SW** option of the **gNodeBParam**.***SyncDiffDetectSwitch*** parameter. This function requires that the base station version be V100R018C10 or later, the MAE version be V100R022C10 or later, and the **GAP Based Sync Difference Detect Switch** be turned on on the MAE.

NOTE

The original function of air-interface-based network-wide synchronization deviation detection will be removed after the MAE is upgraded to V100R024C10 or later.

After air-interface-based network-wide synchronization deviation detection is enabled, missing detection may occur in the following scenarios:

- No neighboring cell relationship has been configured between a base station and its surrounding base stations, or neighboring cell relationships have been configured but air interface transmission is interrupted. In this case, sequence detection cannot be performed between this base station and its surrounding base stations.
- The number of out-of-synchronization base stations providing continuous coverage and managed by the same OSS is greater than or equal to **Out of Sync NE Num Upper Limit**.
- A base station serves both synchronized and out-of-synchronization cells.
- Out-of-synchronization has occurred before function activation.

NOTE

- The characteristic sequence used for this function occupies one gap symbol. If there are strict requirements on the gap length, it is recommended that one gap symbol be added for this function.
- The air-interface-based network-wide synchronization deviation detection period is relevant to the number of sites. If the number of sites involved in the detection task is less than or equal to 10,000, the analysis is completed within 30 minutes after the task is started. If the number of sites involved in the detection task is greater than 10,000, the analysis is completed within 4 hours after the task is started.
- If an NE is added to or deleted from an OSS, the modification can be updated within 5 hours.
- Air-interface-based network-wide synchronization deviation detection and multi-site joint clock fault diagnosis cannot be started at the same time. If they are enabled at the same time, the former function starts 10 minutes later.

12.1.1.2 X2/Xn-based Network-wide Synchronization Deviation Detection

Air-interface-based network-wide synchronization deviation detection cannot be performed in some scenarios, for example, when transmission between base stations is interrupted or the inter-base-station synchronization deviation exceeds one symbol. To solve this issue, the X2/Xn-based network-wide synchronization deviation detection function is introduced to measure inter-base-station synchronization deviation, improving the fault detection rate in scenarios with a synchronization deviation of hundreds of microseconds.

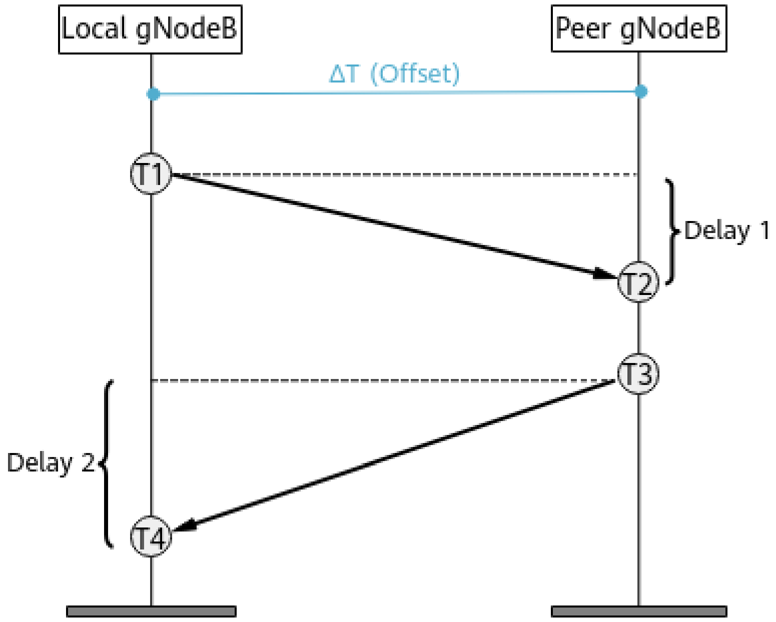
With this function, the inter-base-station synchronization deviation can be calculated using the time information exchanged between gNodeBs, which is obtained based on the IP PM packets transmitted and received over the X2 or Xn interface. The calculation method is similar to that for measuring the synchronization deviation between a base station and an IEEE 1588V2 clock server, as shown in **Figure 12-3**.

T2 – T1 = Delay 1 + Offset

T4 – T3 = Delay 2 – Offset

If Delay 1 equals Delay 2 (indicating symmetrical bidirectional delay), the synchronization deviation is calculated as follows: Offset = [(T2 – T1) – (T4 – T3)]/2.

Figure 12-3 X2/Xn-based network-wide synchronization deviation detection



X2/Xn-based network-wide synchronization deviation detection is also implemented by the OSS and base stations on the entire network. On the base station side, this function is enabled by setting the **gNBInterfaceParam**.***X2XnSyncDiffDetectSwitch*** to **ON**. On the OSS side, this function is controlled by **X2/Xn Based Sync Difference Detect Switch** on the MAE. X2/Xn-based network-wide synchronization deviation detection takes effect only when it is enabled on both the base station and OSS sides.

X2/Xn-based synchronization deviation detection requires that transmission over the X2 or Xn interface between base stations be normal and that the user-plane interface have been set up upon an inter-base-station handover. Factors such as delay variation over the X2 or Xn interface affects measurement deviation. Therefore, the precision of inter-base-station synchronization deviation measured through the X2 or Xn interface is lower than that measured through the air interface.

NOTE

In NSA networking, this function is not supported between gNodeBs when multiple NR operators work in RAN sharing mode and operator-specific NCI configuration is enabled. For details about the operator-specific NCI configuration function, see *Multi-Operator Sharing*.

12.1.2 Network Analysis

12.1.2.1 Benefits

Synchronization deviation detection helps quickly identify base stations with synchronization deviations, which reduces service costs.

12.1.2.2 Impacts

The impacts between this function and other functions are described in **12.1.3.2.3 Function Impacts**.

This function has no impact on the network.

12.1.3 Requirements

12.1.3.1 Licenses

None

12.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

12.1.3.2.1 Prerequisite Functions

None

12.1.3.2.2 Mutually Exclusive Functions

None

12.1.3.2.3 Function Impacts

The following table describes the function impacts of air-interface-based network-wide synchronization deviation detection.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*	Air-interface-based network-wide synchronization deviation detection does not take effect in hyper cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** and **NRDUCellMultiTrp**.***DataTransMode*** set to **BASIC_MODE** or **DPS_MODE**	*Cell Combination*	Air-interface-based network-wide synchronization deviation detection does not take effect in combined cells.
Low-frequency TDD	Combined-RRU cell	None	*Cell Management*	Air-interface-based network-wide synchronization deviation detection does not take effect in combined-RRU cells.
Low-frequency TDD	Timing carrier shutdown	**TIMING_CARRIER_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	If sequence detection is performed on all cells of a to-be-detected base station during the period when timing carrier shutdown takes effect, the sequence detection fails. In this case, the possible clock fault of the base station may fail to be detected.
Low-frequency TDD	RF module deep dormancy	**RRU**.***DORMANCYSW*** set to **ON** or **ON_DEEP_SUPER**	*Multi-RAT Coordinated Energy Saving*	If sequence detection is performed on all cells of a to-be-detected base station during the period when RF module deep dormancy takes effect, the sequence detection fails. In this case, the possible clock fault of the base station may fail to be detected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF module super dormancy	**RRU**.***DORMANCYSW*** set to **ON_SUPER** or **ON_DEEP_SUPER**	*Multi-RAT Coordinated Energy Saving*	If sequence detection is performed on all cells of a to-be-detected base station during the period when RF module super dormancy takes effect, the sequence detection fails. In this case, the possible clock fault of the base station may fail to be detected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Extended Cell Range (TDD)	**NRDUCell**.***CellRadius*** set to a value greater than 14500 m and less than or equal to 60000 m	*Extended Cell Range*	If the switch for air-interface-based network-wide synchronization deviation detection and the switch for time-domain interference avoidance in certain scenarios (controlled by the corresponding option of the **gNodeBAlgo**.***TimeIntrfAvoidSw*** parameter) are turned on for cells with Extended Cell Range enabled, the positions for sending and detecting synchronization sequences will be adjusted. As a result, sequence detection will fail between the base station serving these cells and any neighboring base station with the switch for time-domain interference avoidance in certain scenarios turned off. Therefore, the switch setting for time-domain interference avoidance must be the same for all cells with Extended Cell Range enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Extended Cell Range of 90 km (TDD)	**NRDUCell**.***CellRadius*** set to a value greater than 60000 m and less than or equal to 90000 m	*Extended Cell Range*	After Extended Cell Range of 90 km (TDD) is enabled in cells with the slot structure being SS518, if the switch for air-interface-based network-wide synchronization deviation detection is turned on and the scenario-specific option of the **gNodeBAlgo**.***TimeIntrfAvoidSw*** parameter is selected, the positions for sending and detecting synchronization sequences will be adjusted in these cells. As a result, sequence detection will fail between the base station serving these cells and any neighboring base station with the scenario-specific option deselected. Therefore, the setting of the following scenario-specific option must be the same for all cells with the slot structure being SS518: ● SS518_4GP_TIME_INTRF_AVOID_SW

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Clock out-of-synchronization detection	**CLK_DETECT_SW** option of the **gNodeBParam**.***ClkOutofsyncDetSwitch*** parameter	*Synchronization*	• Clock out-of-synchronization detection algorithm 1.0: There is a possibility that sequence resource conflicts occur between network-wide synchronization deviation detection and synchronization sequence detection using clock out-of-synchronization detection algorithm 1.0. When a sequence resource conflict occurs, the base station allocates the sequence resources in a first-come first-served manner, which increases the missing detection rate and false detection rate of clock out-of-synchronization detection. • Clock out-of-synchronization detection algorithm 2.0: none

The following table describes the function impacts of X2/Xn-based network-wide synchronization deviation detection.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Operator-specific NCI configuration	**gNBSharingMode**.***MultipleNCISwitch***	*Multi-Operator Sharing*	In NSA networking, when multiple NR operators work in RAN sharing mode and operator-specific NCI configuration is enabled, X2/Xn-based network-wide synchronization deviation detection does not take effect.

12.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with the BBU5900, BBU5910, or BBU5900A.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

The BookBBU5901 and BookBBU5901a do not support this function.

RF Modules

All NR TDD-capable 4T4R/8T8R/32T32R/64T64R low-frequency RF modules, except for certain ones, support network-wide synchronization deviation detection.

The following NR TDD-capable 4T4R/8T8R/32T32R/64T64R low-frequency RF modules do not support network-wide synchronization deviation detection:

- RRU5254
- RRU5258
- RRU5836E

12.1.3.4 Networking

Air-interface-based network-wide synchronization deviation detection:

- Before enabling air-interface-based network-wide synchronization deviation detection, ensure that the time of the MAE-Access is aligned with that of base stations. You can log in to the MAE-Access, run the **DSP TIME** command to query the time of any base station, and then compare the queried time with that of the MAE-Access. The time difference must be less than or equal to 2 minutes. In addition, the difference between the time of the MAE-Access and UTC time must be less than or equal to 2 minutes.
- NR TDD base stations must be configured with a proper inter-site intra-frequency neighbor relationship. Otherwise, inter-site sequence detection cannot be performed with neighboring base stations.
- This function cannot be enabled on scattered base stations, as they may fail to receive the characteristic sequence from each other due to the large inter-site distance. It is recommended that this function be enabled on all gNodeBs managed by the same OSS.
- Only the base stations working in frequency bands n38, n40, n41, n77, n78, and n79 support sequence detection.
- Only the following slot structures in 4:1, 8:2, and 7:3 slot configuration support sequence detection:

  SS2, SS3, SS4, SS5, SS6, SS18, SS52, SS53, SS54, SS55, SS56, SS518, SS82, SS83, SS84, SS85, SS86, SS818, SS102, SS103, SS104, SS105, SS106, and SS1017
- When second-value hopping exists between the to-be-detected base stations and their neighboring base stations, the actual positions where data is transmitted and received differ from the indicated ones. Consequently, the to-be-detected base stations with second-value hopping are incorrectly considered as synchronized base stations of their neighboring base stations for which detection is not enabled, causing missing detection of suspected out-of-synchronization base stations.
- If 1588 time synchronization is used on the entire network under the same OSS, select at least three geographically dispersed sites to deploy the GNSS time synchronization source.
- Air-interface-based network-wide synchronization deviation detection cannot be enabled for base stations in tunnels (including subways and high-speed railways).

NOTICE

If air-interface-based network-wide synchronization deviation detection is enabled for a base station in a tunnel, the RF module of the base station may be damaged.

X2/Xn-based network-wide synchronization deviation detection:

- In NSA networking, this function is not supported between gNodeBs when multiple NR operators work in RAN sharing mode and operator-specific NCI configuration is enabled. For details about the operator-specific NCI configuration function, see *Multi-Operator Sharing*.
- This function is not recommended on a newly deployed network where the conditions for enabling X2/Xn-based measurement are not met. This is because there may be no handovers between most base stations and the X2/Xn user plane is not set up.

- If a base station with X2/Xn-based network-wide synchronization deviation detection enabled is migrated from the original OSS to a new OSS, the detection results of this function may be incomplete for the migrated base station or the number of running base stations may be inaccurate on the original OSS.

12.1.3.5 Others

None

12.1.4 Operation and Maintenance

12.1.4.1 Data Configuration

12.1.4.1.1 Data Preparation

Air-Interface-based Network-wide Synchronization Deviation Detection

Table 12-1 describes the parameters used for function activation on the base station side.

Table 12-1 Parameters used for activating air-interface-based network-wide synchronization deviation detection on the base station side

Parameter Name	Parameter ID	Option	Setting Notes
Synchronization Difference Detect Switch	**gNodeBParam**.***SyncDiffDetectSwitch***	GAP_BASED_SYNC_DIFF_DET_SW	The default value is recommended.

Table 12-2 describes the parameters used for function activation on the OSS side.

Table 12-2 Parameters used for activating air-interface-based network-wide synchronization deviation detection on the OSS side

Parameter Name	Setting Notes
GAP Based Sync Difference Detect Switch	Set this parameter to **ON**.
NARFCN Of To-be-detected gNodeB	The default value is recommended.
Update Time Of To-be-detected gNodeB	This parameter can be set to a value ranging from 00:00:00 to 23:59:59. The default value is 8:00:00. It is recommended that this parameter be set to a value outside the range of power saving periods.

NOTE

The following parameters for air-interface-based network-wide synchronization deviation detection support only default values and cannot be configured:

- NE Num Threshold for Trig Intelligent Analysis: The default value is 10 and cannot be changed. If the total number of base stations to be detected on the entire network is less than or equal to this threshold, this function does not take effect.
- Synchronization Difference Threshold: The default value is 3 µs and cannot be changed.
- Out of Sync NE Num Upper Limit: The value is automatically calculated, which is 30% of the base stations on which air-interface-based network-wide synchronization deviation detection is executed. In addition, the value must range from 10 to 1000. Therefore, this value cannot be changed.
- Update Period Of To-be-detected gNodeB: The default value is one day and cannot be changed.

X2/Xn-based Network-wide Synchronization Deviation Detection

Table 12-3 describes the parameters used for function activation on the base station side.

Table 12-3 Parameters used for activating X2/Xn-based network-wide synchronization deviation detection on the base station side

Parameter Name	Parameter ID	Setting Notes
X2Xn Sync Difference Detect Switch	**gNBInterfacePar-am**.***X2XnSyncDiffDetect Switch***	The default value is recommended.

Table 12-4 describes the parameters used for function activation on the OSS side.

Table 12-4 Parameters used for activating X2/Xn-based network-wide synchronization deviation detection on the OSS side

Parameter Name	Setting Notes
X2/Xn Based Sync Difference Detect Switch	Set this parameter to **ON**.

12.1.4.1.2 Using MML Commands

Before using MML commands, refer to **12.1.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Network-wide synchronization deviation detection takes effect only when the switches are turned on on both the base station and OSS sides.

On the base station side

```
//Turning on the switch for air-interface-based network-wide synchronization deviation detection on the base station side
MOD GNODEBPARAM: SyncDiffDetectSwitch=GAP_BASED_SYNC_DIFF_DET_SW-1;
//Turning on the switch for X2/Xn-based network-wide synchronization deviation detection on the base station side
MOD GNBINTERFACEPARAM: X2XnSyncDiffDetectSwitch=ON;
```

On the OSS side

- Air-interface-based network-wide synchronization deviation detection

  Choose **MAE-Access** > **Maintenance** > **Clock Management** > **5G** > **Clock Fault Diagnosis Settings**. Specify the parameters listed in **Table 12-2** to enable air-interface-based network-wide synchronization deviation detection.
- X2/Xn-based network-wide synchronization deviation detection

  Choose **MAE-Access** > **Maintenance** > **Clock Management** > **5G** > **Clock Fault Diagnosis Settings** > **Synchronization Difference Detect**. Set **X2/Xn Based Sync Difference Detect Switch** to **ON** to enable X2/Xn-based network-wide synchronization deviation detection.

Deactivation Command Examples

On the OSS side

- Air-interface-based network-wide synchronization deviation detection

  Choose **MAE-Access** > **Maintenance** > **Clock Management** > **5G** > **Clock Fault Diagnosis Settings**. Set the **GAP Based Sync Difference Detect Switch** to **OFF** to disable air-interface-based network-wide synchronization deviation detection.
- X2/Xn-based network-wide synchronization deviation detection

  Choose **MAE-Access** > **Maintenance** > **Clock Management** > **5G** > **Clock Fault Diagnosis Settings** > **Synchronization Difference Detect**. Set **X2/Xn Based Sync Difference Detect Switch** to **OFF** to disable X2/Xn-based network-wide synchronization deviation detection.

On the base station side

```
//(Optional) Disabling air-interface-based network-wide synchronization deviation detection on the base station side if this function is not disabled on the OSS
MOD GNODEBPARAM: SyncDiffDetectSwitch = GAP_BASED_SYNC_DIFF_DET_SW-0;
//(Optional) Disabling X2/Xn-based network-wide synchronization deviation detection on the base station side if this function is not disabled on the OSS
MOD GNBINTERFACEPARAM: X2XnSyncDiffDetectSwitch=OFF;
```

12.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

12.1.4.2 Activation Verification

- The MAE displays the lists of sites with an excessive synchronization deviation and suspected out-of-synchronization sites detected by air-interface-based network-wide synchronization deviation detection. If the list file exported from the MAE contains the data of a certain base station, the base station has started an air-interface-based network-wide synchronization deviation detection task. The list file contains the clock detection results and inter-site synchronization deviations of all to-be-detected base stations on the entire network.
- The list file exported from the MAE contains a result table of X2/Xn-based network-wide synchronization deviation detection, in which inter-site synchronization deviations detected by this function are displayed.

12.1.4.3 Network Monitoring

None

12.2 Multi-site Joint Clock Fault Diagnosis

12.2.1 Principles

The single-site clock fault diagnosis function is available in NR TDD time synchronization. However, this function may fail to detect regional faults. The multi-site joint clock fault diagnosis function is introduced on the FMA to solve the issue of undetected clock faults. It introduces the following diagnosis actions compared with single-site clock fault diagnosis:

- Identifies sites with abnormal clocks, especially those sites that can hardly be identified by single-site clock fault diagnosis, for example, sites with an excessive phase deviation.
- Provides diagnosis information about whether sites with abnormal clocks have caused interference to neighboring sites and whether the abnormal sites provide continuous coverage. The information helps determine whether to shut down the cells of these sites.
- Provides the root cause of any abnormal clocks, detailing whether the clock source is faulty or the internal system clock is abnormal.
- In the case of phase deviation faults or frequency offset faults of the clock source: Provides diagnosis information about whether the faults are caused by clock source faults or internal errors in single reference source scenarios, and information about whether the faults are caused by the active or standby clock source faults in active/standby clock source scenarios. In addition, the diagnosis results are delivered to base stations.
- Identifies pseudolites based on the comparison of GNSS satellite card parameters, and delivers the pseudolite information to base stations.
- Provides simple fault location for sites with abnormal clock sources.
  – Identifies the sites with pseudolite interference and locates the GNSS interference source.

   – Locates the faults of sites with IEEE 1588V2 clock faults. The fault location results tell whether the IEEE 1588V2 clock source faults are introduced by clock server faults or the transport network.

NOTE

- An alarm cause indicating pseudolite interference faults has been added to ALM-26124 GNSS Satellite Card Clock Output Unavailable. This alarm is reported when pseudolite is detected by the GNSS clock source.
- Multi-site joint clock fault diagnosis and air-interface-based network-wide synchronization deviation detection cannot be started at the same time. If they are enabled at the same time, the latter function starts 10 minutes later.

The implementation is as follows:

1. Jointly performs quick scanning of the clock deviation of multiple sites. The inter-site deviation is calculated based on the air interface, and the wireless network is quickly scanned to provide quantitative data of the inter-site phase deviation.
2. Jointly analyzes the multidimensional clock data during clock fault diagnosis. The clock data of independent sites (clock adjustment amount, GNSS satellite card output, clock alarms, comparison of the phase difference between different sources, and IEEE 1588V2 clock topology), service interference data, and topology information of the local and neighboring sites are cross verified and comprehensively analyzed. In this way, clock faults are identified, and the fault source is determined.
3. Locates the GNSS interference source based on the GNSS pseudolite information. The position of the pseudolite is located based on the pseudorange between the pseudolite and each base station.

The multidimensional clock data is summarized on the OSS side as follows:

- Clock status on the base station side and abnormal changes
- Service interference situations
- Out-of-synchronization status and cell status
- Satellite card decision results
- Phase deviation measurement results over the air interface
- Comprehensive diagnosis results of the local and neighboring base stations on the OSS side

The sites with abnormal clocks are identified after the joint analysis and cross verification of the independent multidimensional clock data of multiple sites. The results are displayed on the OSS side, and delivered to base stations.

For a base station that is identified with clock faults on the OSS, this function measures the air interface phase deviation and checks whether the time sequence of the base station over the air interface is abnormal. The measurement mechanism of the air interface phase deviation is the same as the mechanism of sequence detection in network-wide synchronization deviation detection.

For an out-of-synchronization base station that has caused interference to neighboring base stations, the base station reports an alarm, facilitating clock maintenance and fault locating.

After the clock fault is rectified, operators need to manually confirm that the out-of-synchronization interference has been eliminated and delete the exception information concerning the base station on the OSS. Then, OSS sends an alarm clearance message to the base station, which will clear the alarm. When deleting base station exception information on the OSS, you are advised not to repeat the operation before the deletion is complete. The deletion takes 3 minutes. Repeated deletion operations may cause the loss of the response message.

The multi-site joint clock fault diagnosis function is controlled by the **TASM**.***MSCLKJOINTDIAGSW*** parameter on the base station side. When this function is enabled on both the base station and OSS sides, detection is automatically performed.

NOTE

- The GNSS pseudolite identification results are accurate only when the **GNSS**.***POSCHECKSW*** parameter is turned on. Therefore, do not turn off this switch.
- The GNSS interference source location requires U-BLOX M8T satellite cards, as only U-BLOX M8T satellite cards support pseudorange reporting.
- The GNSS interference source location supports the location of only spoofing pseudolites and does not support the location of the interference source caused by GNSS signal suppression.
- To improve the effectiveness of the detection, it is recommended that the **CLK_DETECT_SW** option of the **gNodeBParam**.***ClkOutofsyncDetSwitch*** parameter be selected on the base station side to enable clock out-of-synchronization detection. This improves the detection accuracy of out-of-synchronization sites.
- It is recommended that the **GAP_BASED_SYNC_DIFF_DET_SW** option of the **gNodeBParam**.***SyncDiffDetectSwitch*** parameter be set to **ON** on the base station side to enable network-wide synchronization deviation detection, improving the detection accuracy of sites with phase deviations.
- In active/standby clock source scenarios, only the faults of the active clock source can be diagnosed. The faults of the standby clock source can be identified and diagnosed only after it becomes the active clock source.
- When out-of-synchronization alarms are reported on multiple base stations and then the faults are rectified, operators need to manually clear the out-of-synchronization alarms on the base station side in the following scenario: Base stations are disconnected from the OSS and then the entries of these base stations are accidentally removed from the list of sites with clock faults on the OSS. Operators can turn off the switch specified by the **TASM**.***MSCLKJOINTDIAGSW*** parameter to clear the alarms.
  - If base stations have reported out-of-synchronization alarms and the entries of these base stations are removed from the list of sites with clock faults due to an unexpected reset or service restart on the OSS, the alarms on the base station side cannot be cleared.
  - If base stations have reported out-of-synchronization alarms and then they are disconnected from the OSS, the clock fault is later rectified, and the entries of these base stations are removed from the list of sites with clock faults on the OSS, the result is that the alarm clearance message cannot be communicated to the base stations and the alarms are not cleared on the base station side.
  - If the base stations are disconnected from the OSS and then the entries of these base stations are accidentally removed from the list of sites with clock faults, the alarm clearance message is discarded, and the alarms on the base station side cannot be cleared.
  - A base station that has gone out of synchronization before the activation of multi-site joint clock fault diagnosis cannot report an alarm after this function is enabled. Such a base station cannot be identified as an out-of-synchronization base station.
- If the base station clock phase is abnormal due to GNSS interference or 1588 clock source faults, the excessive phase deviation alarm and the alarm indicating an excessive phase deviation between the active and standby clock sources may be reported far apart in time in active/standby clock source scenarios. In this case, these two related faults are handled as two independent faults, and whether the clock fault occurs on the active or standby clock source cannot be correctly determined. The misjudgment can be corrected by measuring the phase deviation over the air interface. This issue is not solved in the current version.

12.2.2 Network Analysis

12.2.2.1 Benefits

The multi-site and joint multidimensional clock fault diagnosis helps quickly locate the base stations with clock faults and locate the source of the faults.

12.2.2.2 Impacts

The impacts between this function and other functions are described in **12.2.3.2.3 Function Impacts**.

In the root cause diagnosis of clock phase deviations/frequency offsets, the phase step deviations are determined as reference source faults and the excessive accumulated phase deviations caused by frequency jump are considered to be caused by internal errors or other reasons based on different fault models. As a result, the root cause of the fault caused by reference source frequency jump is determined as an internal error or other reasons, and the root cause of the phase deviation is incorrect.

If a base station that has reported an out-of-synchronization alarm is migrated from the original OSS to a new OSS, the alarm cannot be automatically cleared. In this case, the operators need to manually clear the alarm on the base station side.

12.2.3 Requirements

12.2.3.1 Licenses

None

12.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

12.2.3.2.1 Prerequisite Functions

None

12.2.3.2.2 Mutually Exclusive Functions

None

12.2.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Network-wide synchronization deviation detection (low-frequency TDD)	**GAP_BASED_SYNC_DIFF_DET_SW** option of the **gNodeBParam**.***SyncDiffDetectSwitch*** parameter	*Synchronization*	Currently, the multi-site joint clock fault diagnosis function switch is turned on by default, and the network-wide synchronization deviation detection function switch is turned off by default. Some diagnostic data of multi-site joint clock fault diagnosis depends on network-wide synchronization deviation detection. When the network-wide synchronization deviation detection function switch is turned off, the proportion of clock fault diagnosis results being unknown increases.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Clock out-of-synchronization detection (low-frequency TDD)	**CLK_DETECT_SW** option of the **gNodeBParam**.***ClkOutofsyncDetSwitch*** parameter	*Synchronization*	Currently, the multi-site joint clock fault diagnosis function switch is turned on by default, and the clock out-of-synchronization detection function switch is turned off by default. Some diagnostic data of multi-site joint clock fault diagnosis depends on clock out-of-synchronization detection. When the clock out-of-synchronization detection function switch is turned off, the proportion of clock fault diagnosis results being unknown increases.

12.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with the BBU5900, BBU5910, or BBU5900A.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

The BookBBU5901 and BookBBU5901a do not support this function.

RF Modules

This function does not depend on RF modules.

12.2.3.4 Networking

For details, see **12.1.3.4 Networking**.

12.2.3.5 Others

None

12.2.4 Operation and Maintenance

12.2.4.1 Data Configuration

12.2.4.1.1 Data Preparation

Table 12-5 describes the parameter used for function activation on the base station side.

Parameters in the **TASM** MO can be configured using the **SET CLKSYNCMODE** command.

Table 12-5 Parameter used for activating multi-site joint clock fault diagnosis on the base station side

Parameter Name	Parameter ID	Setting Notes
Multi-station clock joint diagnostic switch	**TASM.*MSCLKJOINTDIAGSW***	Retain the default value.

Table 12-6 describes the parameters used for function activation on the OSS.

Table 12-6 Parameters used for activating multi-site joint clock fault diagnosis on the OSS

Parameter Name	Setting Notes
Clock Fault Diagnosis Switch	It is recommended that this switch be turned on.
Fault Post-Processing	It is recommended that this switch be turned on when out-of-synchronization alarms need to be reported. Before turning on this switch, you need to turn on **Clock Fault Diagnosis Switch**.

Parameter Name	Setting Notes
Minimum Number of Detected Base Stations	The value range of this parameter is from 3 to 5000. The default value is 100. This parameter value can be adjusted as required.
Mode Decision Ratio	The value range of this parameter is from 1% to 99%. The default value is 80%. This parameter value can be adjusted as required. A smaller value of this parameter results in less accurate GNSS decision results.
gNodeB Name	Select the sites for GNSS interference source location as required. There is no default value for this parameter. This parameter needs to be set only when the GNSS interference source is manually located.
Pseudo Satellite No	Set this parameter as required. There is no default value for this parameter. This parameter needs to be set only when the GNSS interference source is manually located.
Max Closed Sites Count[a]	The default value of 3 is recommended.
a: The fault post-processing switch needs to be turned on.	

12.2.4.1.2 Using MML Commands

Before using MML commands, refer to **12.2.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Multi-site joint clock fault diagnosis is activated only when the related switches are turned on on both the base station and OSS sides.

On the base station side

```
//Turning on the multi-site joint clock fault diagnosis switch
SET CLKSYNCMODE: MSCLKJOINTDIAGSW = ON,
MSCLKDIAGPROCESSSW=CLK_SRC_PROC_SW-1&CLK_OUTOFSYNC_PROC_SW-0;
```

On the OSS side

Step 1 Choose **MAE-Access > Maintenance > Clock Management > 5G > Clock Fault Diagnosis Settings**. On the **Clock Fault Diagnosis Settings** tab page, set **Clock Fault Diagnosis Switch** to **ON**.

Step 2 (Optional) Click the **GNSS Signal Analysis** tab. Set **Minimum Number of Detected Base Stations** and **Mode Decision Ratio**, and click **Apply**.

Step 3 (Optional) Click the **GNSS Interference Positioning** tab. Set **gNodeB Name** and **Pseudo Satellite No**, and then click **Apply**.

Step 4 (Optional) Click the **Clock Fault Diagnosis Settings** tab. Set **Max Closed Sites Count** and click **Apply**.

----End

The setting notes for these parameters are described in **Table 12-6**.

After **Fault Post-Processing** is turned on on the OSS, the out-of-synchronization detection results are delivered to the base stations, which then report the out-of-synchronization alarm.

Deactivation Command Examples

On the OSS side

Choose **MAE-Access > Maintenance > Clock Management > 5G > Clock Fault Diagnosis Settings**. On the **Clock Fault Diagnosis Settings** tab page, set **Clock Fault Diagnosis Switch** to **OFF** to disable multi-site joint clock fault diagnosis.

On the base station side

```
//(Optional) Disabling multi-site joint clock fault diagnosis on the base station side if it is not disabled on the OSS side
SET CLKSYNCMODE: MSCLKJOINTDIAGSW = OFF;
```

12.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

12.2.4.2 Activation Verification

- Run the **LST CLKSYNCMODE** command on the base station side to query the execution status of the multi-site joint clock fault diagnosis function. If **Multi-station clock joint diagnostic switch** is **ON**, the multi-site joint clock fault diagnosis function has been enabled on the base station side.
- Export the list of sites with clock faults, GNSS satellite card diagnosis results, and 1588 clock fault demarcation results on the MAE. If the files exported from the MAE contain the data of a specific base station, the multi-site joint clock fault diagnosis has been started for the base station. The exported files also contain the analysis results of the base station.
  - The exported GNSS satellite card diagnosis results are described as follows:
    - The satellite installation status and satellite status are displayed by the satellite number.

  - The satellite installation status consists of 32 digits, corresponding to 32 satellites. The value 1 indicates that the satellite with the corresponding number is traced. The value 0 indicates that the satellite with the corresponding number is not traced.
  - The satellite status consists of 32 digits, corresponding to 32 satellites. The value 1 indicates that the satellite with the corresponding number is abnormal. The value 0 indicates that the satellite with the corresponding number is normal or is not traced.
  - The pseudolite ID is obtained by performing an AND operation on the values of the satellite installation status and satellite status.
- The exported 1588 clock fault demarcation results are described as follows:
  - If base stations with the same clock server ID or upper-level transmission device ID all have clock faults, the clock signal transmission of the upper-level transmission device is abnormal.
  - Base stations with the same clock server GM ID are classified into the same group. The base station group corresponds to different upper-level transmission device. If base stations in a group all have clock faults, the clock signal transmission of the clock server is abnormal.
  - If the server ID and upper-level transmission device ID are both 0, the 1588 clock synchronization link on the base station side is interrupted and the network topology identifier cannot be obtained. Alternatively, it indicates that the IEEE 1588V2 link of the NE has been deleted when an earlier OSS version is used to manage the NE. These base stations are classified into one group, and the clock faults of these base stations are caused by link interruption.
  - The IEEE 1588V2 clock is faulty because the previous-hop transmission device is abnormal. When the GNSS and IEEE 1588V2 clocks work in active/standby mode, the IEEE 1588V2 clock can function as the active or standby clock source, and faults on the previous-hop transmission device are considered as transport network faults.
- The base station that is considered out-of-synchronization on the MAE reports ALM-26262 External Clock Reference Problem with the value of **Specific Problem** being **Out of Clock Synchronization**. After the clock fault is rectified, you must clear the alarm by manually deleting the entry of the base station from the list of sites with clock faults on the MAE.
- The base station that is considered in the holdover state on the MAE continues to report ALM-26262 External Clock Reference Problem after the holdover duration elapses and the reference clock fault is not rectified. The cells are not out of service, and the value of **Long Hold Status** is **Activated** in the **DSP CLKSTAT** command output.

12.2.4.3 Network Monitoring

None

12.3 Self-isolation of Clock Faults

12.3.1 Principles

The multi-site joint clock fault diagnosis function supports post-processing. After identifying an abnormal base station, this function instructs the base station to report alarms and perform post-processing.

- When the post-processing switch is turned on and the air interface phase deviation detection results of at least one period are available, this function performs post-processing for the abnormal base station in the following scenarios:
  - The base station with clock faults has become out-of-synchronization and has caused interference to neighboring base stations: If a standby clock source is available, the base station switches to the standby clock source. If no standby clock source is available and the base station clock is in the holdover state, the base station clock forcibly traces the clock source. If no standby clock source is available and the base station clock is tracing the clock source, the TDD cells go out of service to avoid interference to neighboring base stations.
  - The phase deviation of the clock source exceeds a threshold but the base station using the clock source has not caused interference to neighboring base stations: If a standby clock source is available, the base station switches to the standby clock source. If no standby clock source is available, this function does not perform post-processing.
- When the post-processing switch is turned off, this function does not perform post-processing for abnormal base stations and only reports alarms.
- Clock drift cannot be easily identified, therefore if the out-of-synchronization fault is caused by clock drift, this root cause cannot be accurately determined. Post-processing is not performed (cells are not automatically shut down). The base station concerned reports ALM-26262 External Clock Reference Problem with the alarm cause "Out of Clock Synchronization".

NOTE

- If a base station with clock faults has caused interference to neighboring base stations and is not configured with a standby clock source, the base station automatically disables its cells if the post-processing switch is turned on for the base station. After the cells are shut down, no air interface detection mechanism is available, and the fault cannot be automatically rectified. Operators must manually confirm the fault rectification, and after that the fault can be automatically recovered.
- When the post-processing switch is turned on, an out-of-synchronization interfering base station cannot automatically go out-of-service if the NR RAT serves as the minor standard. This is because the out-of-synchronization interference is not caused by the NR clock.

12.3.2 Network Analysis

12.3.2.1 Benefits

Differentiated processing can be performed based on the clock fault diagnosis result. This way, the impact of clock faults on services can be reduced through clock source switching, forced clock source tracing, and base station shutdown.

12.3.2.2 Impacts

The impacts between this function and other functions are described in **12.3.3.2.3 Function Impacts**.

In the root cause diagnosis of clock phase deviations/frequency offsets, the phase step deviations are determined as reference source faults and the excessive accumulated phase deviations caused by frequency jump are considered to be caused by internal errors or other reasons based on different fault models. As a result, the root cause of the fault caused by reference source frequency jump is determined as an internal error or other reasons, and the root cause of the phase deviation is incorrect.

If a base station for which an out-of-synchronization alarm is reported and the TDD cells become out of service accordingly is migrated from the original OSS to a new OSS, the alarm cannot be cleared and the TDD cells cannot be restored automatically. In this case, operators need to manually clear the alarm and restore the TDD cells on the base station side.

12.3.3 Requirements

12.3.3.1 Licenses

None

12.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

12.3.3.2.1 Prerequisite Functions

None

12.3.3.2.2 Mutually Exclusive Functions

None

12.3.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Network-wide synchronization deviation detection (low-frequency TDD)	**GAP_BASED_SYNC_DIFF_DET_SW** option of the **gNodeBParam**.***SyncDiffDetectSwitch*** parameter	*Synchronization*	Currently, the multi-site joint clock fault diagnosis function switch is turned on by default, and the network-wide synchronization deviation detection function switch is turned off by default. Some diagnostic data of multi-site joint clock fault diagnosis depends on network-wide synchronization deviation detection. When the network-wide synchronization deviation detection function switch is turned off, the proportion of clock fault diagnosis results being unknown increases.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Clock out-of-synchronization detection (low-frequency TDD)	**CLK_DETECT_SW** option of the **gNodeBParam**.***ClkOutofsyncDetSwitch*** parameter	*Synchronization*	Currently, the multi-site joint clock fault diagnosis function switch is turned on by default, and the clock out-of-synchronization detection function switch is turned off by default. Some diagnostic data of multi-site joint clock fault diagnosis depends on clock out-of-synchronization detection. When the clock out-of-synchronization detection function switch is turned off, the proportion of clock fault diagnosis results being unknown increases.

12.3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with the BBU5900, BBU5910, or BBU5900A.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable 4T4R/8T8R/32T32R/64T64R low-frequency RF modules, except for certain ones, support self-isolation of clock faults.

The following NR TDD-capable 4T4R/8T8R/32T32R/64T64R low-frequency RF modules do not support self-isolation of clock faults:

- RRU5254
- RRU5258
- RRU5836E

12.3.3.4 Networking

For details, see **12.1.3.4 Networking**.

12.3.3.5 Others

None

12.3.4 Operation and Maintenance

12.3.4.1 Data Configuration

12.3.4.1.1 Data Preparation

Table 12-7 describes the parameters used for function activation on the base station side.

Parameters in the **TASM** MO can be configured using the **SET CLKSYNCMODE** command.

Table 12-7 Parameters used for activating multi-site joint clock fault diagnosis on the base station side

Parameter Name	Parameter ID	Setting Notes
Multi-Site Clock Diagnosis Post-processing Switch	**CLK_SRC_PROC_SW** option of the **TASM**.***MSCLKDIAGPROCESSSW*** parameter	It is recommended that this option be selected.
Multi-Site Clock Diagnosis Post-processing Switch	**CLK_OUTOFSYNC_PROC_SW** option of the **TASM**.***MSCLKDIAGPROCESSSW*** parameter	It is recommended that this option be deselected.

Table 12-8 describes the parameters used for function activation on the OSS.

Table 12-8 Parameters used for activating multi-site joint clock fault diagnosis on the OSS

Parameter Name	Setting Notes
Clock Fault Diagnosis Switch	It is recommended that this switch be turned on.
Fault Post-Processing	It is recommended that this switch be turned on when out-of-synchronization alarms need to be reported. Before turning on this switch, you need to turn on **Clock Fault Diagnosis Switch**.
Minimum Number of Detected Base Stations	The value range of this parameter is from 3 to 5000. The default value is 100. This parameter value can be adjusted as required.
Mode Decision Ratio	The value range of this parameter is from 1% to 99%. The default value is 80%. This parameter value can be adjusted as required. A smaller value of this parameter results in less accurate GNSS decision results.
gNodeB Name	Select the sites for GNSS interference source location as required. There is no default value for this parameter. This parameter needs to be set only when the GNSS interference source is manually located.
Pseudo Satellite No	Set this parameter as required. There is no default value for this parameter. This parameter needs to be set only when the GNSS interference source is manually located.
Max Closed Sites Count[a]	The default value of 3 is recommended.
a: The fault post-processing switch needs to be turned on.	

12.3.4.1.2 Using MML Commands

Before using MML commands, refer to **12.3.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

This function is activated only when the related switches are turned on on both the base station and OSS sides.

On the base station side

```
//Turning on the multi-site joint clock fault diagnosis switch
SET CLKSYNCMODE: MSCLKJOINTDIAGSW = ON,
MSCLKDIAGPROCESSSW=CLK_SRC_PROC_SW-1&CLK_OUTOFSYNC_PROC_SW-0;
```

On the OSS side

Step 1 Choose **MAE-Access > Maintenance > Clock Management > 5G > Clock Fault Diagnosis Settings**. On the **Clock Fault Diagnosis Settings** tab page, set **Clock Fault Diagnosis Switch** to **ON**.

Step 2 (Optional) Click the **GNSS Signal Analysis** tab. Set **Minimum Number of Detected Base Stations** and **Mode Decision Ratio**, and click **Apply**.

Step 3 (Optional) Click the **GNSS Interference Positioning** tab. Set **gNodeB Name** and **Pseudo Satellite No**, and then click **Apply**.

Step 4 (Optional) Click the **Clock Fault Diagnosis** tab. Set **Max Closed Sites Count** and click **Apply**.

----End

The setting notes for these parameters are described in **Table 12-8**.

After **Fault Post-Processing** is turned on on the OSS, the out-of-synchronization detection results are delivered to the base stations, which then report the out-of-synchronization alarm.

Deactivation Command Examples

On the OSS side

Choose **MAE-Access > Maintenance > Clock Management > 5G > Clock Fault Diagnosis Settings**. On the **Clock Fault Diagnosis Settings** tab page, set **Clock Fault Diagnosis Switch** to **OFF** to disable multi-site joint clock fault diagnosis.

On the base station side

```
//(Optional) Disabling multi-site joint clock fault diagnosis on the base station side if it is not disabled on the OSS side
SET CLKSYNCMODE: MSCLKJOINTDIAGSW = ON,
MSCLKDIAGPROCESSSW=CLK_SRC_PROC_SW-0&CLK_OUTOFSYNC_PROC_SW-0;
```

12.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

12.3.4.2 Activation Verification

- Run the **LST CLKSYNCMODE** command on the base station side to query the execution status of the multi-site joint clock fault diagnosis function. If **Multi-station clock joint diagnostic switch** is **ON**, the multi-site joint clock fault diagnosis function has been enabled on the base station side.
- Export the list of sites with clock faults, GNSS satellite card diagnosis results, and 1588 clock fault demarcation results on the MAE. If the files exported from the MAE contain the data of a specific base station, the multi-site joint clock fault diagnosis has been started for the base station. The exported files also contain the analysis results of the base station.
  - The exported GNSS satellite card diagnosis results are described as follows:
    - The satellite installation status and satellite status are displayed by the satellite number.
    - The satellite installation status consists of 32 digits, corresponding to 32 satellites. The value 1 indicates that the satellite with the corresponding number is traced. The value 0 indicates that the satellite with the corresponding number is not traced.
    - The satellite status consists of 32 digits, corresponding to 32 satellites. The value 1 indicates that the satellite with the corresponding number is abnormal. The value 0 indicates that the satellite with the corresponding number is normal or is not traced.
    - The pseudolite ID is obtained by performing an AND operation on the values of the satellite installation status and satellite status.
  - The exported 1588 clock fault demarcation results are described as follows:
    - Based on the upper-level transmission equipment ID (ID of the directly connected transmission equipment), base stations connected to the same transmission equipment are classified into the same group. If the base stations connected to the same transmission equipment all have clock faults, the clock signal transmission of the upper-level transmission equipment is abnormal.
    - Base stations with the same clock server GM ID are classified into the same group. The base station group corresponds to different upper-level transmission device. If base stations in a group all have clock faults, the clock signal transmission of the clock server is abnormal.
    - If the server ID and upper-level transmission device ID are both 0, the 1588 clock synchronization link on the base station side is interrupted and the network topology identifier cannot be obtained. Alternatively, it indicates that the IEEE 1588V2 link of the NE has been deleted when an earlier OSS version is used to manage the NE. These base stations are classified into one group, and the clock faults of these base stations are caused by link interruption.
    - The IEEE 1588V2 clock is faulty because the previous-hop transmission device is abnormal. When the GNSS and IEEE 1588V2 clocks work in active/standby mode, the IEEE 1588V2 clock can

function as the active or standby clock source, and faults on the previous-hop transmission device are considered as transport network faults.

– The base station that is considered out-of-synchronization on the MAE reports ALM-26262 External Clock Reference Problem with the value of **Specific Problem** being **Out of Clock Synchronization**. After the clock fault is rectified, you must clear the alarm by manually deleting the entry of the base station from the list of sites with clock faults on the MAE.
– The base station that is considered in the holdover state on the MAE continues to report ALM-26262 External Clock Reference Problem after the holdover duration elapses and the reference clock fault is not rectified. The cells are not out of service, and the value of **Long Hold Status** is **Activated** in the **DSP CLKSTAT** command output.

12.3.4.3 Network Monitoring

None

12.4 Ultra-Long Holdover for Sites with Abnormal Clock Sources

12.4.1 Principles

For a base station that is diagnosed with faulty clock source (by multi-site joint clock fault diagnosis on the MAE) and enters the holdover state, clock reference of surrounding synchronized base stations can be measured through the air interface and sent to the faulty base station. The faulty base station can then continue to provide services for a maximum of 14 x 24 hours.

Ultra-long holdover for sites with abnormal clock sources is implemented by the OSS and base stations on the entire network. On the base station side, this function is enabled by selecting the **CLK_OTASYNC_PROC_SW** option of the **TASM**.***MSCLKDIAGPROCESSSW*** parameter. On the OSS side, this function is enabled by turning on the **Ultra-Long Hold Switch** and **Ultra-Long Hold Enhancement Switch** on the MAE. Ultra-long holdover for sites with abnormal clock sources takes effect only when it is enabled on both the base station and OSS sides.

After this function takes effect, a base station can enter the ultra-long holdover state if either of the following conditions is met:

- The base station has entered the holdover state as the clock source is lost and unavailable for the base station.
- The base station has entered the holdover state as the phase deviation of the clock source is too large and the base station clock is isolated from this clock source.

After a base station enters the ultra-long holdover state, it will exit this state if any of the following conditions is met:

- Ultra-long holdover is disabled on the OSS side or on the base station side (by deselecting the **CLK_OTASYNC_PROC_SW** option of the **TASM**.***MSCLKDIAGPROCESSSW*** parameter).
- The **GAP_BASED_SYNC_DIFF_DET_SW** option of the **gNodeBParam**.***SyncDiffDetectSwitch*** parameter is deselected for the base station in the ultra-long holdover state or all its synchronized neighboring base stations.
- The OSS detects that the base station in the ultra-long holdover state still suffers from clock out-of-synchronization interference.
- The restoration duration of the clock source is more than 2 minutes, and the base station clock has locked the clock source again.
- Changes in air interface configurations (such as the frequency and slot configuration) or clock configurations (such as the synchronization mode and major standard for clock mutual lock) of the base station do not meet the conditions for ultra-long holdover.
- The base station has not received the ultra-long holdover clock adjustment message from the OSS for eight consecutive hours.
- The base station is upgraded or reset.

Ultra-long holdover is not supported by any of the following base stations:

- The base station clock is tracing the clock source, and the clock has phase deviation or is out of synchronization.
- The base station clock is abnormal due to internal faults.
- The holdover duration of the base station has expired, and its cells go out of service.
- The base station does not have neighboring base stations, or the signal quality over the air interface between this base station and its neighboring base stations is poor.
- The clocks are abnormal for all level-1 and level-2 neighboring base stations of this base station. Level-1 neighboring base stations refer to the neighboring base stations of this base station, and level-2 neighboring base stations refer to the neighboring base stations of these level-1 neighboring base stations.

12.4.2 Network Analysis

12.4.2.1 Benefits

For base stations with abnormal clock sources, the holdover duration after the clock source becomes faulty can be extended. This provides more time for fault rectification.

12.4.2.2 Impacts

The impacts between this function and other functions are described in **12.4.3.2.3 Function Impacts**.

The holdover duration can be extended only for some gNodeBs with clock faults. In scenarios where LTE and NR share a common clock, the holdover duration on the NR side cannot be shared with LTE.

If a base station that has entered the ultra-long holdover state is migrated from the original OSS to a new OSS, it will exit the ultra-long holdover state 8 hours after the migration.

12.4.3 Requirements

12.4.3.1 Licenses

None

12.4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

12.4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	Multi-site joint clock fault diagnosis	**TASM**.***MSCLKJOINTDIAGSW***	*Synchronization*
Low-frequency TDD	Network-wide synchronization deviation detection	**GAP_BASED_SYNC_DIFF_DET_SW** option of the **gNodeBParam**.***SyncDiffDetectSwitch*** parameter	*Synchronization*

12.4.3.2.2 Mutually Exclusive Functions

None

12.4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Timing carrier shutdown	**TIMING_CARRIER_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	Assume that the **Ultra-Long Hold Switch** is turned on but the **Ultra-Long Hold Enhancement Switch** is turned off. If sequence detection is performed in all cells on a to-be-detected base station during the period when timing carrier shutdown takes effect, the sequence detection fails. In this case, the base station cannot enter the ultra-long holdover state even if its clock is faulty, or it exits the ultra-long holdover state as sequence detection cannot be performed for a long time due to energy saving. If the **Ultra-Long Hold Enhancement Switch** is turned on, timing carrier shutdown does not take effect for the base station in the ultra-long holdover state or its reference neighboring base stations.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intelligent carrier shutdown	**INTRA_GNB_MULTI_CARR_SD_SW** or **INTER_GNB_MULTI_CARR_SD_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	Assume that the **Ultra-Long Hold Switch** is turned on but the **Ultra-Long Hold Enhancement Switch** is turned off. If sequence detection is performed in all cells on a to-be-detected base station during the period when intelligent carrier shutdown takes effect, the sequence detection fails. In this case, the base station cannot enter the ultra-long holdover state even if its clock is faulty, or it exits the ultra-long holdover state as sequence detection cannot be performed for a long time due to energy saving. If the **Ultra-Long Hold Enhancement Switch** is turned on, intelligent carrier shutdown does not take effect for the base station in the ultra-long holdover state or its reference neighboring base stations.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE and NR intelligent carrier shutdown	• LTE: **CellAlgoExtSwitch**.***LnrSmartCarrier-ShutdownSw*** • NR: **LNR_SMART_CARRIER_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Multi-RAT Coordinated Energy Saving*	Assume that the **Ultra-Long Hold Switch** is turned on but the **Ultra-Long Hold Enhancement Switch** is turned off. If sequence detection is performed in all cells on a to-be-detected base station during the period when LTE and NR intelligent carrier shutdown takes effect, the sequence detection fails. In this case, the base station cannot enter the ultra-long holdover state even if its clock is faulty, or it exits the ultra-long holdover state as sequence detection cannot be performed for a long time due to energy saving. If the **Ultra-Long Hold Enhancement Switch** is turned on, LTE and NR intelligent carrier shutdown does not take effect for the base station in the ultra-long holdover state or its reference neighboring base stations.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF module deep dormancy	**RRU**.***DORMANCYSW*** set to **ON** or **ON_DEEP_SUPER**	*Multi-RAT Coordinated Energy Saving*	Assume that the **Ultra-Long Hold Switch** is turned on but the **Ultra-Long Hold Enhancement Switch** is turned off. If sequence detection is performed in all cells on a to-be-detected base station during the period when RF module deep dormancy takes effect, the sequence detection fails. In this case, the base station cannot enter the ultra-long holdover state even if its clock is faulty, or it exits the ultra-long holdover state as sequence detection cannot be performed for a long time due to energy saving. If the **Ultra-Long Hold Enhancement Switch** is turned on, RF module deep dormancy does not take effect for the base station in the ultra-long holdover state or its reference neighboring base stations.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF module super dormancy	**RRU**.***DORMANCYSW (LTE eNodeB, 5G gNodeB)*** set to **ON_SUPER** or **ON_DEEP_SUPER**	*Multi-RAT Coordinated Energy Saving*	Assume that the **Ultra-Long Hold Switch** is turned on but the **Ultra-Long Hold Enhancement Switch** is turned off. If sequence detection is performed in all cells on a to-be-detected base station during the period when RF module super dormancy takes effect, the sequence detection fails. In this case, the base station cannot enter the ultra-long holdover state even if its clock is faulty, or it exits the ultra-long holdover state as sequence detection cannot be performed for a long time due to energy saving. If the **Ultra-Long Hold Enhancement Switch** is turned on, RF module super dormancy does not take effect for the base station in the ultra-long holdover state or its reference neighboring base stations.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Frequency band-based energy saving of wideband modules	**RRUEXTINFO**.***FREQBANDCTRLSW***	*Multi-RAT Coordinated Energy Saving*	Assume that the **Ultra-Long Hold Switch** is turned on but the **Ultra-Long Hold Enhancement Switch** is turned off. If sequence detection is performed in all cells on a to-be-detected base station during the period when "frequency band-based energy saving of wideband modules" takes effect, the sequence detection fails. In this case, the base station cannot enter the ultra-long holdover state even if its clock is faulty, or it exits the ultra-long holdover state as sequence detection cannot be performed for a long time due to energy saving. If the **Ultra-Long Hold Enhancement Switch** is turned on, "frequency band-based energy saving of wideband modules" does not take effect for the base station in the ultra-long holdover state or its reference neighboring base stations.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	Capacity-layer cells: • **INTRA_GNB_MULTI_CARR_SD_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter (intra-base-station multi-carrier scenarios) or **INTER_GNB_MULTI_CARR_SD_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter (inter-base-station multi-carrier scenarios) • **FAST_CARRIER_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingExtSwitch*** parameter Basic-layer cells: **FAST_CARR_SHUT_SERV_ASSUR_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingExtSwitch*** parameter	*Energy Conservation and Emission Reduction*	Assume that the **Ultra-Long Hold Switch** is turned on but the **Ultra-Long Hold Enhancement Switch** is turned off. If sequence detection is performed in all cells on a to-be-detected base station during the period when fast carrier shutdown takes effect, the sequence detection fails. In this case, the base station cannot enter the ultra-long holdover state even if its clock is faulty, or it exits the ultra-long holdover state as sequence detection cannot be performed for a long time due to energy saving. If the **Ultra-Long Hold Enhancement Switch** is turned on, fast carrier shutdown does not take effect for the base station in the ultra-long holdover state or its reference neighboring base stations.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Extended Cell Range (TDD)	**NRDUCell**.***CellRadius*** set to a value greater than 14500 m and less than or equal to 60000 m	*Extended Cell Range*	If the switch for ultra-long holdover for sites with abnormal clock sources and the switch for time-domain interference avoidance in certain scenarios (controlled by the corresponding option of the **gNodeBAlgo**.***TimeIntrfAvoidSw*** parameter) are turned on for cells with Extended Cell Range enabled, the positions for sending and detecting synchronization sequences will be adjusted. As a result, sequence detection will fail between the base station serving these cells and any neighboring base station with the switch for time-domain interference avoidance in certain scenarios turned off. Therefore, the switch setting for time-domain interference avoidance must be the same for all cells with Extended Cell Range enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Extended Cell Range of 90 km (TDD)	**NRDUCell**.***CellRadius*** set to a value greater than 60000 m and less than or equal to 90000 m	*Extended Cell Range*	After Extended Cell Range of 90 km (TDD) is enabled in cells with the slot structure being SS518, if the switch for ultra-long holdover for sites with abnormal clock sources is turned on and the scenario-specific option in the gNodeBAlgo.***TimeIntrfAvoidSw*** parameter is selected, the positions for sending and detecting synchronization sequences will be adjusted in these cells. As a result, sequence detection will fail between the base station serving these cells and any neighboring base station with the scenario-specific option deselected. Therefore, the setting of the following scenario-specific option must be the same for all cells with the slot structure being SS518: ● SS518_4GP_TIME_INTRF_AVOID_SW

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF module fast deep dormancy	**FAST_DEEP_DORM_SW** option of the **RRUEXTINFO**.***RRUDOROPTFEATSW*** parameter	*Multi-RAT Coordinated Energy Saving*	Assume that the **Ultra-Long Hold Switch** is turned on but the **Ultra-Long Hold Enhancement Switch** is turned off. If sequence detection is performed in all cells on a to-be-detected base station during the period when RF module fast deep dormancy takes effect, the sequence detection fails. In this case, the base station cannot enter the ultra-long holdover state even if its clock is faulty, or it exits the ultra-long holdover state as sequence detection cannot be performed for a long time due to energy saving. If the ultra-long hold enhancement switch is turned on, RF module fast deep dormancy does not take effect for the base station in the ultra-long holdover state or its reference neighboring base stations.

12.4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with the BBU5900, BBU5910, or BBU5900A.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable 4T4R/8T8R/32T32R/64T64R low-frequency RF modules, except for certain ones, support ultra-long holdover for sites with abnormal clock sources.

The following NR TDD-capable 4T4R/8T8R/32T32R/64T64R low-frequency RF modules do not support ultra-long holdover for sites with abnormal clock sources:

- RRU5254
- RRU5258
- RRU5836E

12.4.3.4 Networking

- Before enabling this function, it is recommended that air-interface-based network-wide synchronization deviation detection be enabled for all base stations managed by the same OSS. This increases the fault detection rate and prevents clock adjustment errors caused by missing detection of abnormal neighboring base stations.
- In separate-MPT clock mutual lock scenarios, this function is not supported if NR serves as the minor standard. Therefore, it is recommended that NR be configured as the major standard for clock mutual lock.
- Other networking requirements are described in **12.3.3.4 Networking**.

12.4.3.5 Others

None

12.4.4 Operation and Maintenance

12.4.4.1 Data Configuration

12.4.4.1.1 Data Preparation

Table 12-9 describes the parameters used for function activation on the base station side.

Parameters in the **TASM** MO can be configured using the **SET CLKSYNCMODE** command.

Table 12-9 Parameters used for activating ultra-long holdover for sites with abnormal clock sources on the base station side

Parameter Name	Parameter ID	Option	Setting Notes
Multi-Site Clock Diagnosis Post-processing Switch	**TASM.*MSCLKDIAGPROCESSSW***	CLK_OTASYNC_PROC_SW	This option is deselected by default. You are advised to set this option based on the network plan.

Table 12-10 describes the parameters used for function activation on the OSS side.

Table 12-10 Parameters used for activating ultra-long holdover for sites with abnormal clock sources on the OSS side

Parameter Name	Setting Notes
Ultra-Long Hold Switch	It is recommended that this switch be turned on.
Ultra-Long Hold Enhancement Switch	It is recommended that this switch be turned on. This switch can be turned on only if the **Ultra-Long Hold Switch** and the switch for air-interface-based network-wide synchronization deviation detection are turned on.

12.4.4.1.2 Using MML Commands

Before using MML commands, refer to **12.4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Ultra-long holdover for sites with abnormal clock sources takes effect only when the switches for multi-site joint clock fault diagnosis, air-interface-based network-wide synchronization deviation detection, ultra-long holdover, and ultra-long holdover enhancement are turned on on both the base station and OSS sides.

On the base station side

```
//Turning on the multi-site joint clock fault diagnosis switch and selecting the CLK_OTASYNC_PROC_SW
option on the base station side
SET CLKSYNCMODE: MSCLKJOINTDIAGSW=ON,
MSCLKDIAGPROCESSSW=CLK_SRC_PROC_SW-1&CLK_OUTOFSYNC_PROC_SW-0&CLK_OTASYNC_PROC_SW-1
;
```

On the OSS side

Step 1 Choose **MAE-Access** > **Maintenance** > **Clock Management** > **5G** > **Clock Fault Diagnosis Settings**. On the **Clock Fault Diagnosis Settings** tab page, set **Clock Fault Diagnosis Switch** and **GAP Based Sync Difference Detect Switch** to **ON**.

Step 2 On the **Clock Fault Diagnosis Settings** tab page, set **Ultra-Long Hold Switch** and **Ultra-Long Hold Enhancement Switch** to **ON**.

----End

Deactivation Command Examples

On the OSS side

Choose **MAE-Access > Maintenance > Clock Management > 5G > Clock Fault Diagnosis Settings**. On the **Clock Fault Diagnosis Settings** tab page, set **Ultra-Long Hold Switch** and **Ultra-Long Hold Enhancement Switch** to **OFF** to disable ultra-long holdover.

On the base station side

```
//(Optional) Disabling ultra-long holdover for sites with abnormal clock sources on the base station side if
it is not disabled on the OSS side
SET CLKSYNCMODE: MSCLKJOINTDIAGSW = ON,
MSCLKDIAGPROCESSSW=CLK_SRC_PROC_SW-1&CLK_OUTOFSYNC_PROC_SW-0&CLK_OTASYNC_PROC_SW-0
;
```

12.4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

12.4.4.2 Activation Verification

Run the **LST CLKSYNCMODE** command on the base station to query whether the **CLK_OTASYNC_PROC_SW** option is selected.

On the OSS side, view the **Clock Fault Diagnosis Result** on the **Clock Fault Diagnosis Settings** tab page. Five minutes after this function is enabled, trigger clock faults such as excessive phase deviation and clock source loss. Perform fault diagnosis (without affecting services), wait for 3 to 13 minutes, and then check the diagnosis results. If the results indicate that clock sources are lost and the base stations are in the holdover state, wait for about 15 minutes and then query the export file to view the measurement results of the base stations in the ultra-long handover state. Base stations in the ultra-long holdover state are those whose measurement results displayed on the MAE contain phase offsets and frequency offsets.

After the IEEE 1588V2 clock source is faulty for about 4 hours or the GNSS clock source is faulty for about 8 hours, the base station starts to synchronize the local clock with the clock signals received over the air interface and enters the state of ultra-long holdover for sites with abnormal clock sources. You can run the **DSP CLKSTAT** command to query the ultra-long holdover status of base stations and the **DSP CLKRECORD** command to query the records of ultra-long holdover status of base stations.

12.4.4.3 Network Monitoring

None

13 Comparison of Synchronization Technologies

Table 13-1 compares the common synchronization solutions for gNodeBs. (NR TDD gNodeBs support only time synchronization.)

Table 13-1 Advantages and disadvantages of common synchronization solutions

Synchronization Solution	Frequency Synchronization	Time Synchronization	Advantage	Disadvantage
GPS/ RGPS/ CGPS/ BeiDou/ Galileo	Supported	Supported	Each gNodeB is configured with a GPS/RGPS/CGPS/BeiDou/Galileo synchronization source. These synchronization sources do not require support of the network.	GPS/RGPS/CGPS/ BeiDou/Galileo hardware and related installation and maintenance incur costs.
IEEE 1588V2	Supported	Supported	1. When IEEE 1588V2 is used only for frequency synchronization, this technology supports transparent transmission across the data bearer network and has low requirements for the intermediate transmission equipment. 2. This solution supports both frequency synchronization and time synchronization. It also meets the clock requirements of the NR TDD mode. 3. IEEE 1588V2 is a standard protocol. Different profiles complying with IEEE 1588V2 support interworking between IEEE 1588V2-compliant equipment of different manufacturers.	1. To achieve time synchronization, all intermediate transmission equipment must be upgraded to support IEEE 1588. 2. The clock recovery quality is susceptible to delay, jitter, and packet loss on the data bearer network.

Synchronization Solution	Frequency Synchronization	Time Synchronization	Advantage	Disadvantage
IEEE 1588V2 and synchronous Ethernet combination	Not supported	Supported	The use of combined synchronization sources enhances time synchronization robustness and improves time holdover performance.	The IEEE 1588V2 server and synchronous Ethernet must extract clock signals from the same synchronization source.

14 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

15 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

16 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

17 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- IEEE_Std_1588-2008[1]
- T-REC-G[1].8265.1-201010
- IEEE 1588V2, "Precision Clock Synchronization Protocol for Networked Measurement and Control Systems"
- ITU-T G.8261, "Timing and Synchronization aspects in Packet Networks"
- ITU-T G.8262, "Timing characteristics of synchronous Ethernet equipment slave clock (EEC)"
- ITU-T G.8264, "Distribution of timing through packet networks"
- ITU-T G. 8265.1, Precision time protocol telecom profile for frequency synchronization
- ITU-T G.8275.1, Precision time protocol telecom profile for phase/time synchronization with full timing support from the network
- Documents in *3900 & 5900 Series Base Station Product Documentation*:
  - *GNSS Satellite Antenna System Quick Installation Guide*
  - *RGPS Satellite Antenna System Quick Installation Guide*
  - *CGPS User Guide*
- *Technical Specifications* in *3900 & 5900 Series Base Station Product Documentation*
- *IPCLK3000 User Guide* in IPCLK3000 product documentation
- *Remote Interference Management (Low-Frequency TDD)*
- *Hyper Cell*
- *Cell Combination*
- *Energy Conservation and Emission Reduction*
- *Multi-RAT Coordinated Energy Saving*
- *Cell Management*
- *Extended Cell Range*
- *Multi-Operator Sharing*
- *Co-MM (Low-Frequency TDD)*

- *Distributed Antenna Solutions (Low-Frequency TDD)*
- *Multi-Frequency Smart Aggregation*
- *License Control Item Lists*
- *Standards Compliance*